

8

7

6

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1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

CK APPD
DATE

6

0001395489

ENGINEERING RELEASED

2012-03-13

SCHEM, MLB, J30

03/12/12

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02/15/2011

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-9058	1	SCHEM, MLB, J30	SCH	CRITICAL	
820-3115	1	PCBF, MLB, J30	PCB	CRITICAL	

DRAWING

TITLE=MLB

ABBREV=DRAWING

LAST_MODIFIED=Thu Mar 13 14:00:17 2012

DRAWING TITLE

SCHEM, MLB, J30

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REVISION

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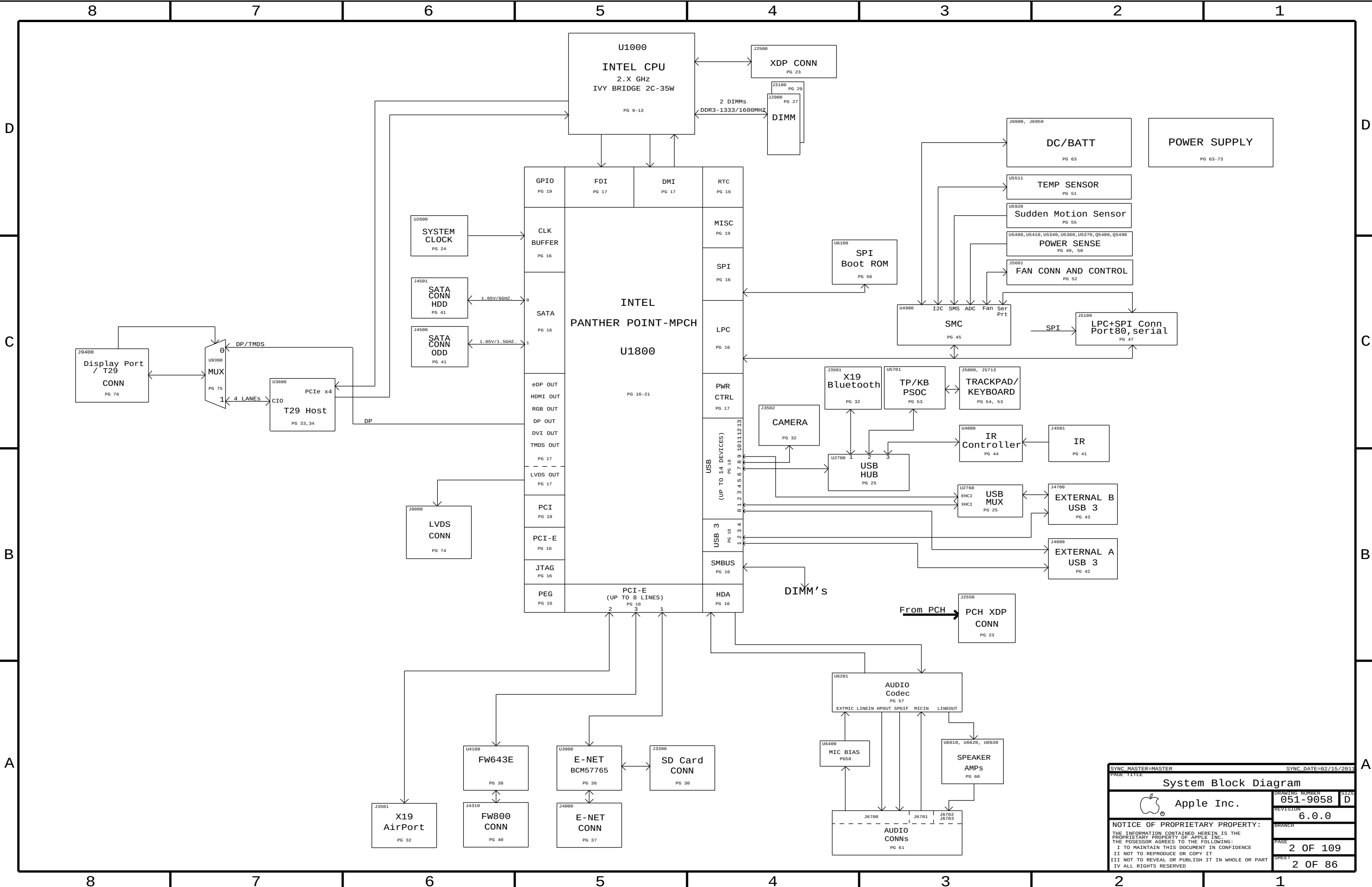
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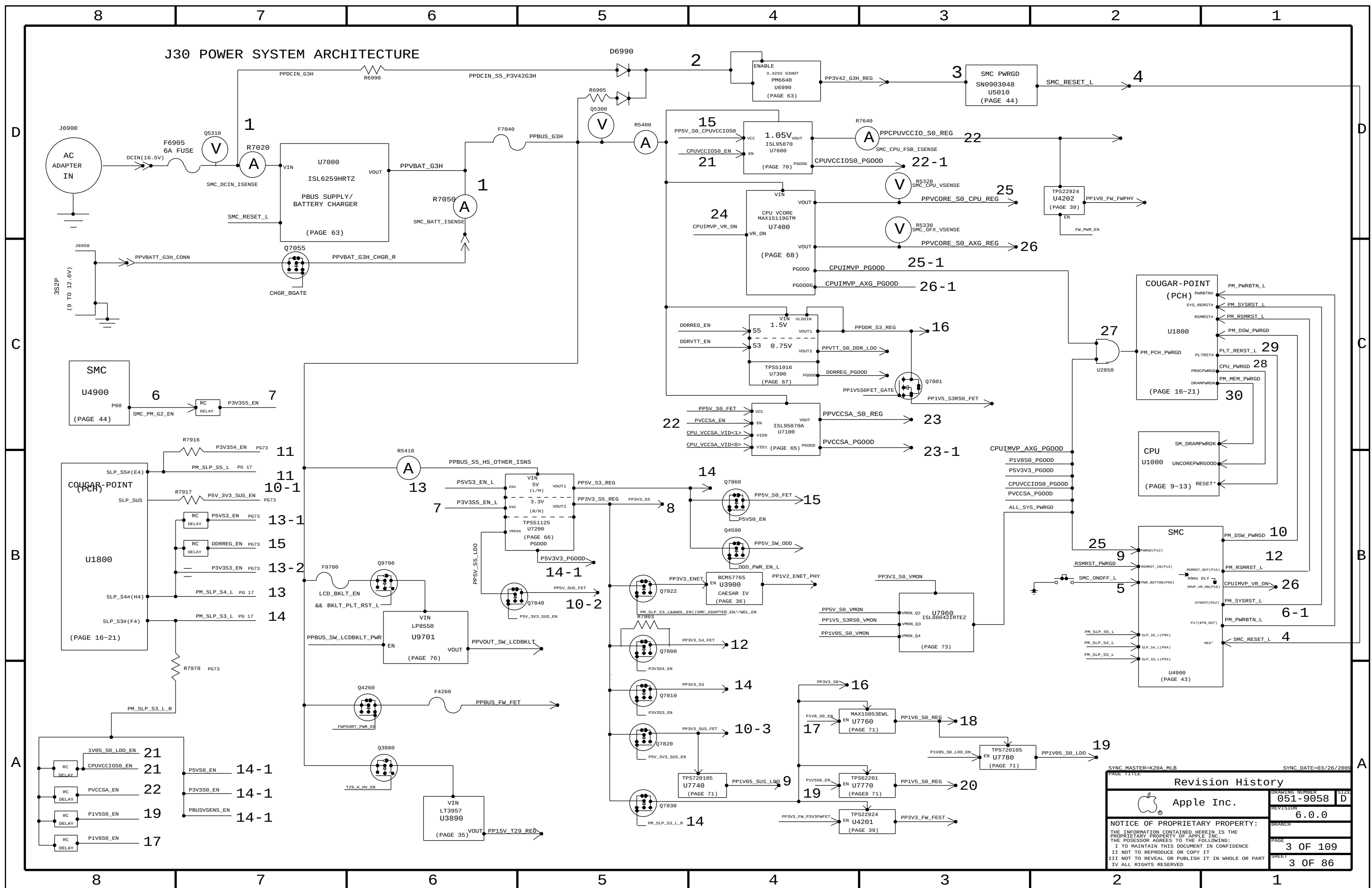
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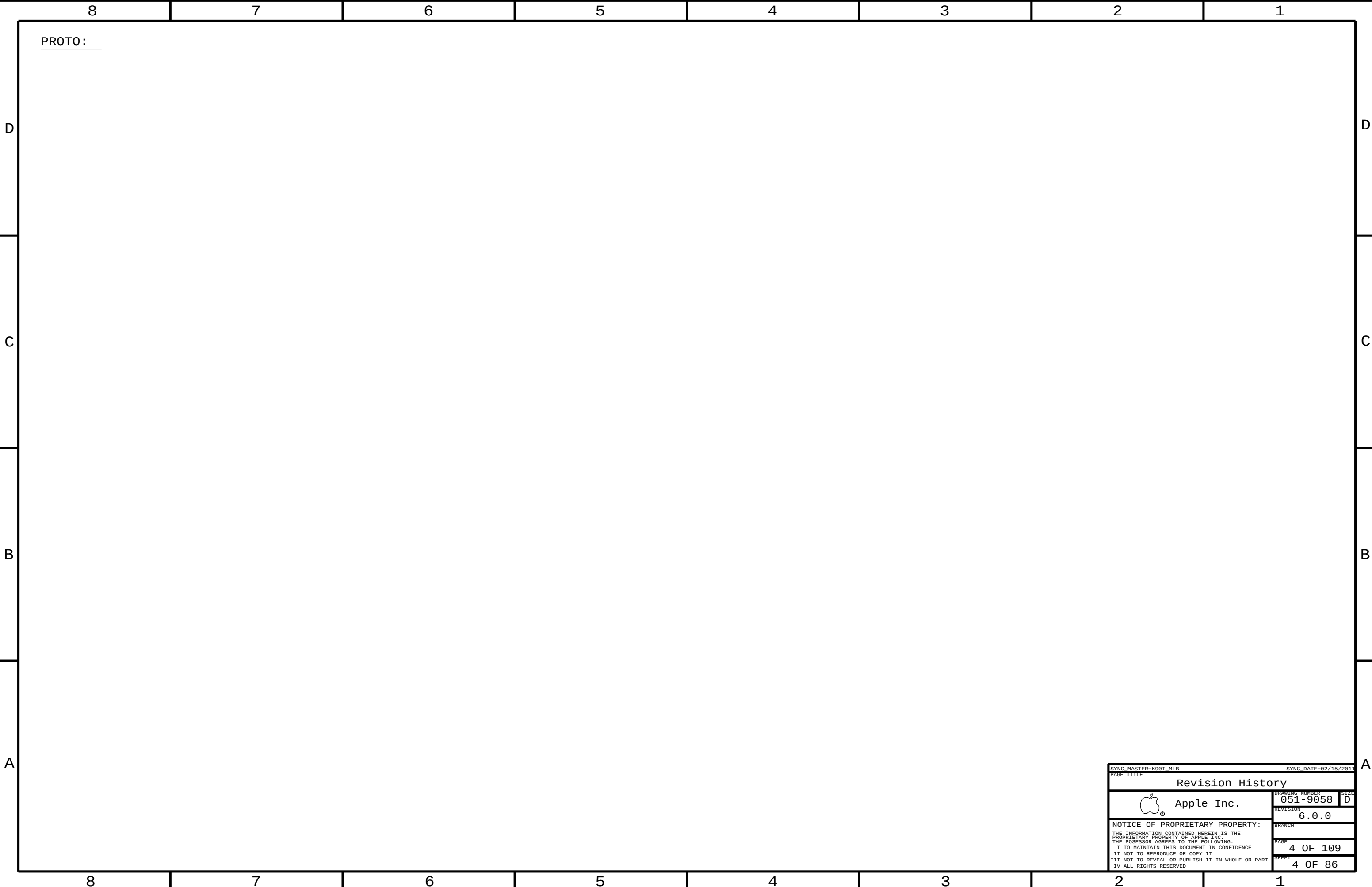
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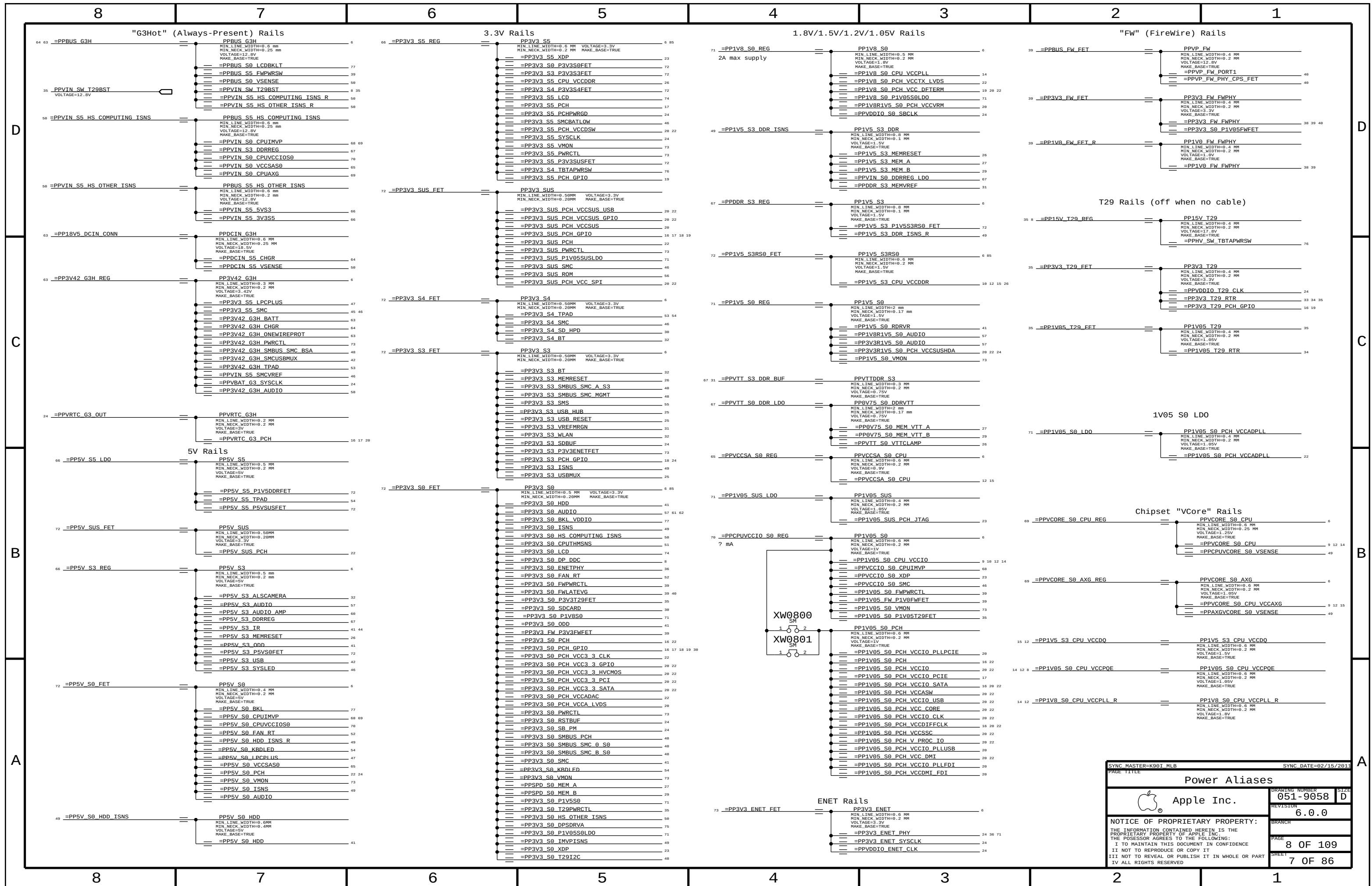
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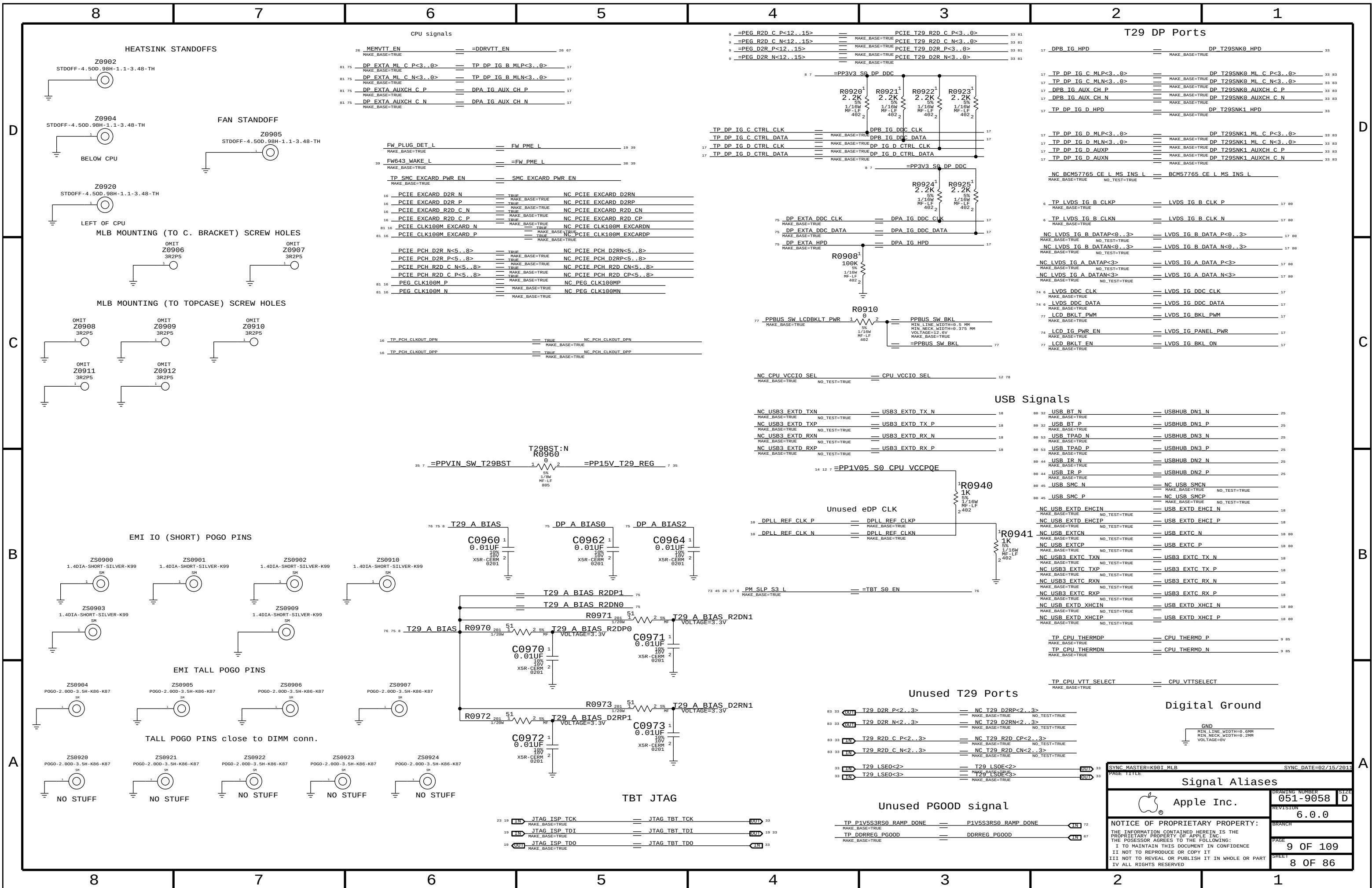
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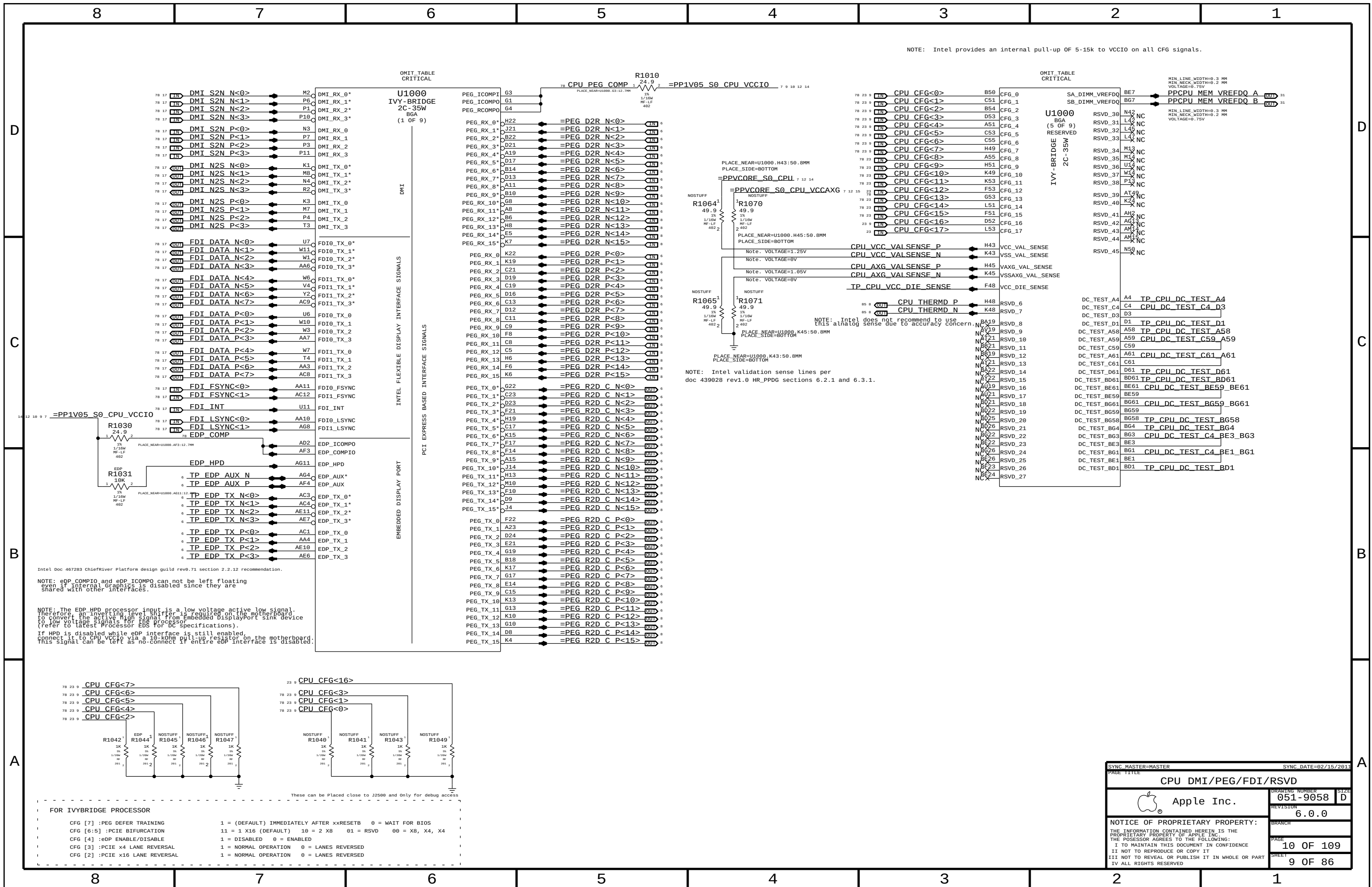
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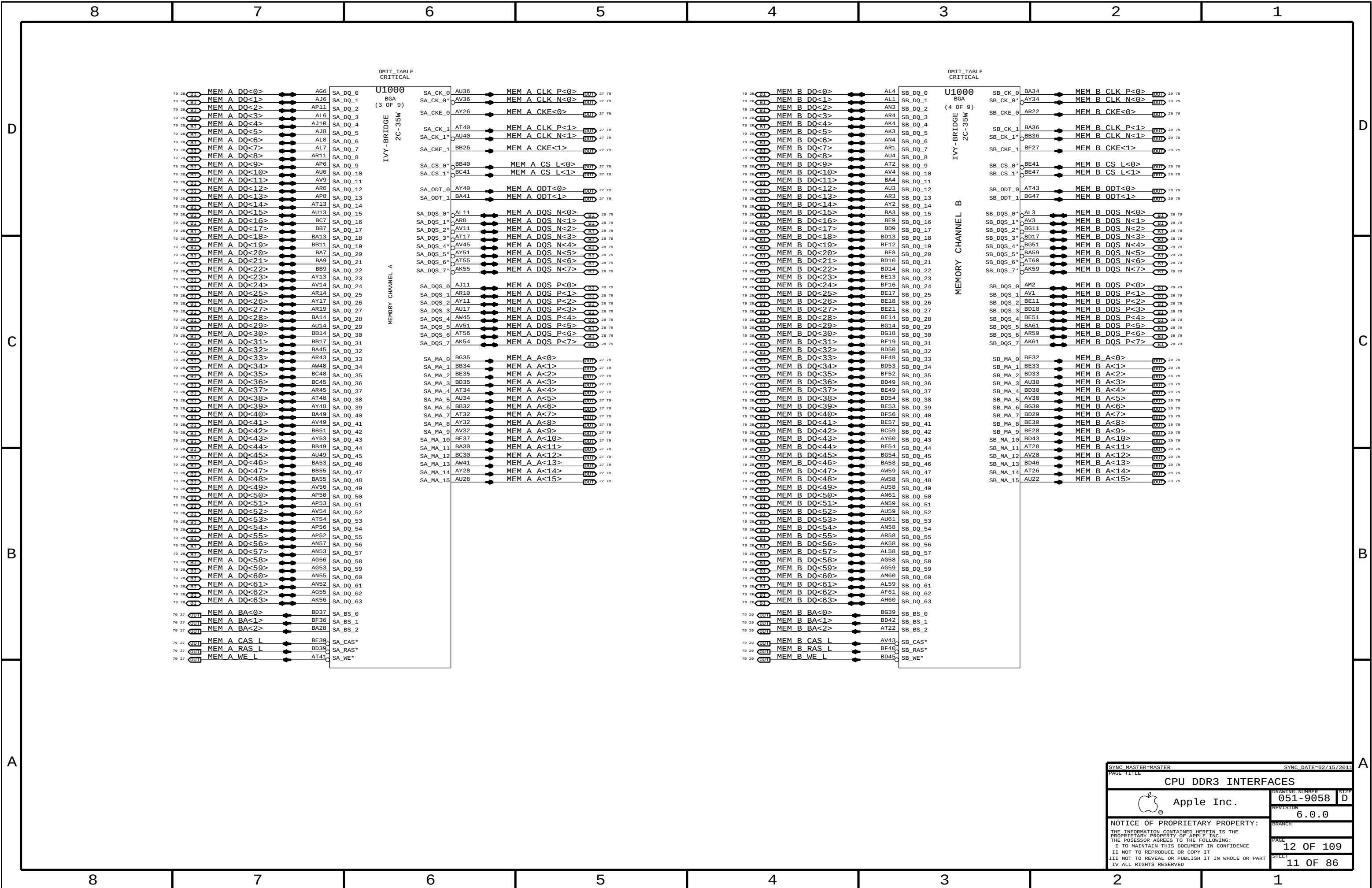
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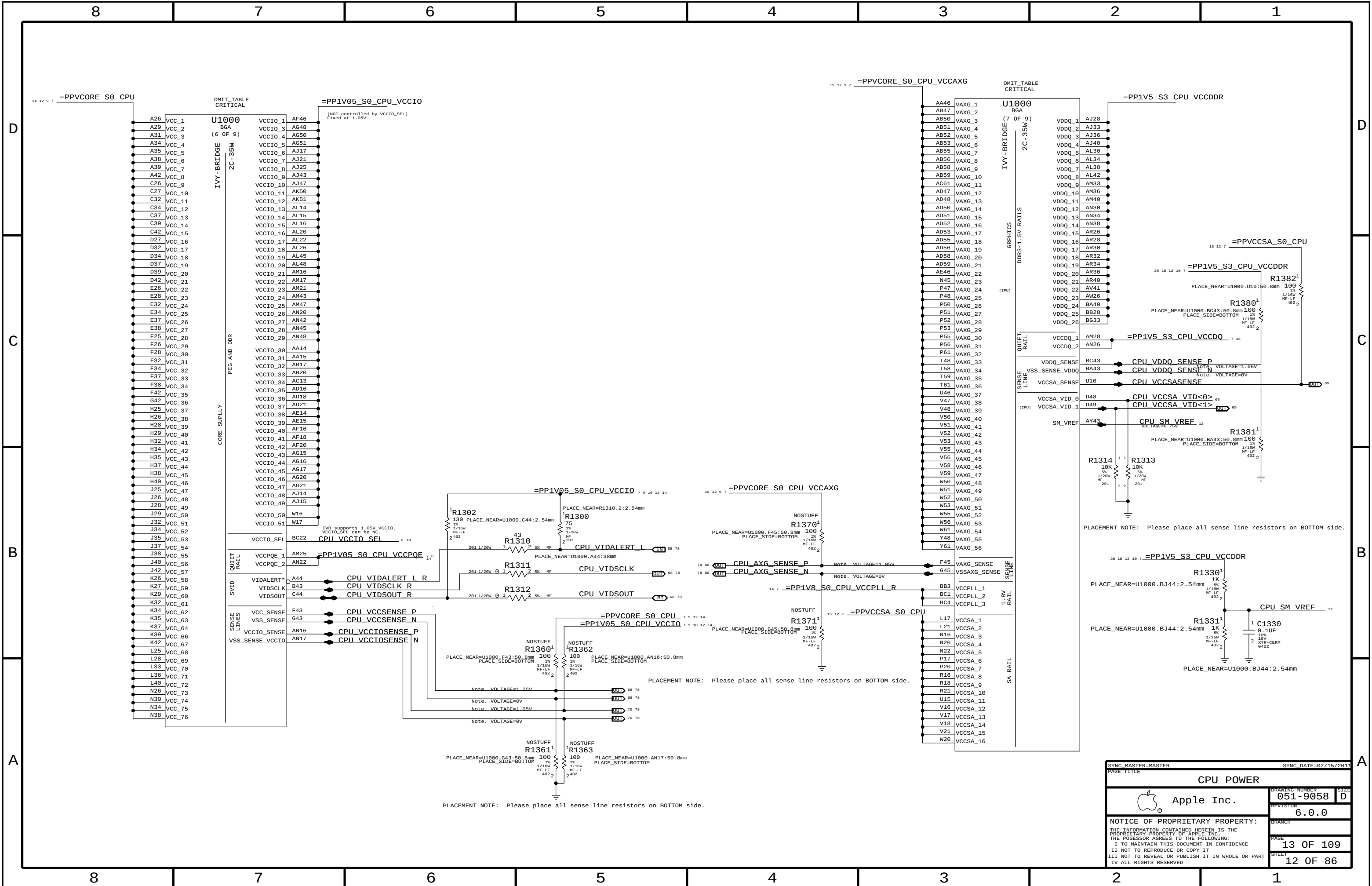
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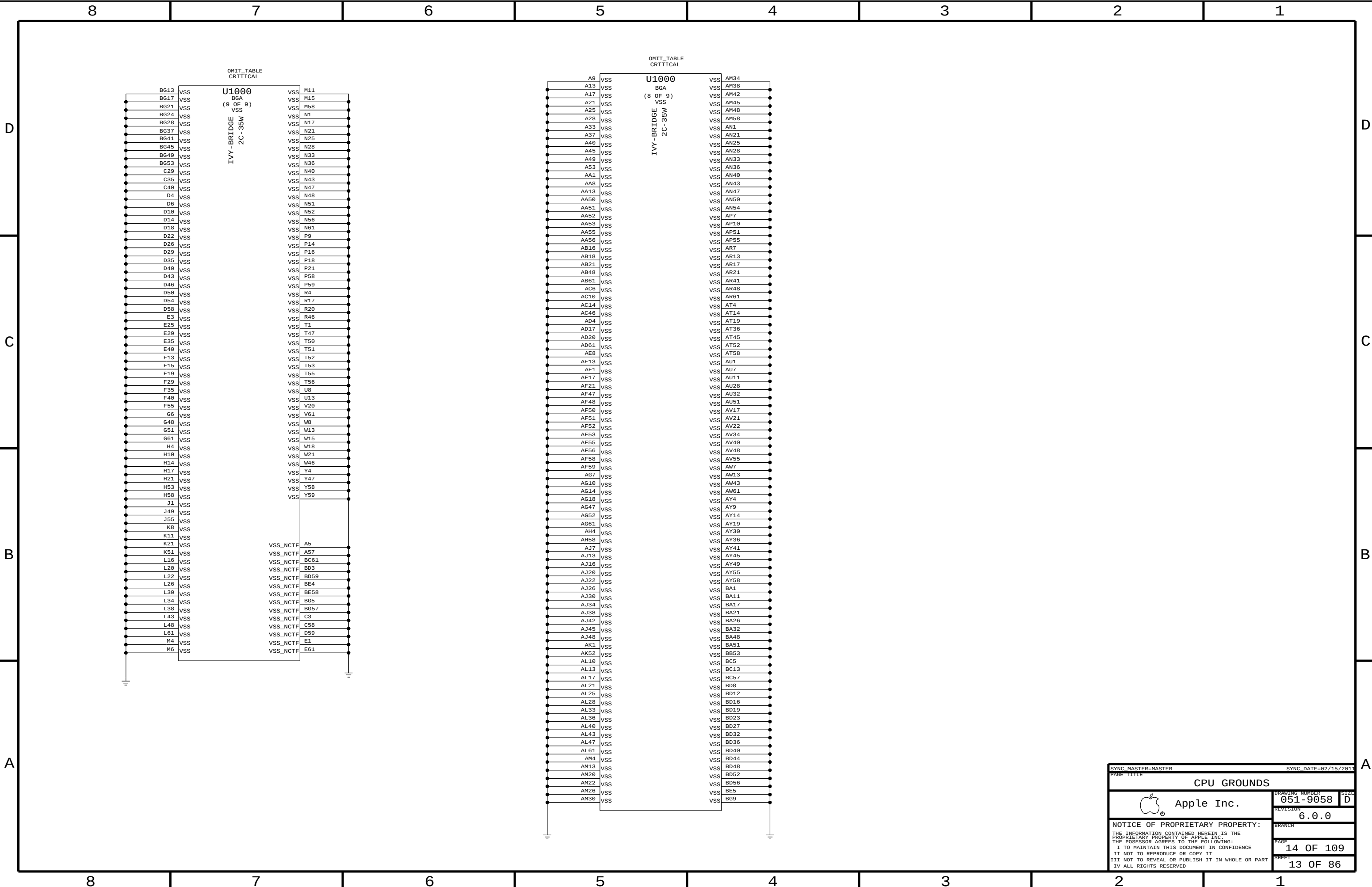






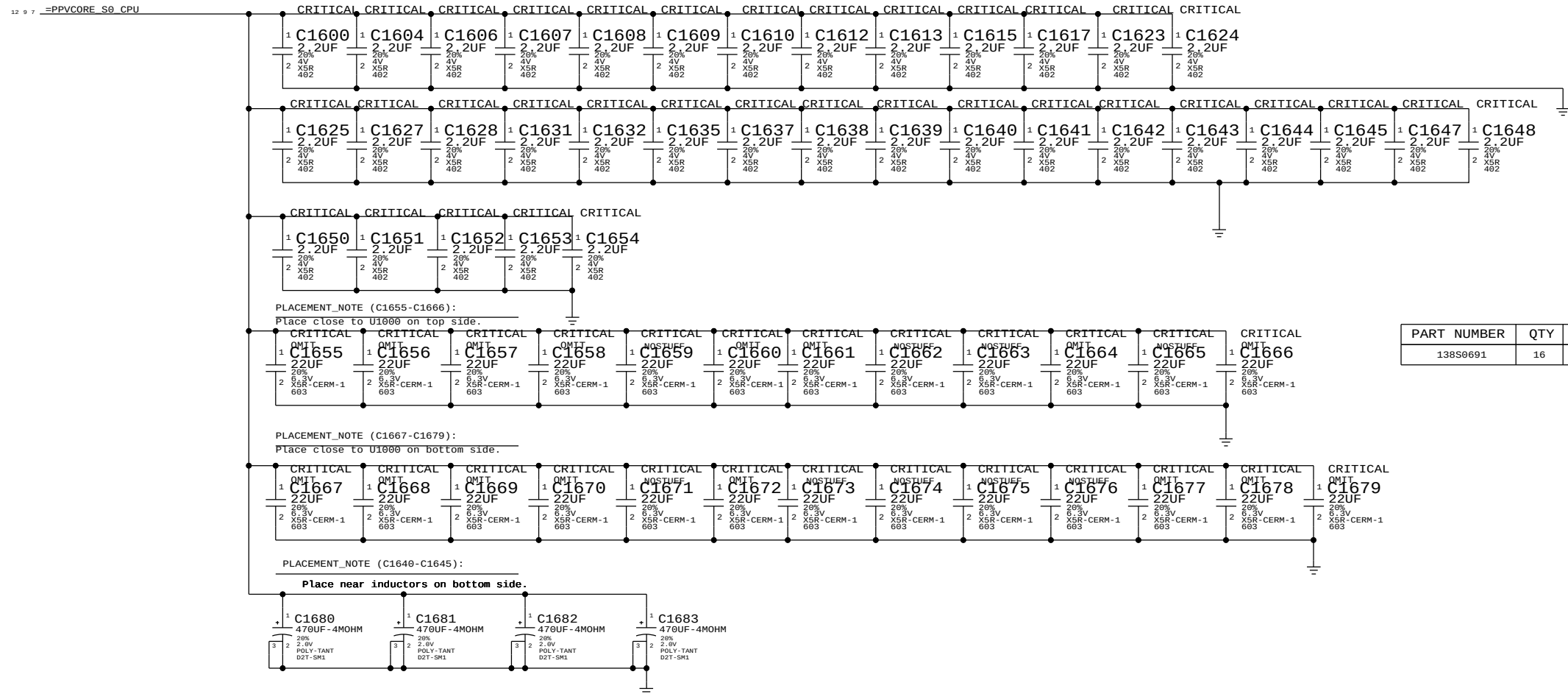


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CPU VCORE DECOUPLING

Intel recommendation (Section 6.2): 35x 2.2uF, 25x 22uF, 4x 470uF



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S9691	16	CAP, CER, X5R, 22uF, 20%, 6.3V, 0603, SAMSUNG		CRITICAL	

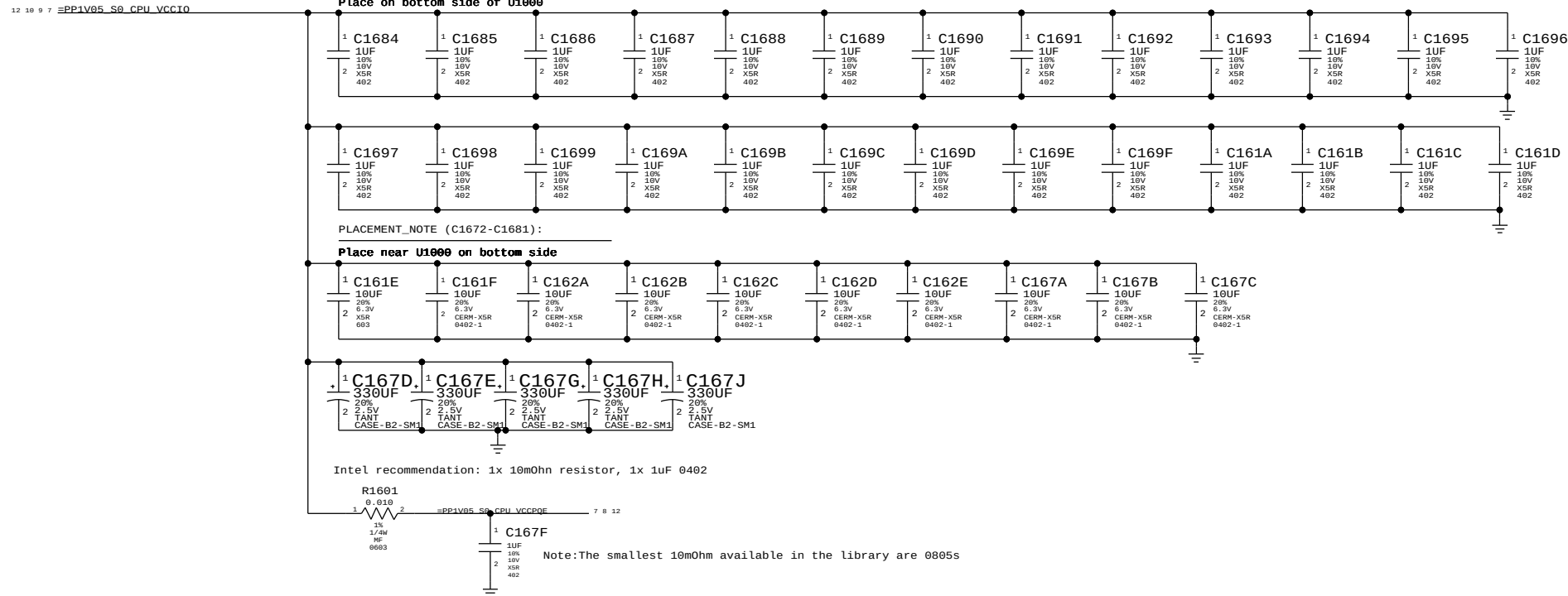
C1655, C1660, C1661, C1664, C1666, C1667, C1670, C1677, C1678, C1679, C1657, C1672, C1658, C1669, C1668, C1656

CPU VCCIO/VCCPQ DECOUPLING

Intel recommendation (Section 6.5): 26x 1uF, 10x 10uF, 2x 330uF

PLACEMENT_NOTE (C1684-C167F):

Place on bottom side of U1000

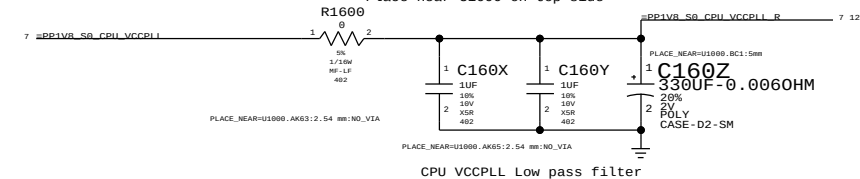


CPU VCCPLL DECOUPLING

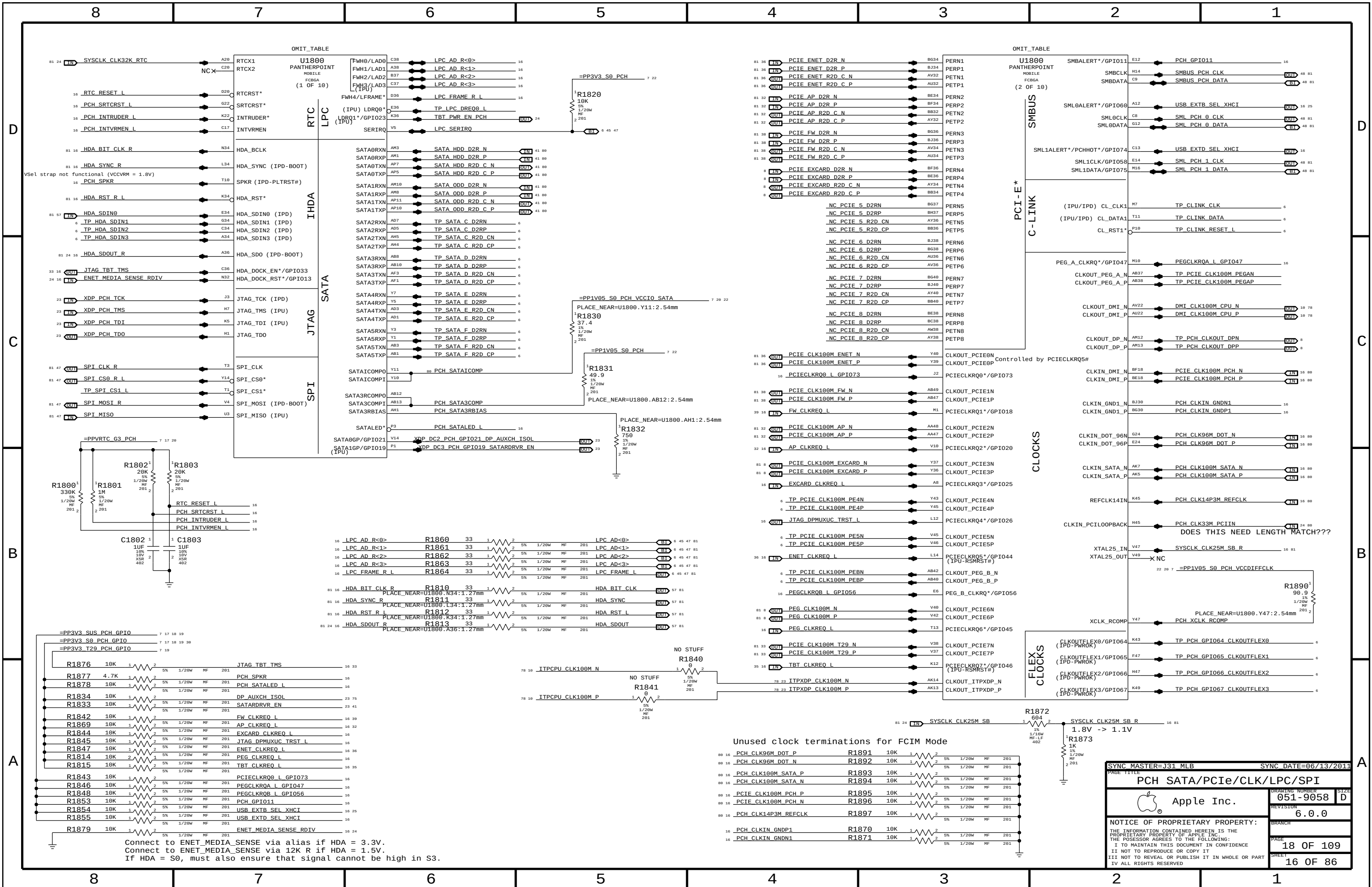
Intel recommendation (section 6.4): 2x 1uF, 1x 330uF

PLACEMENT_NOTE (C1646-C1671):

Place near U1000 on top side



SYMC MASTER-JACK J30		SYMC DATE-09/27/2011	
PAGE TITLE			
CPU DECOUPLING-I			
	Apple Inc.	DRAWING NUMBER	051-9058
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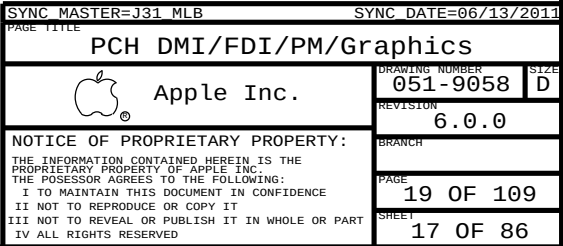


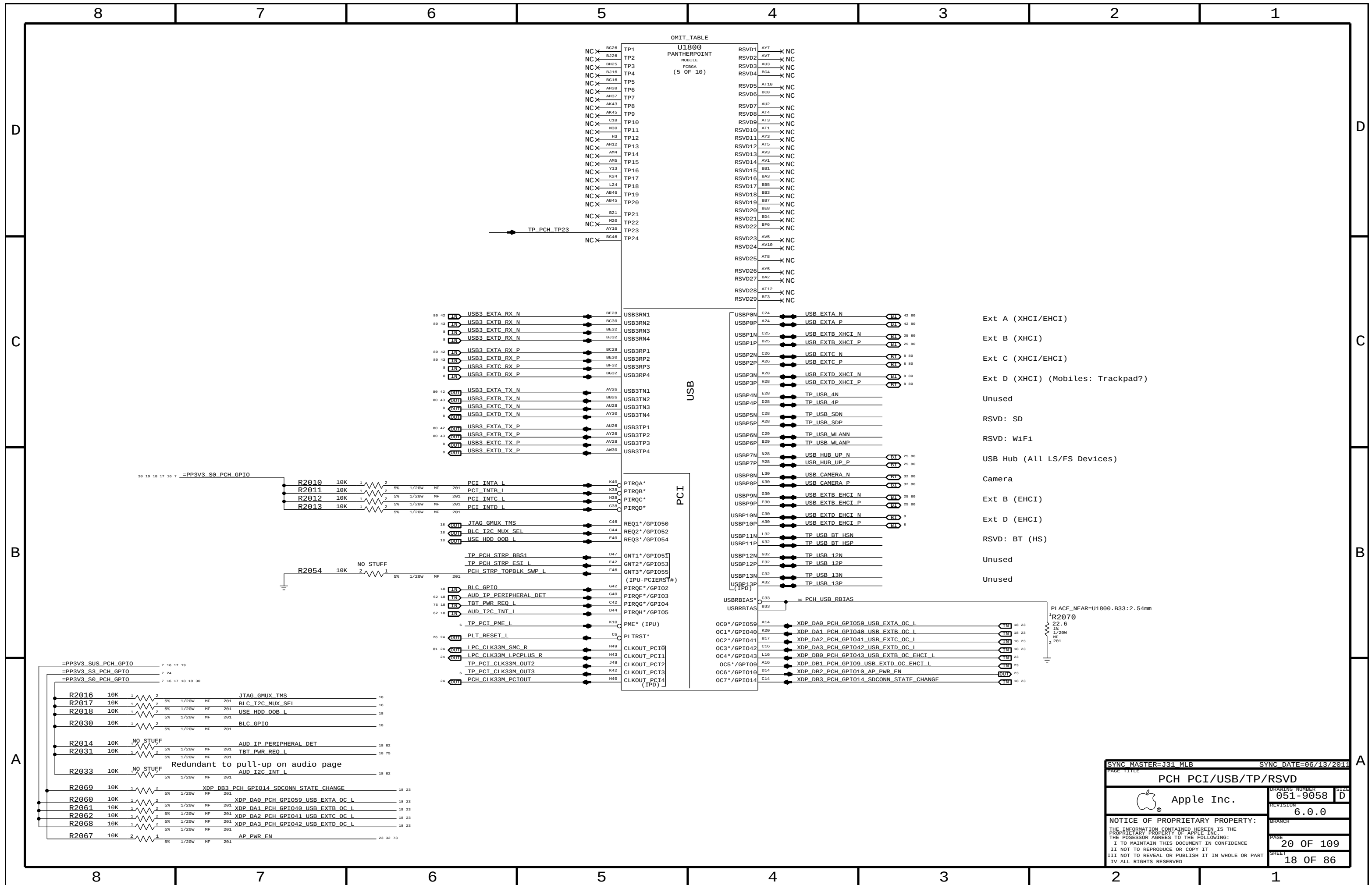
Connect to ENET_MEDIA_SENSE via alias if HDA = 3.3V.
Connect to ENET_MEDIA_SENSE via 12K R if HDA = 1.5V.
If HDA = S0, must also ensure that signal cannot be high in S3.

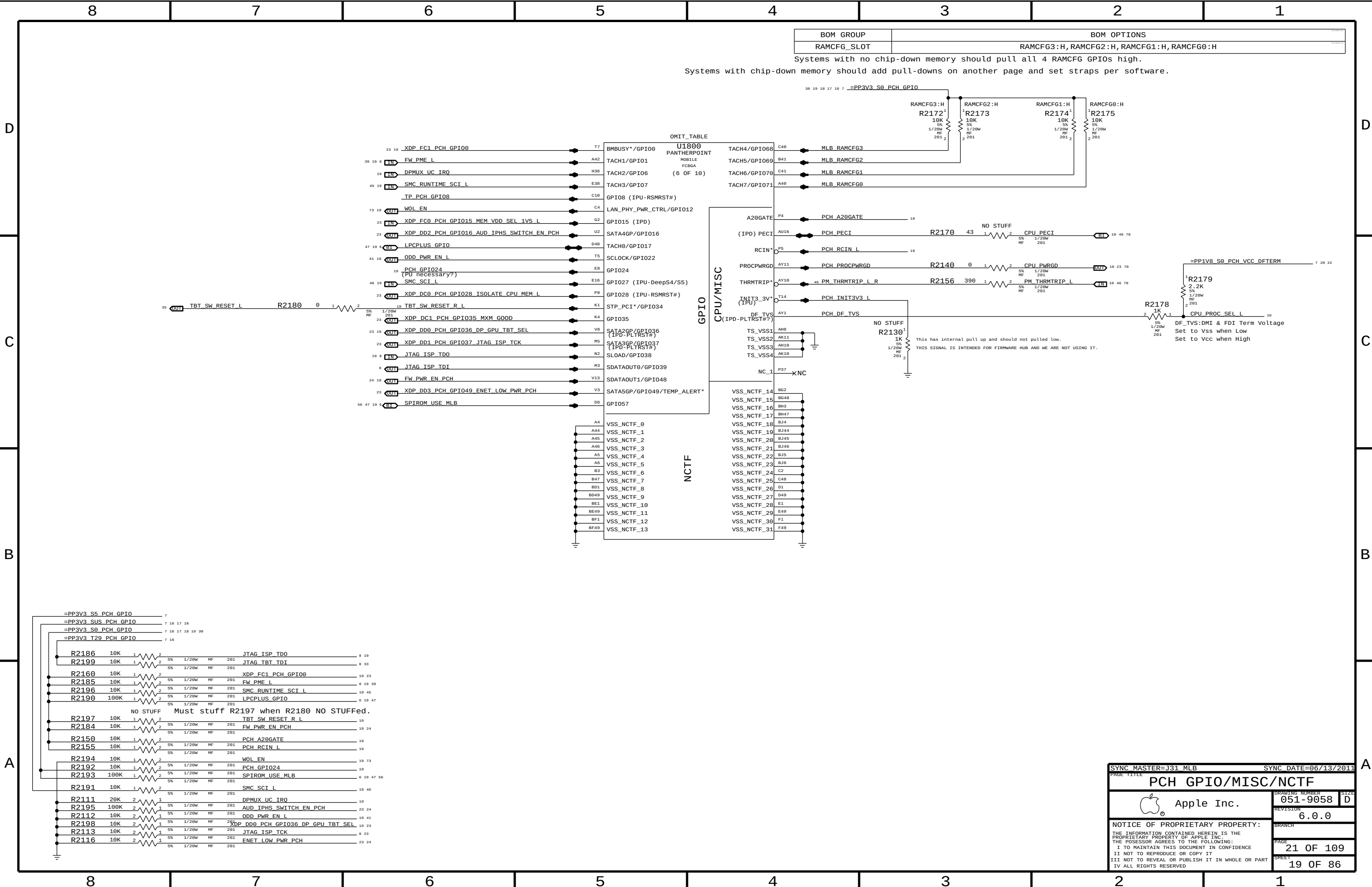
Unused clock terminations for FCIM Mode

88 16	PCH CLK96M_DOT_P	R1891	10K	1	2	5%	1/20W	MF	201
88 16	PCH CLK96M_DOT_N	R1892	10K	1	2	5%	1/20W	MF	201
88 16	PCH CLK100M_SATA_P	R1893	10K	1	2	5%	1/20W	MF	201
88 16	PCH CLK100M_SATA_N	R1894	10K	1	2	5%	1/20W	MF	201
88 16	PCI_E_CLK100M_PCH_P	R1895	10K	1	2	5%	1/20W	MF	201
88 16	PCI_E_CLK100M_PCH_N	R1896	10K	1	2	5%	1/20W	MF	201
88 16	PCH CLK14P3M_REFCLK	R1897	10K	1	2	5%	1/20W	MF	201
16	PCH CLKIN_GNDP1	R1870	10K	1	2	5%	1/20W	MF	201
16	PCH CLKIN_GNDN1	R1871	10K	1	2	5%	1/20W	MF	201

SYNC MASTER=J31 MLB		SYNC DATE=06/13/2011	
PAGE TITLE		PCH SATA/PCIe/CLK/LPC/SPI	
Apple Inc.		DRAWING NUMBER	051-9058
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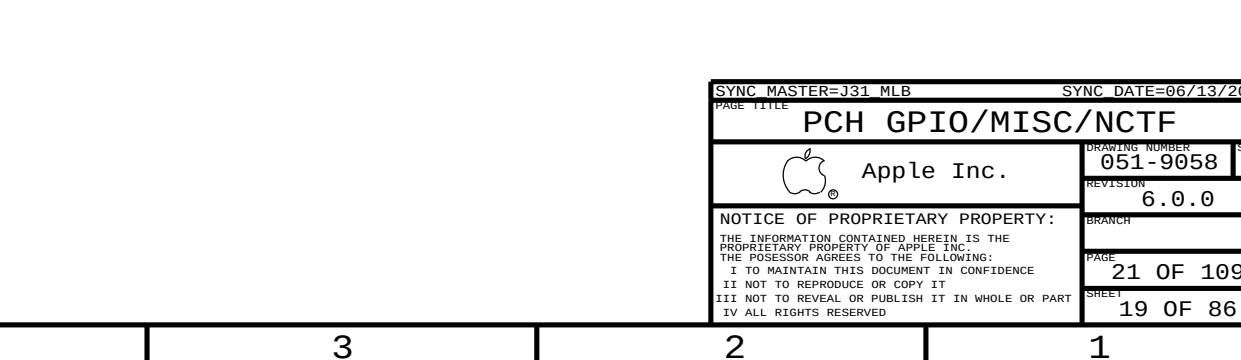
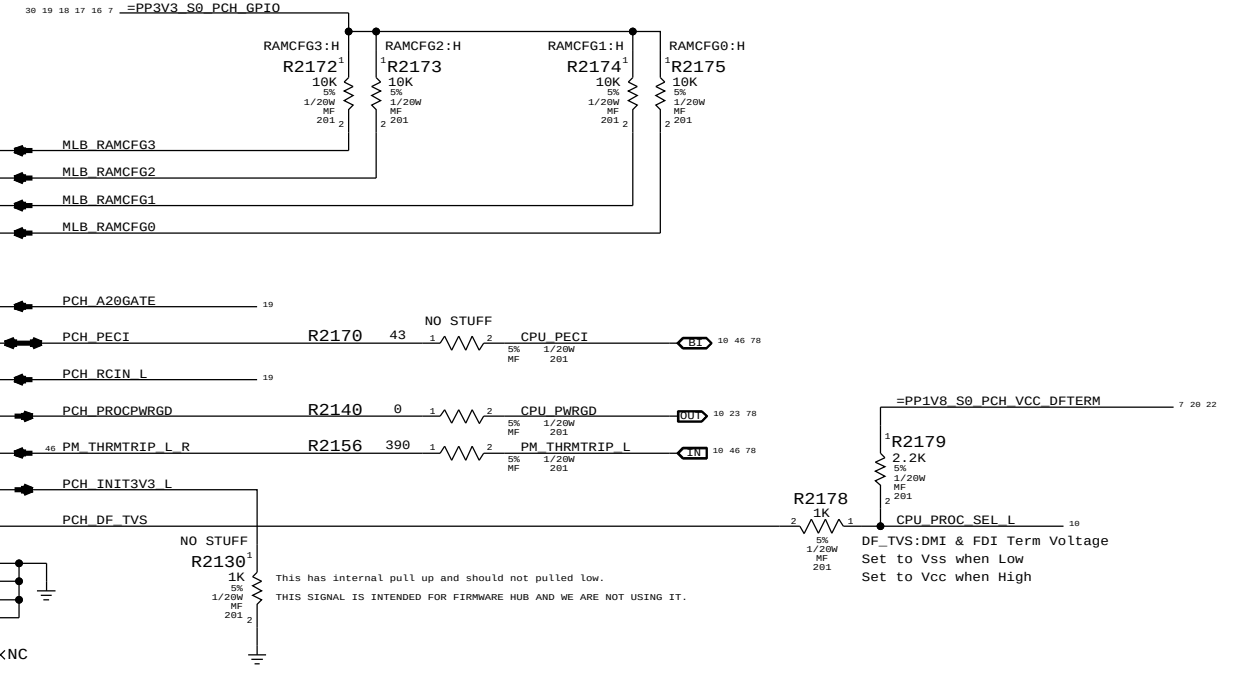






BOM GROUP	BOM OPTIONS
RAMCFG_SL0T	RAMCFG3:H, RAMCFG2:H, RAMCFG1:H, RAMCFG0:H

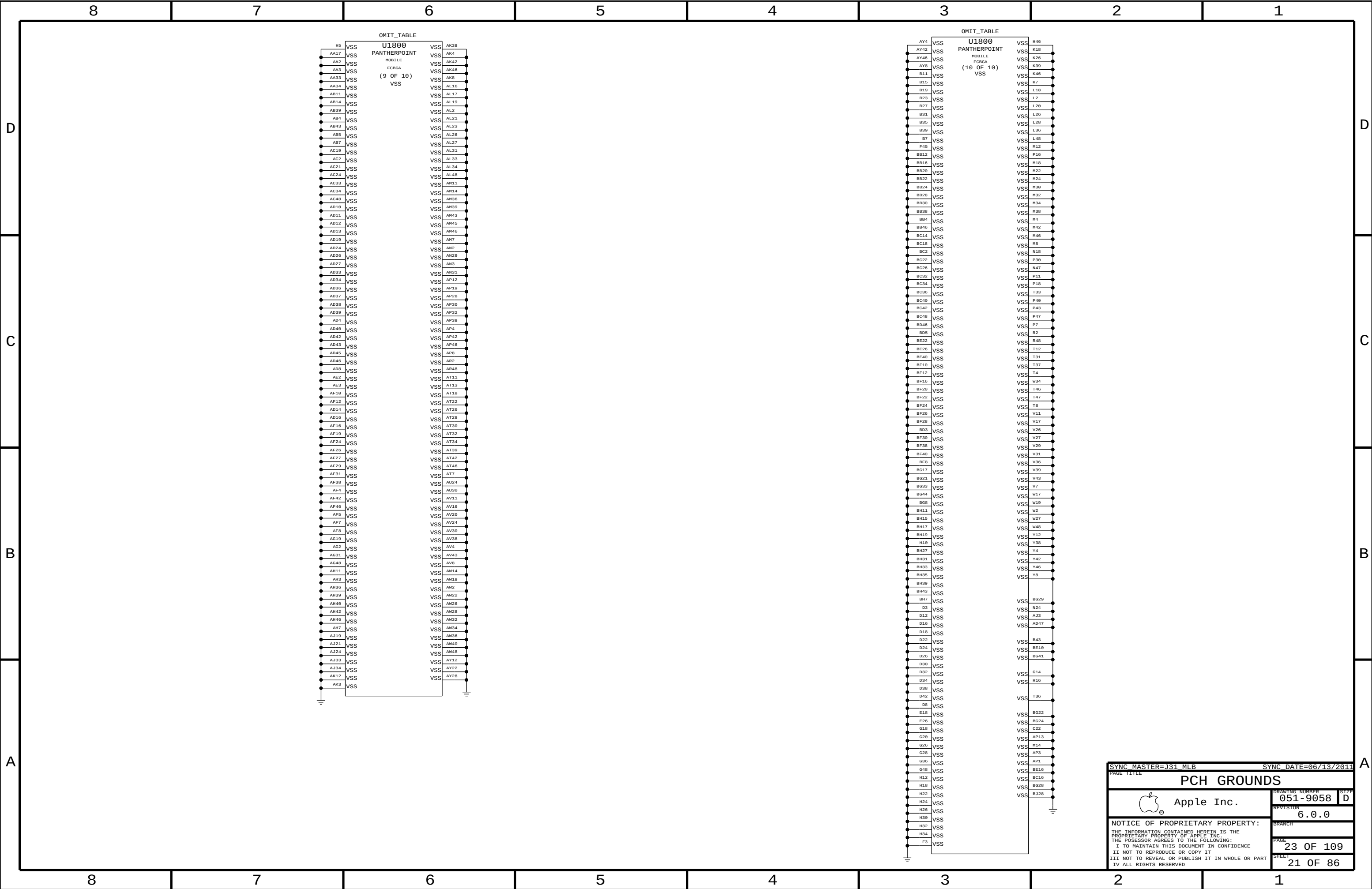
Systems with no chip-down memory should pull all 4 RAMCFG GPIOs high.
Systems with chip-down memory should add pull-downs on another page and set straps per software.



8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

7	=PP3V3_S5_PCH_GPIO	
7 16 17 18	=PP3V3_SUS_PCH_GPIO	
7 16 17 18 19 30	=PP3V3_S0_PCH_GPIO	
7 16	=PP3V3_T29_PCH_GPIO	
8 19	R2186 10K 1 2	JTAG ISP TDO
8 33	R2199 10K 1 2	JTAG TBT TDI
19 23	R2160 10K 1 2	XDP_FC1_PCH_GPIO0
8 19 39	R2185 10K 1 2	FW PME L
19 45	R2196 10K 1 2	SMC_RUNTIME_SCI_L
6 19 47	R2190 100K 1 2	LPCPLUS_GPIO
19	R2197 10K 1 2	TBT_SW_RESET_R_L
19 24	R2184 10K 1 2	FW_PWR_EN_PCH
19	R2150 10K 1 2	PCH_A20GATE
19	R2155 10K 1 2	PCH_RCIN_L
19 73	R2194 10K 1 2	WOL_EN
19	R2192 10K 1 2	PCH_GPIO24
6 19 47 56	R2193 100K 1 2	SPIROM_USE_MLB
19 46	R2191 10K 1 2	SMC_SCI_L
19	R2111 20K 2 1	DPMUX_UC_IRQ
23 24	R2195 100K 2 1	AUD_IPHS_SWITCH_EN_PCH
19 41	R2112 10K 2 1	ODD_PWR_EN_L
19 23	R2198 10K 2 1	XDP_DD0_PCH_GPIO36_DP_GPU_TBT_SEL
8 23	R2113 10K 2 1	JTAG_ISP_TCK
23 24	R2116 10K 2 1	ENET_LOW_PWR_PCH

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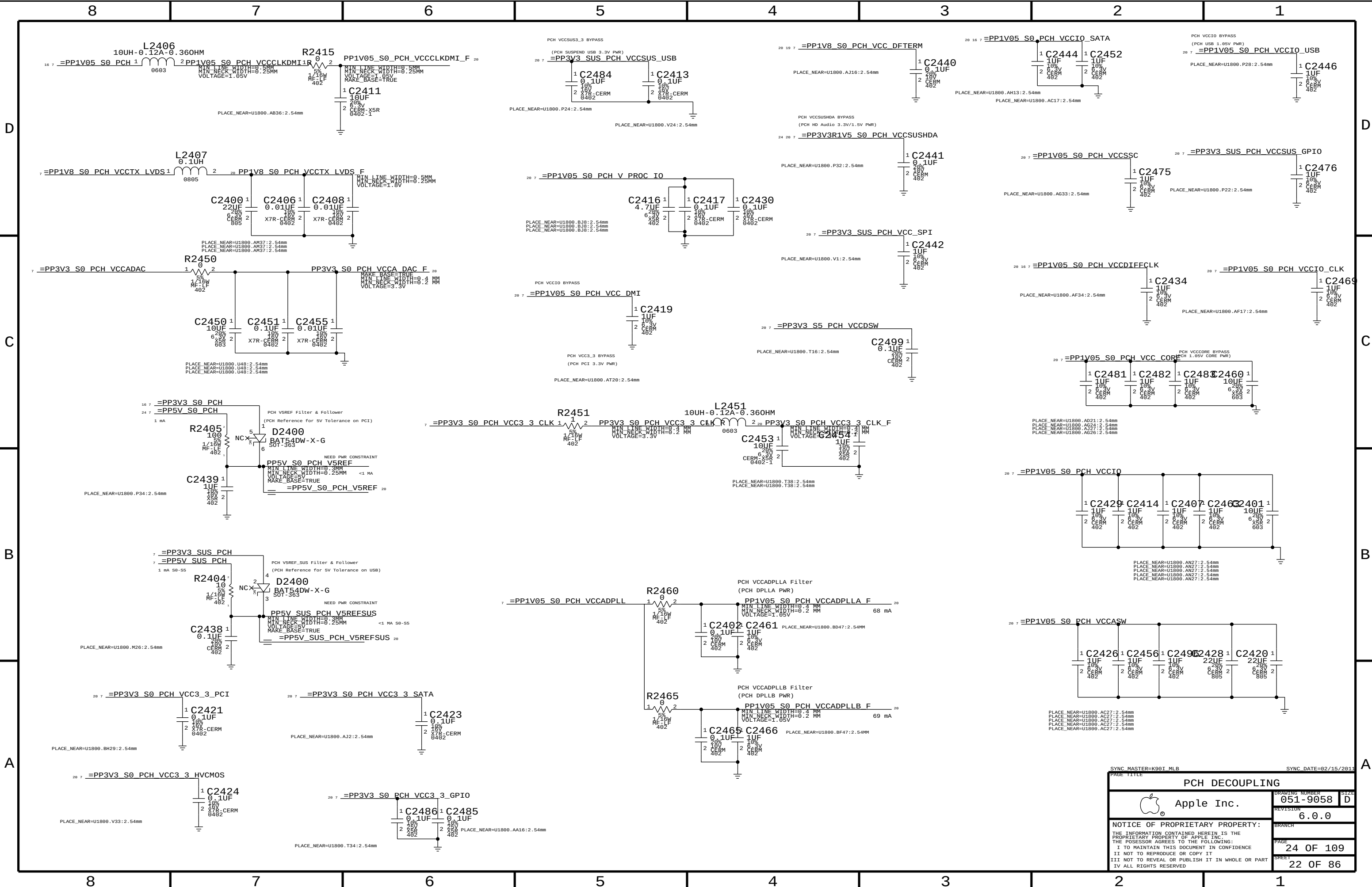
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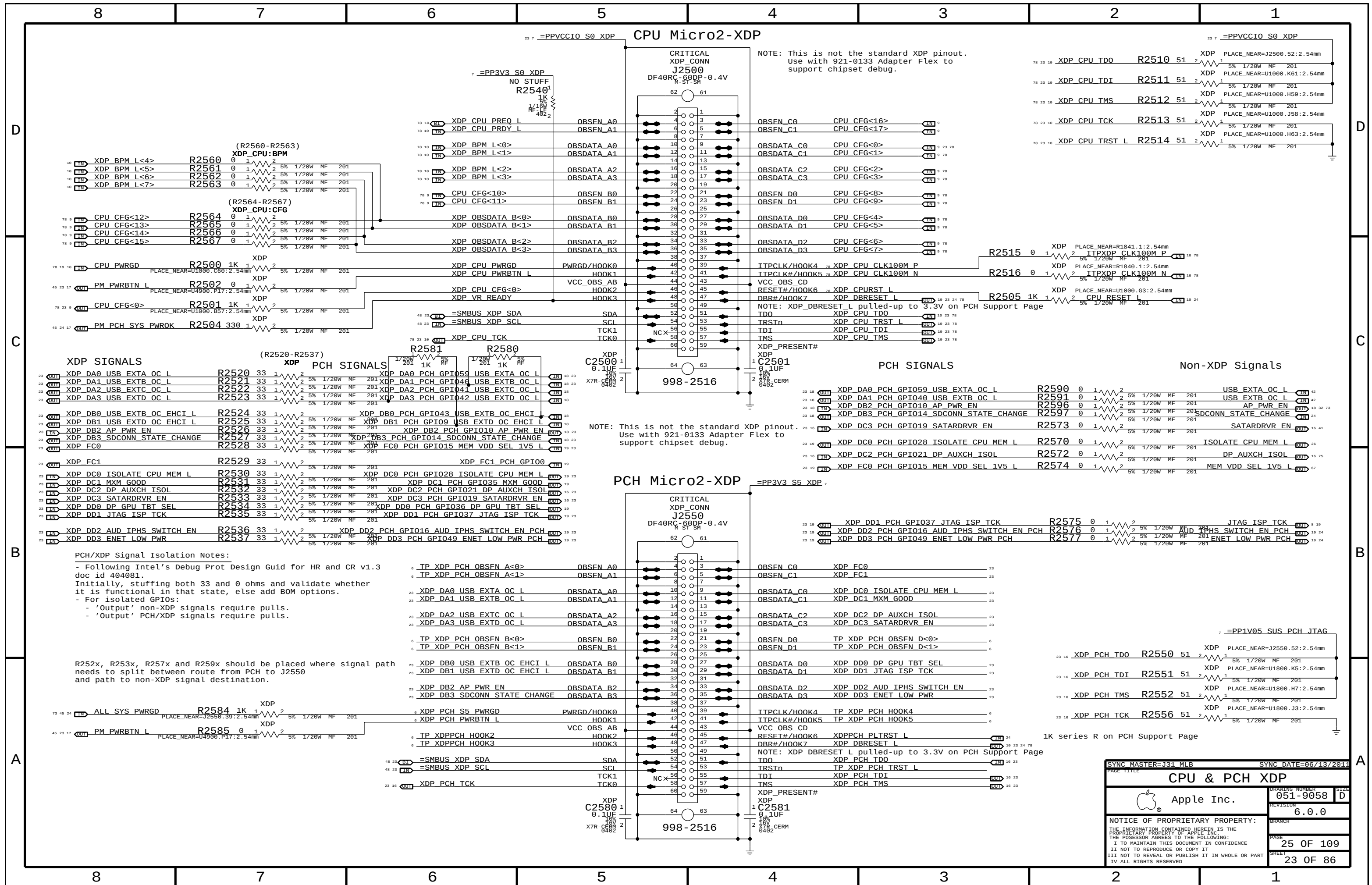
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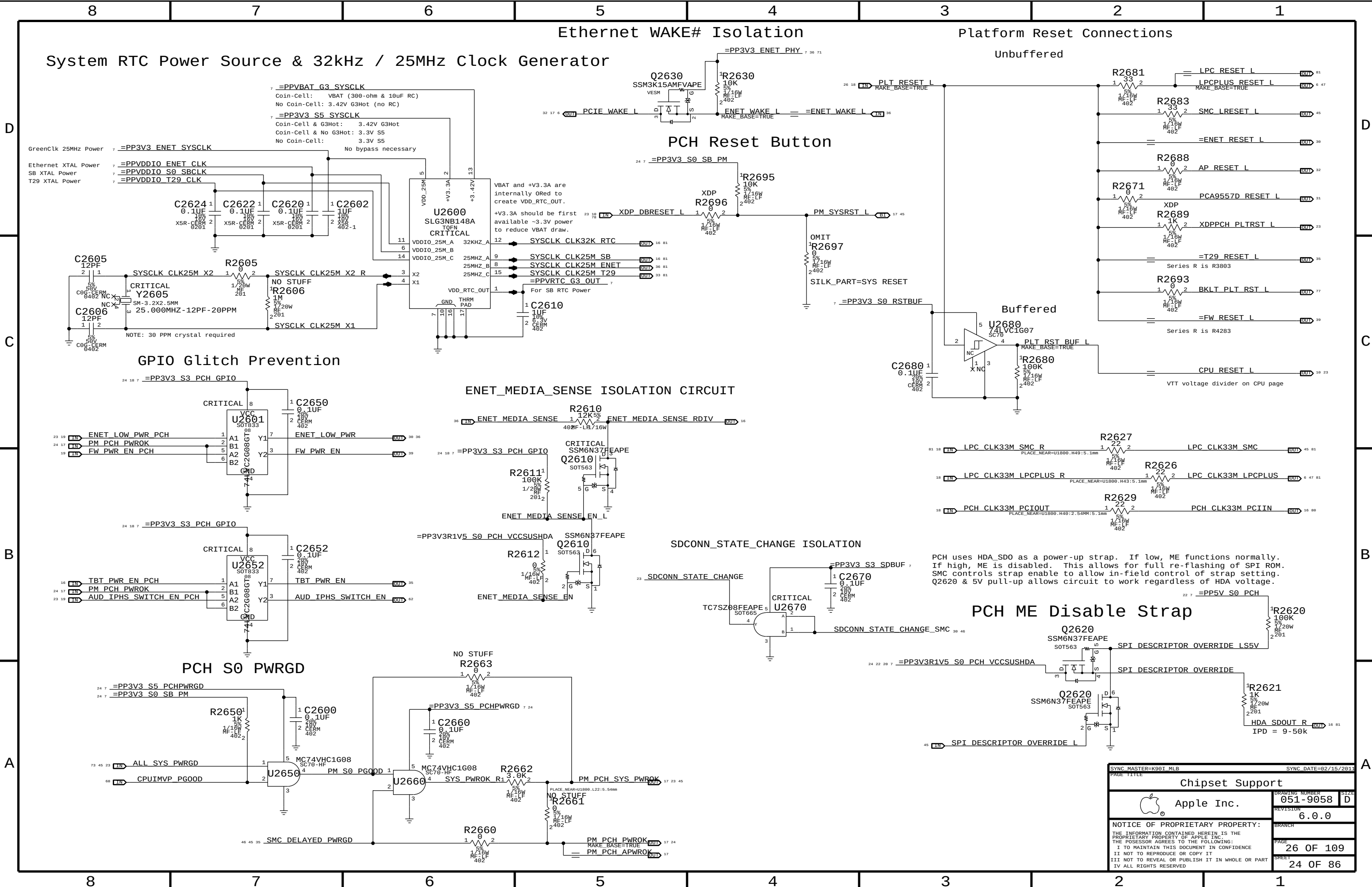
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---	---	---	---	---	---	---	---

```

NON_REM 1 :  NON_REM 0      STRAP PIN CFG
0           :  0             ALL PORTS ARE REMOVABLE
1           :  0             PORT 1 IS NON REMOVABLE
0           :  1             PORT 1&2 ARE NON REMOVABLE
1           :  1             PORT 1&2&3 ARE NON REMOVABLE

```

CANNOT INDICATE ALL 4 PORTS ARE NON REMOVABLE ON USB2514B VIA STARPPING, PROGRAM NON_REMOVABLE_DEVICE_REGISTER 09H

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33850824	1	USB HUB 2514B	U2700	CRITICAL	USBHUB2514B
33850923	1	USB HUB 2513B	U2700	CRITICAL	USBHUB2513B
33850983	1	USB HUB 2512B	U2700	CRITICAL	USBHUB2512B

D

D

C

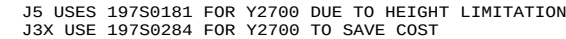
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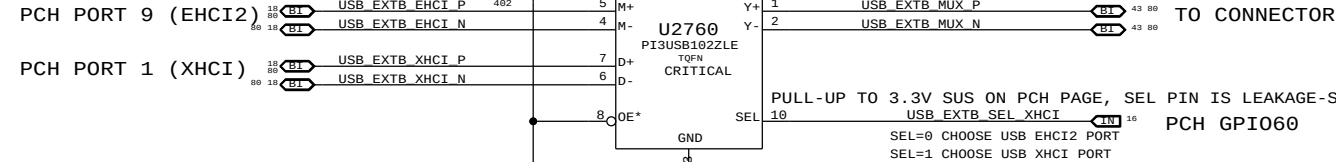
B

A

A



USB XHCI/EHCI2 PORT MUX FOR EXT B



SYNC MASTER=LINDA J30		SYNC DATE=09/19/2011	
PAGE TITLE			
USB HUB & MUX			
 Apple Inc.		DRAWING NUMBER 051-9058	
		SIZE D	
		REVISION 6.0.0	
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The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3<->S0 transitions determines behavior of signals.

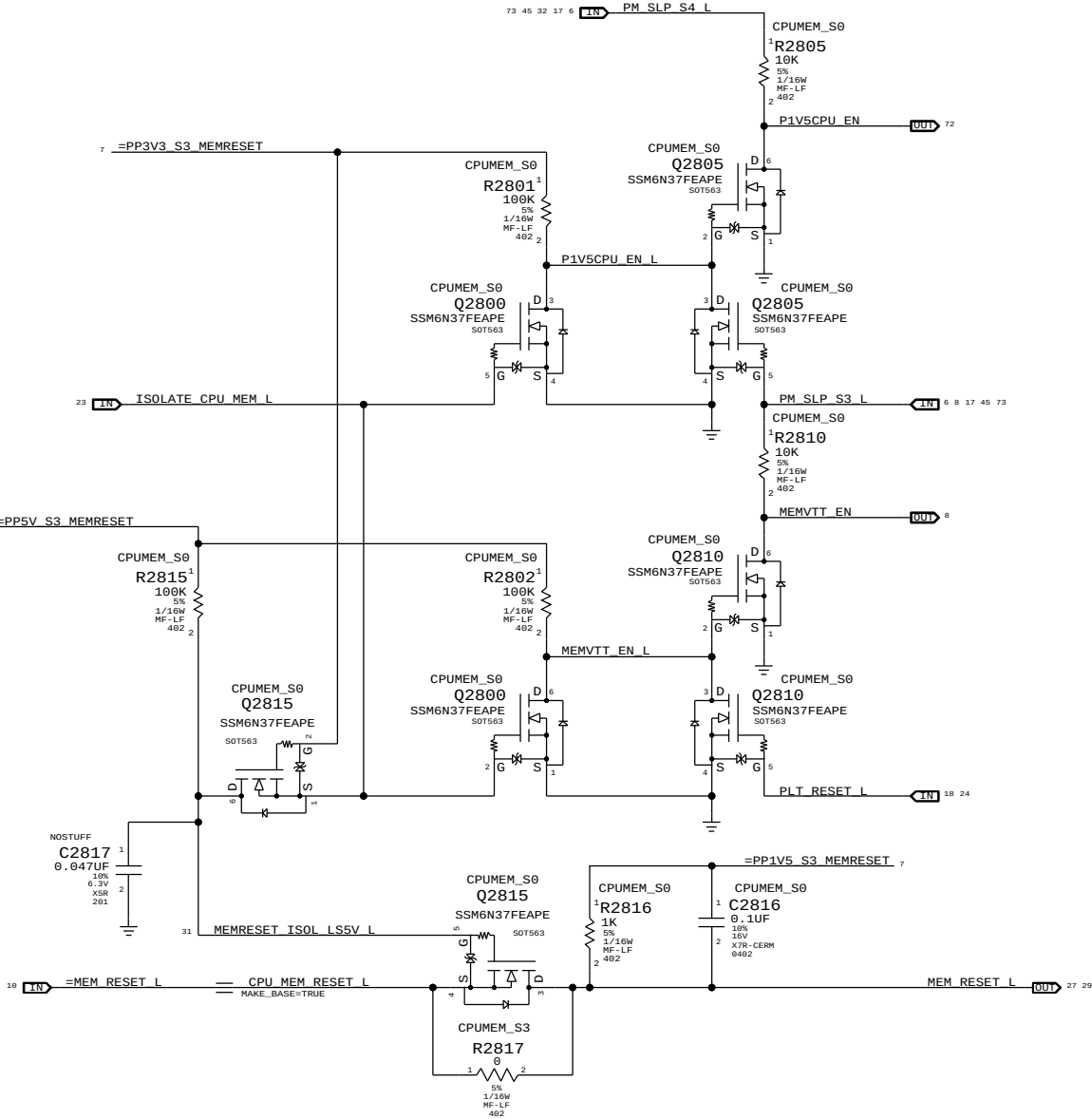
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.

WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

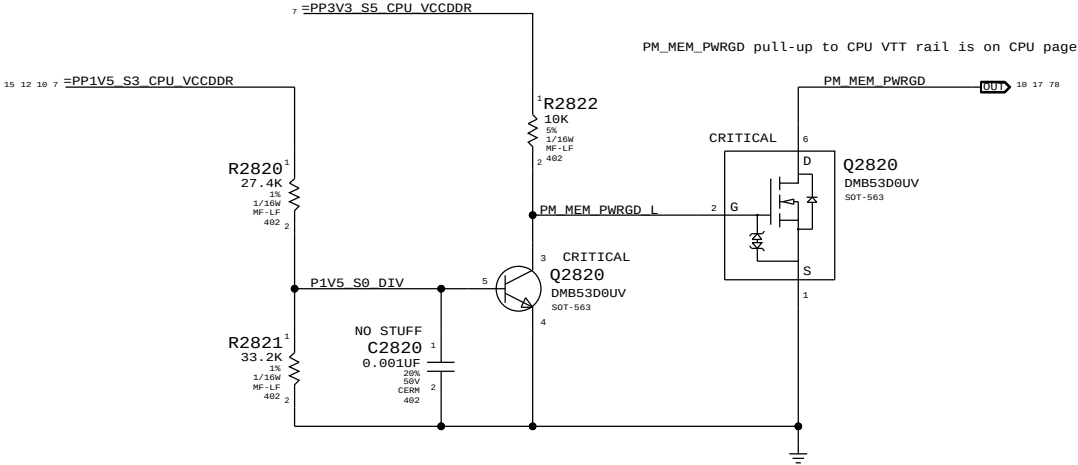
$P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L$

$MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L$

$MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L$

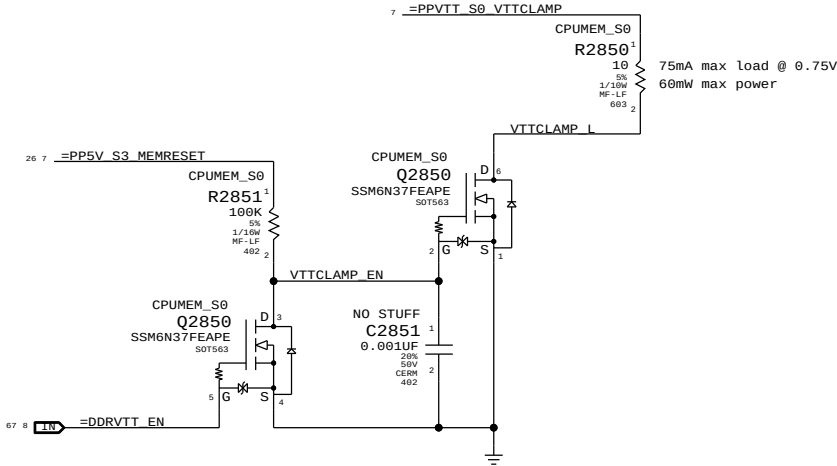


1V5 S0 "PGOOD" for CPU



MEMVTT Clamp

Ensures CKE signals are held low in S3

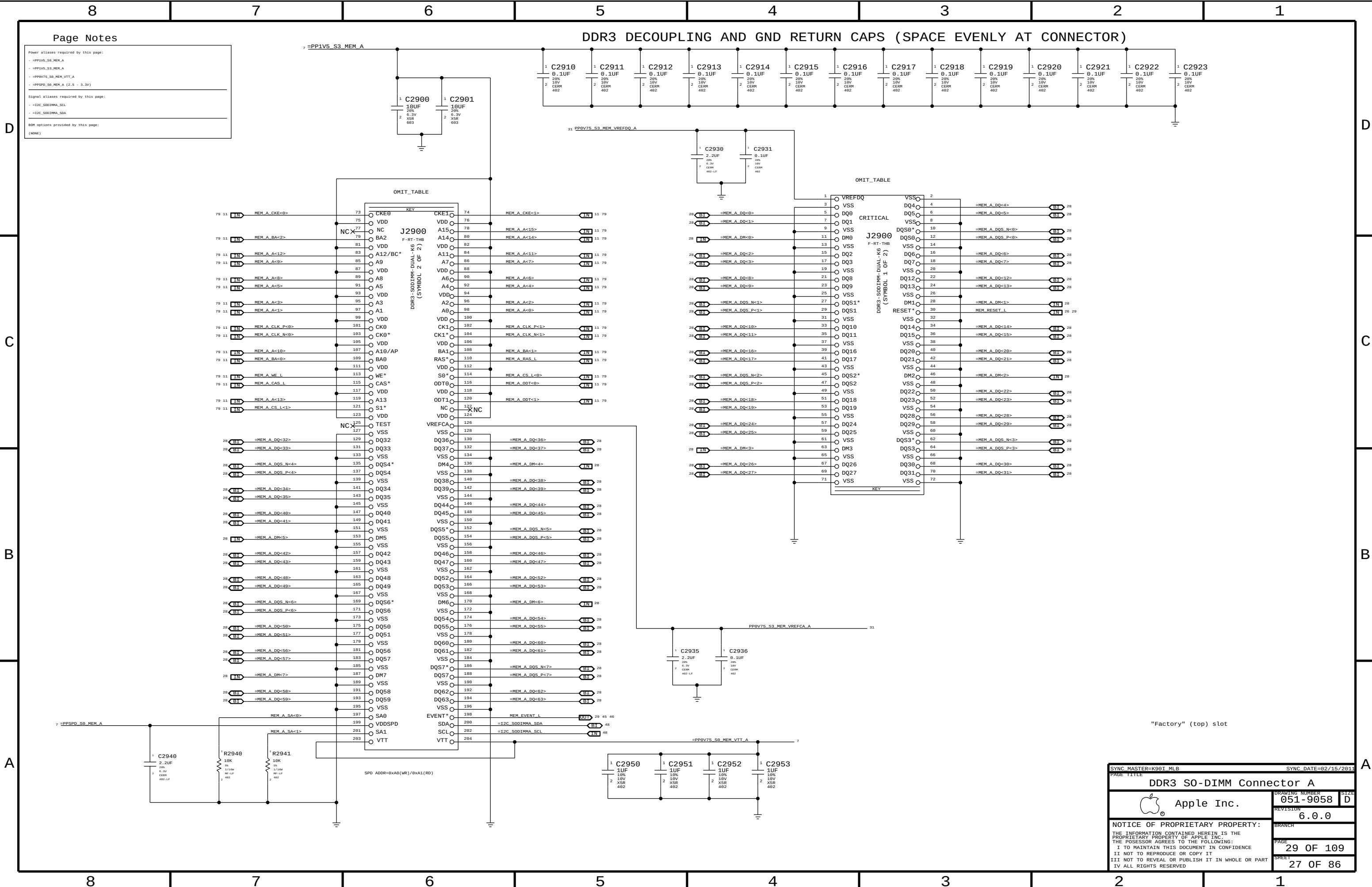


Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPU_MEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPU_MEM_RESET_L	1	1
1	1	0	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
4	0	0	1	1	X	1	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPU_MEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC MASTER=K981 MLB		SYNC DATE=02/15/2011	
PAGE TITLE		CPU Memory S3 Support	
Apple Inc.		DRAWING NUMBER	051-9058
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DDR3 DECOUPLING AND GND RETURN CAPS (SPACE EVENLY AT CONNECTOR)

Page Notes

Power aliases required by this page:

- =PP1V5_S3_MEM_A
- =PP1V5_S3_MEM_A
- =PP0V75_S3_MEM_VTT_A
- =PPSPD_S3_MEM_A (2.5 - 3.3V)

Signal aliases required by this page:

- =I2C_S0DIMM_SCL
- =I2C_S0DIMM_SDA

BOM options provided by this page:

(NONE)

OMIT_TABLE

OMIT_TABLE

CRITICAL

F-RT-TNB

DDR3-S0DIMM-DUAL-KG

(SYMBOL 1 OF 2)

DDR3-S0DIMM-DUAL-KG

(SYMBOL 2 OF 2)

KEY

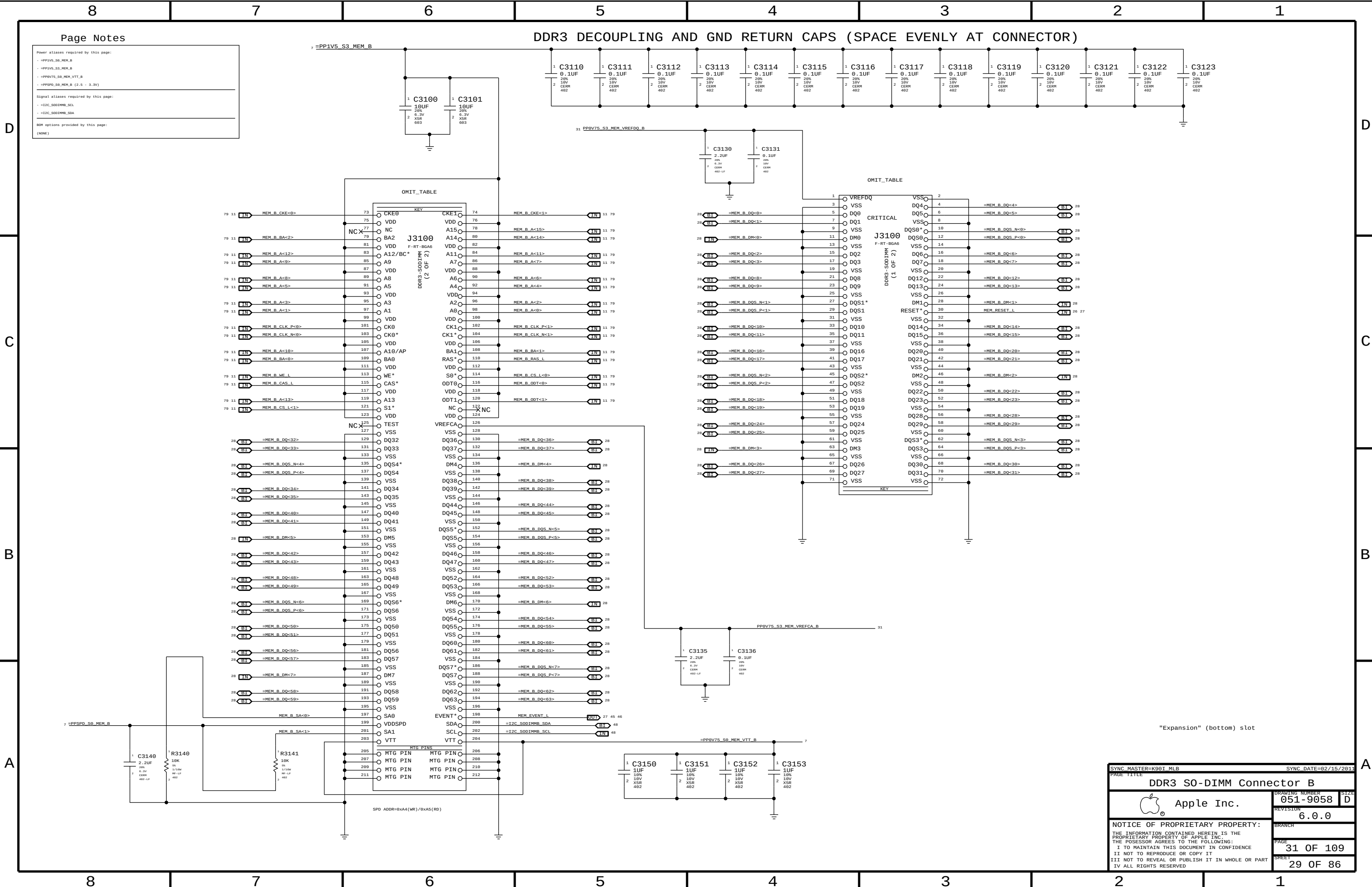
"Factory" (top) slot

SYNC MASTER=K901 MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
DDR3 SO-DIMM Connector A			
 Apple Inc.		DRAWING NUMBER	051-9058
		SIZE	
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		PAGE	29 OF 109
		SHEET	27 OF 86

8	7	6	5	4	3	2	1
D	CPU CHANNEL A DQS 0 -> DIMM A DQS 0		CPU CHANNEL B DQS 0 -> DIMM B DQS 0				
	79 11 MEM_A_DQS_N<0>	==MEM_A_DQS_N<0> 27	79 11 MEM_B_DQS_N<0>	==MEM_B_DQS_N<0> 29			
	79 11 MEM_A_DQS_P<0>	==MEM_A_DQS_P<0> 27	79 11 MEM_B_DQS_P<0>	==MEM_B_DQS_P<0> 29			
	MAKE_BASE=TRUE	==MEM_A_DM<0> 27	MAKE_BASE=TRUE	==MEM_B_DM<0> 29			
	79 11 MEM_A_DQ<7>	==MEM_A_DQ<3> 27	79 11 MEM_B_DQ<7>	==MEM_B_DQ<3> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<6> 27	MAKE_BASE=TRUE	==MEM_B_DQ<2> 29			
	79 11 MEM_A_DQ<6>	==MEM_A_DQ<1> 27	79 11 MEM_B_DQ<6>	==MEM_B_DQ<0> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<5> 27	MAKE_BASE=TRUE	==MEM_B_DQ<4> 29			
	79 11 MEM_A_DQ<5>	==MEM_A_DQ<2> 27	79 11 MEM_B_DQ<5>	==MEM_B_DQ<7> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<7> 27	MAKE_BASE=TRUE	==MEM_B_DQ<6> 29			
C	CPU CHANNEL A DQS 1 -> DIMM A DQS 1		CPU CHANNEL B DQS 1 -> DIMM B DQS 1				
	79 11 MEM_A_DQS_N<1>	==MEM_A_DQS_N<1> 27	79 11 MEM_B_DQS_N<1>	==MEM_B_DQS_N<1> 29			
	79 11 MEM_A_DQS_P<1>	==MEM_A_DQS_P<1> 27	79 11 MEM_B_DQS_P<1>	==MEM_B_DQS_P<1> 29			
	MAKE_BASE=TRUE	==MEM_A_DM<1> 27	MAKE_BASE=TRUE	==MEM_B_DM<1> 29			
	79 11 MEM_A_DQ<15>	==MEM_A_DQ<11> 27	79 11 MEM_B_DQ<15>	==MEM_B_DQ<15> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<10> 27	MAKE_BASE=TRUE	==MEM_B_DQ<14> 29			
	79 11 MEM_A_DQ<14>	==MEM_A_DQ<12> 27	79 11 MEM_B_DQ<14>	==MEM_B_DQ<13> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<9> 27	MAKE_BASE=TRUE	==MEM_B_DQ<8> 29			
	79 11 MEM_A_DQ<13>	==MEM_A_DQ<15> 27	79 11 MEM_B_DQ<13>	==MEM_B_DQ<11> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<14> 27	MAKE_BASE=TRUE	==MEM_B_DQ<10> 29			
B	CPU CHANNEL A DQS 2 -> DIMM A DQS 2		CPU CHANNEL B DQS 2 -> DIMM B DQS 2				
	79 11 MEM_A_DQS_N<2>	==MEM_A_DQS_N<2> 27	79 11 MEM_B_DQS_N<2>	==MEM_B_DQS_N<2> 29			
	79 11 MEM_A_DQS_P<2>	==MEM_A_DQS_P<2> 27	79 11 MEM_B_DQS_P<2>	==MEM_B_DQS_P<2> 29			
	MAKE_BASE=TRUE	==MEM_A_DM<2> 27	MAKE_BASE=TRUE	==MEM_B_DM<2> 29			
	79 11 MEM_A_DQ<23>	==MEM_A_DQ<23> 27	79 11 MEM_B_DQ<23>	==MEM_B_DQ<23> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<22> 27	MAKE_BASE=TRUE	==MEM_B_DQ<18> 29			
	79 11 MEM_A_DQ<22>	==MEM_A_DQ<21> 27	79 11 MEM_B_DQ<22>	==MEM_B_DQ<16> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<20> 27	MAKE_BASE=TRUE	==MEM_B_DQ<17> 29			
	79 11 MEM_A_DQ<21>	==MEM_A_DQ<18> 27	79 11 MEM_B_DQ<21>	==MEM_B_DQ<22> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<19> 27	MAKE_BASE=TRUE	==MEM_B_DQ<19> 29			
A	CPU CHANNEL A DQS 3 -> DIMM A DQS 3		CPU CHANNEL B DQS 3 -> DIMM B DQS 3				
	79 11 MEM_A_DQS_N<3>	==MEM_A_DQS_N<3> 27	79 11 MEM_B_DQS_N<3>	==MEM_B_DQS_N<3> 29			
	79 11 MEM_A_DQS_P<3>	==MEM_A_DQS_P<3> 27	79 11 MEM_B_DQS_P<3>	==MEM_B_DQS_P<3> 29			
	MAKE_BASE=TRUE	==MEM_A_DM<3> 27	MAKE_BASE=TRUE	==MEM_B_DM<3> 29			
	79 11 MEM_A_DQ<31>	==MEM_A_DQ<26> 27	79 11 MEM_B_DQ<31>	==MEM_B_DQ<26> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<24> 27	MAKE_BASE=TRUE	==MEM_B_DQ<30> 29			
	79 11 MEM_A_DQ<30>	==MEM_A_DQ<28> 27	79 11 MEM_B_DQ<30>	==MEM_B_DQ<28> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<25> 27	MAKE_BASE=TRUE	==MEM_B_DQ<29> 29			
	79 11 MEM_A_DQ<29>	==MEM_A_DQ<31> 27	79 11 MEM_B_DQ<29>	==MEM_B_DQ<27> 29			
	MAKE_BASE=TRUE	==MEM_A_DQ<27> 27	MAKE_BASE=TRUE	==MEM_B_DQ<31> 29			

NOTE: Ivybridge does not use DM signals per doc 460452 CR SFF DG Section 2.6.14

SYNC MASTER=K98T MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
DDR3 Byte/Bit Swaps			
 Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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		PAGE	30 OF 109
		SHEET	28 OF 86



Page Notes

Power aliases required by this page:

- PP1V5_S3_MEM_B
- PP1V5_S3_MEM_B
- PP8V75_S3_MEM_VTT_B
- PPSPD_S0_MEM_B (2.5 - 3.3V)

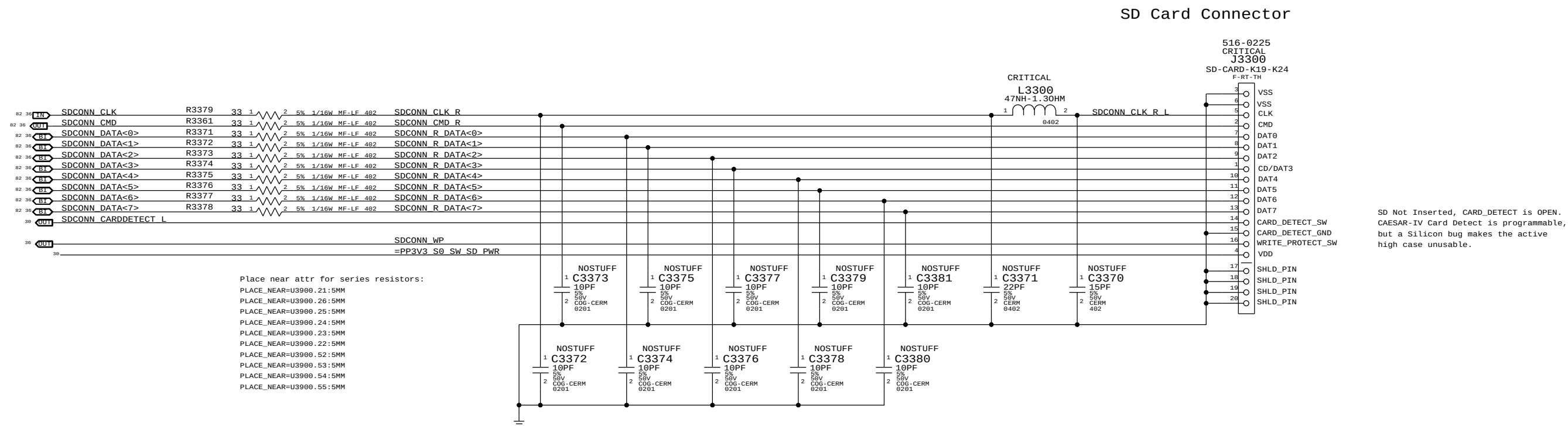
Signal aliases required by this page:

- I2C_S0DIMM_SCL
- I2C_S0DIMM_SDA

BOM options provided by this page:

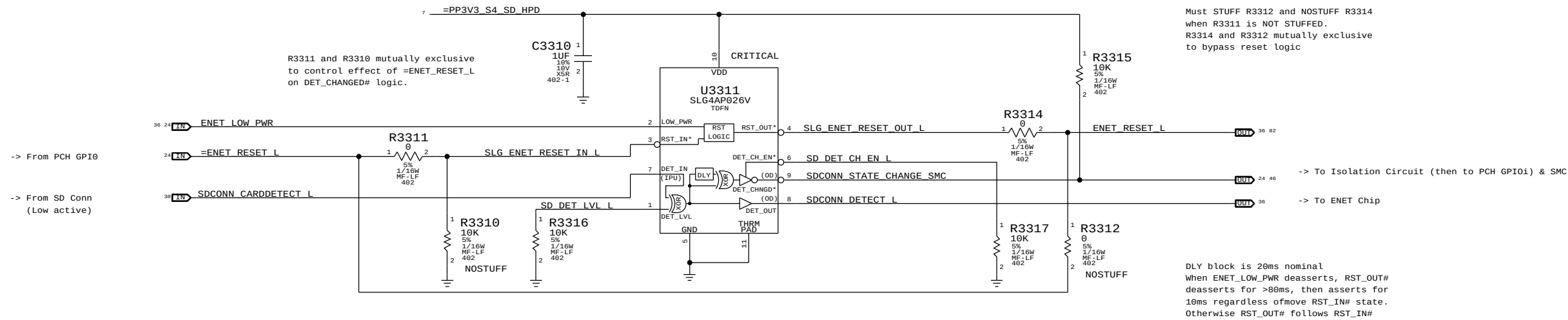
(NONE)

SYNC MASTER=K98T MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
DDR3 S0-DIMM Connector B			
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		REVISION	6.0.0
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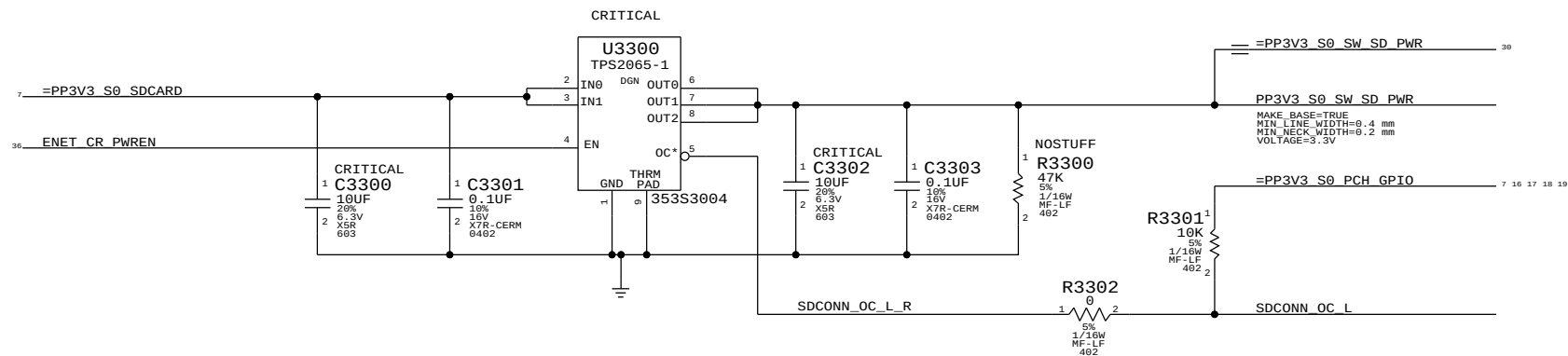
SD Detect & Reset Logic

SDCONN_DETECT Debounce, Inversion, Detect-Changed PCH GPIO Latch Circuit
Converts SDCONN from active-low level signal to active-high pulses.

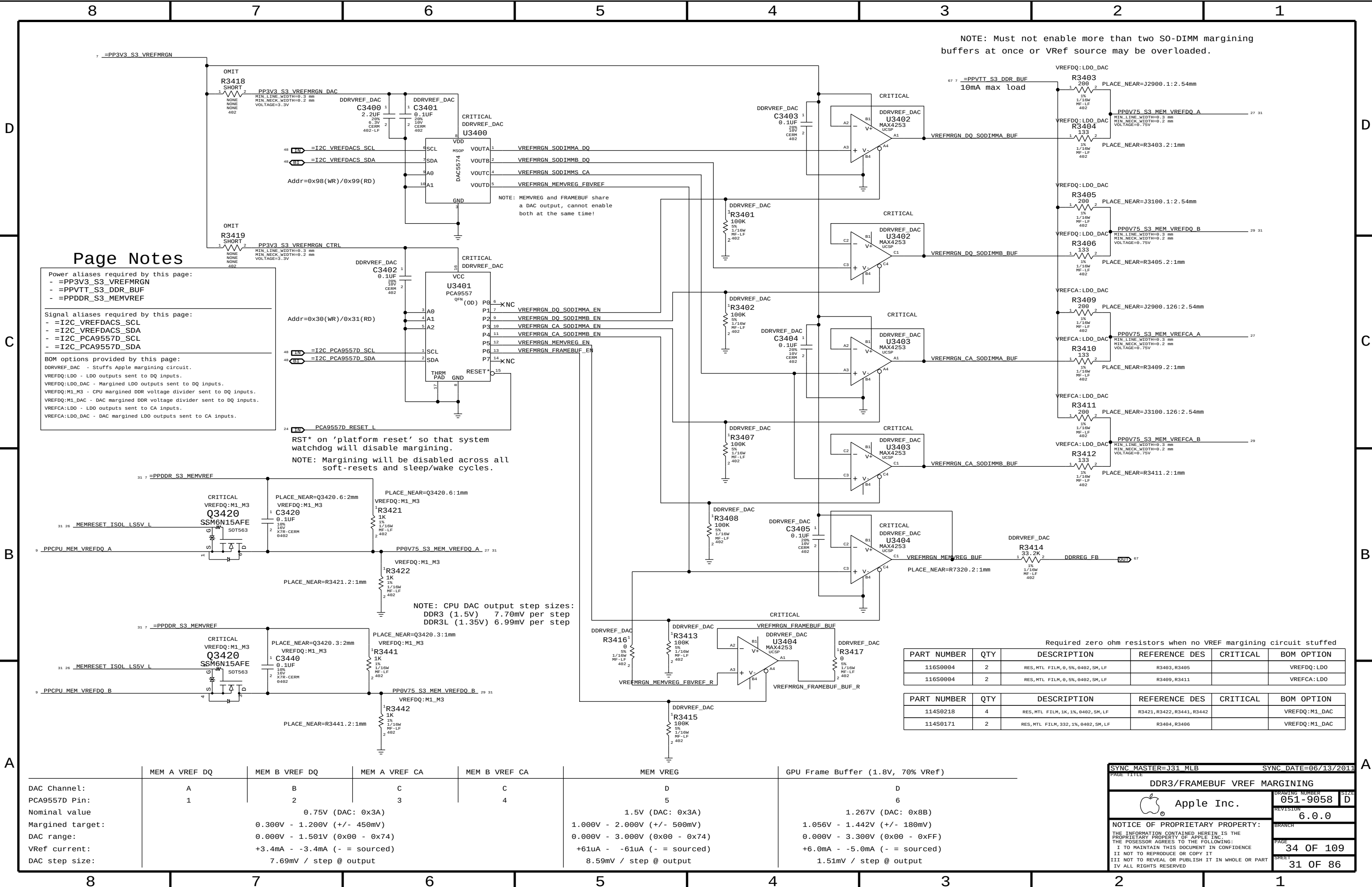


SD Card 3.3V Overcurrent Protection

```
TPS2065-1 (1.0A limit) has active load discharge so R4810 is NOSTUFF.
```



SYNC MASTER=YONAS J30		SYNC DATE=11/03/2011	
PAGE TITLE			
SD Card Connector			
 Apple Inc.		DRAWING NUMBER	SIZE
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		PAGE	33 OF 109
		SHEET	30 OF 86



Page Notes

Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
DDRREF_DAC - Stuffs Apple margining circuit.
VREFDQ:LDO - LDO outputs sent to DQ inputs.
VREFDQ:LDO_DAC - Margined LDO outputs sent to DQ inputs.
VREFDQ:M1_M3 - CPU margined DDR voltage divider sent to DQ inputs.
VREFDQ:M1_DAC - DAC margined DDR voltage divider sent to DQ inputs.
VREFCA:LDO - LDO outputs sent to CA inputs.
VREFCA:LDO_DAC - DAC margined LDO outputs sent to CA inputs.

RST* on 'platform reset' so that system watchdog will disable margining.
NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

NOTE: Must not enable more than two S0-DIMM margining buffers at once or VRef source may be overloaded.

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3403, R3405		VREFDQ:LDO
116S0004	2	RES, MTL FILM, 0, 5%, 0402, SM, LF	R3409, R3411		VREFCA:LDO

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
114S0218	4	RES, MTL FILM, 1K, 1%, 0402, SM, LF	R3421, R3422, R3441, R3442		VREFDQ:M1_DAC
114S0171	2	RES, MTL FILM, 332, 1%, 0402, SM, LF	R3404, R3406		VREFDQ:M1_DAC

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% VRef)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

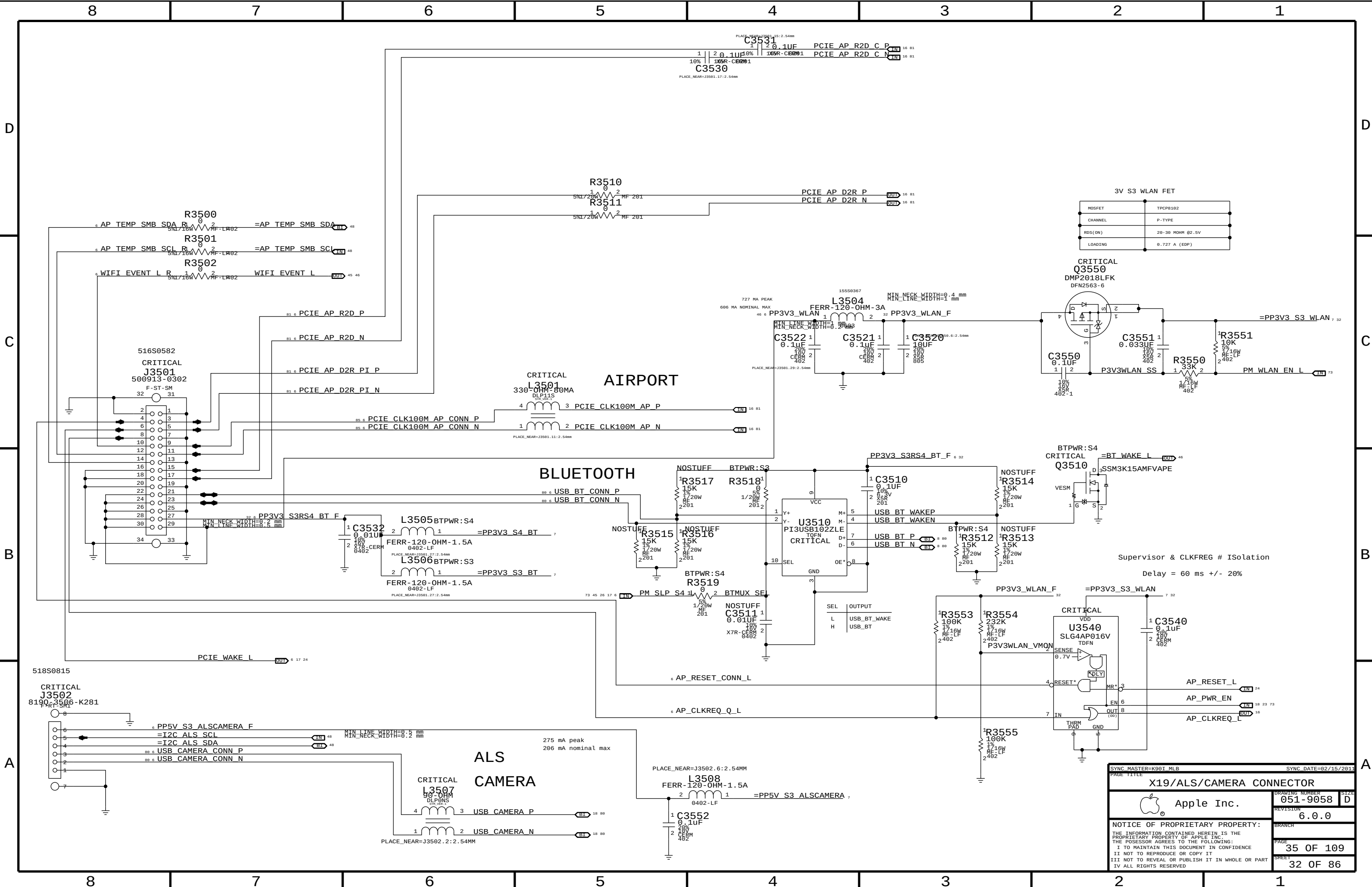
SYNC MASTER=J31 MLB
PAGE TITLE
DDR3/FRAMEBUF VREF MARGINING

Apple Inc.

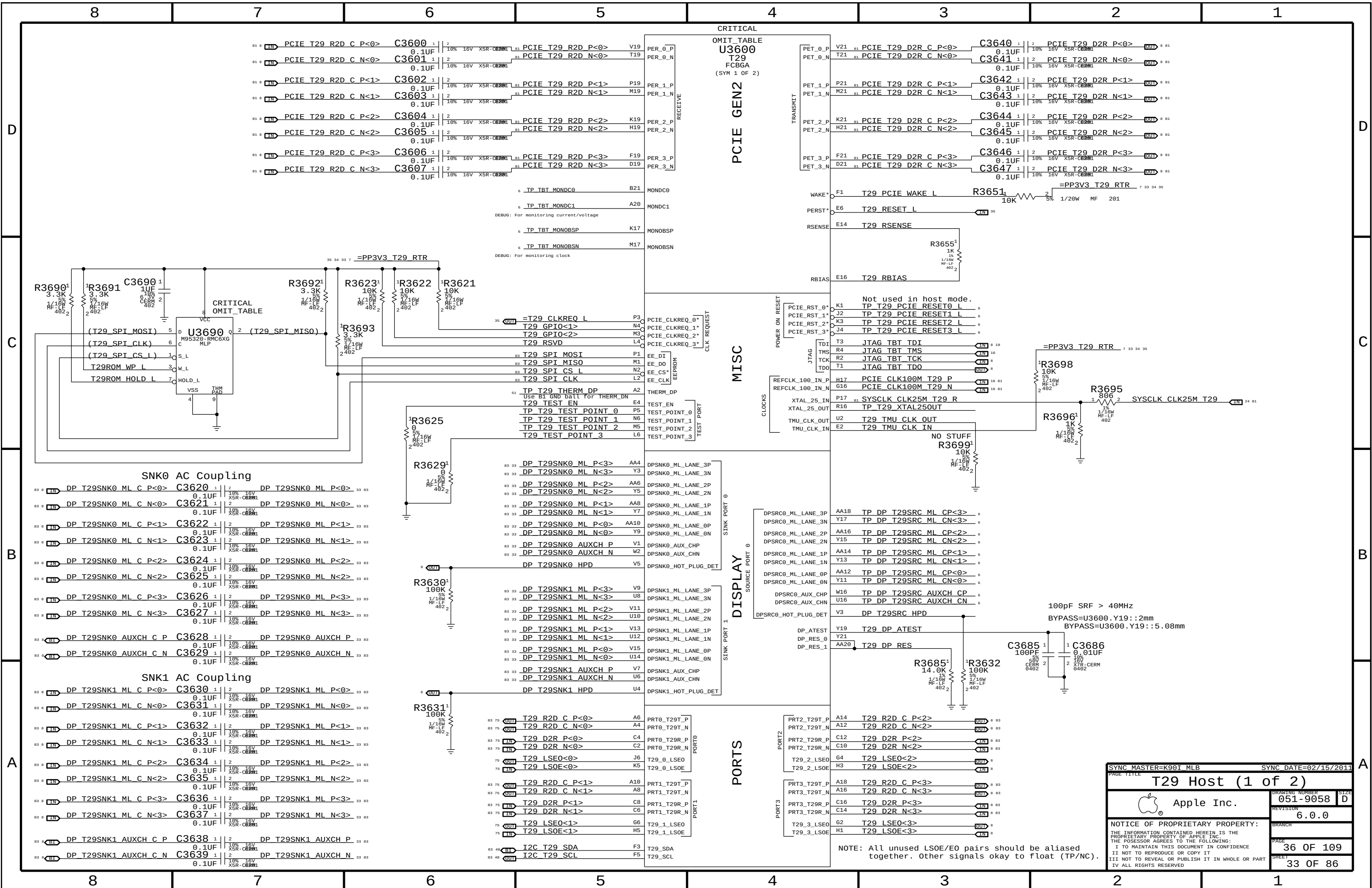
051-9058
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6.0.0

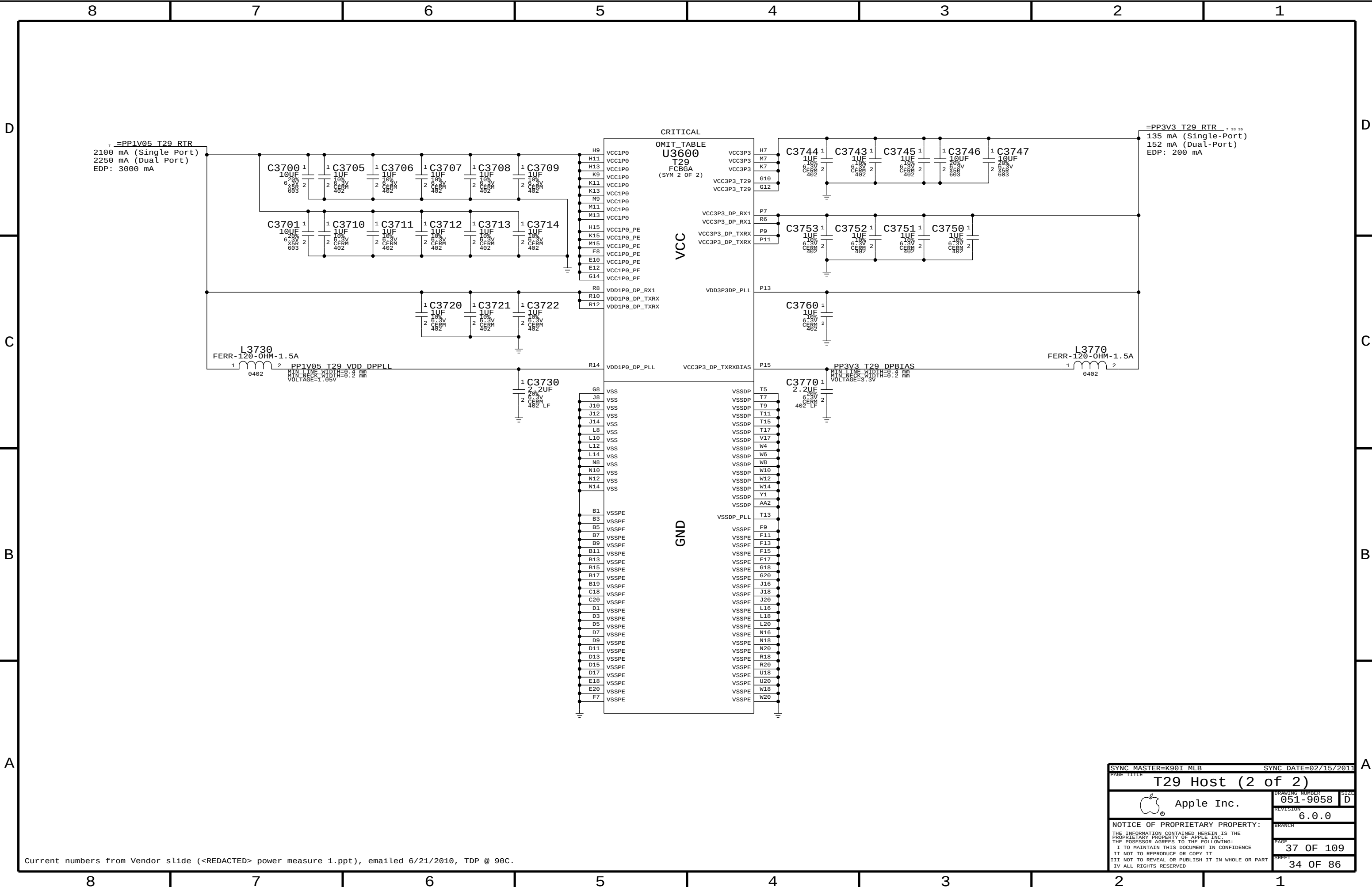
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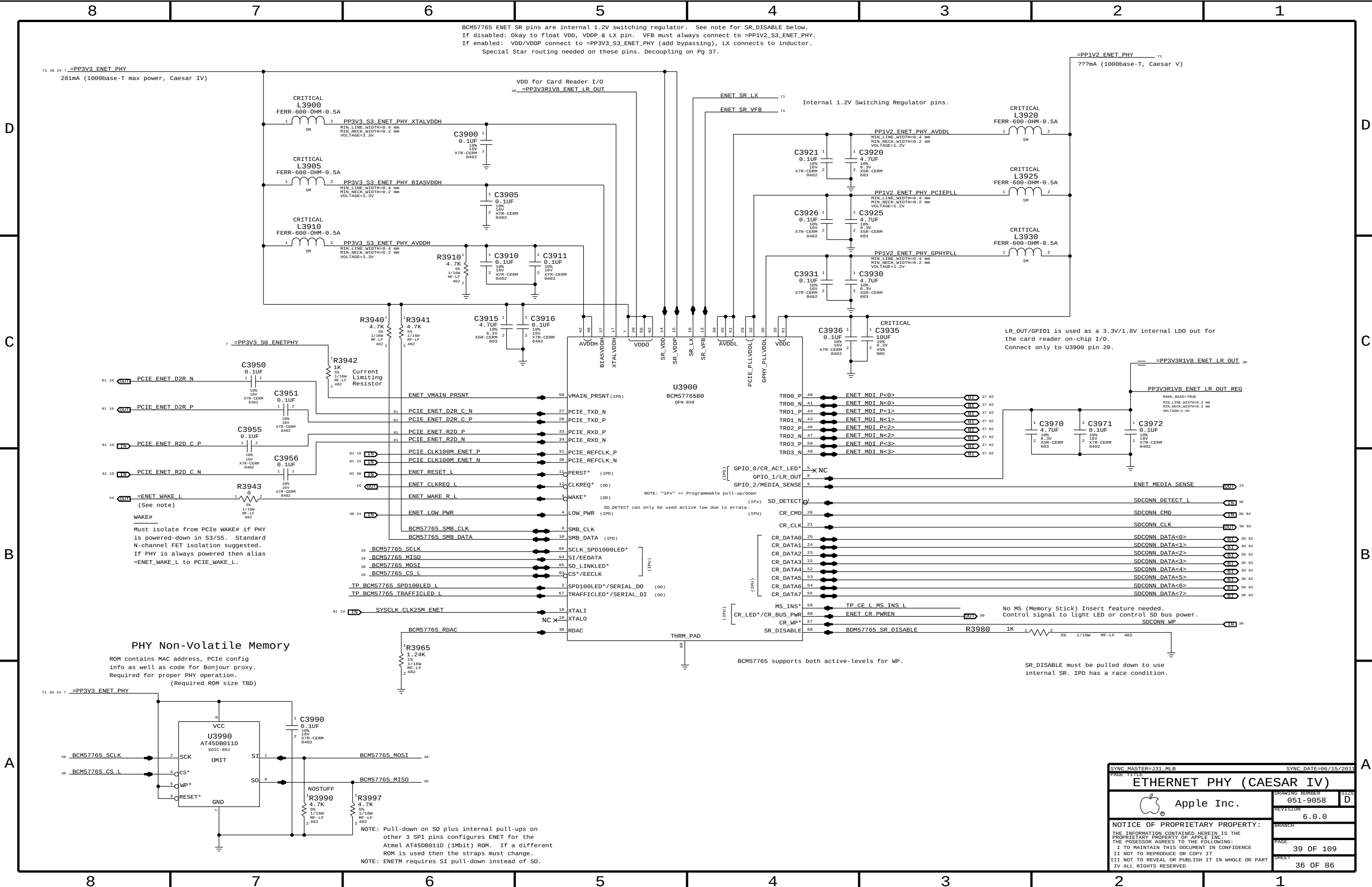
SYNC MASTER=K981 MLB		SYNC DATE=02/15/2011	
PAGE TITLE		SIZE	
X19/ALS/CAMERA CONNECTOR		051-9058 D	
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Current numbers from Vendor slide (<REDACTED> power measure 1.ppt), emailed 6/21/2010, TDP @ 90C.

SYNC MASTER=K90I MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
T29 Host (2 of 2)			
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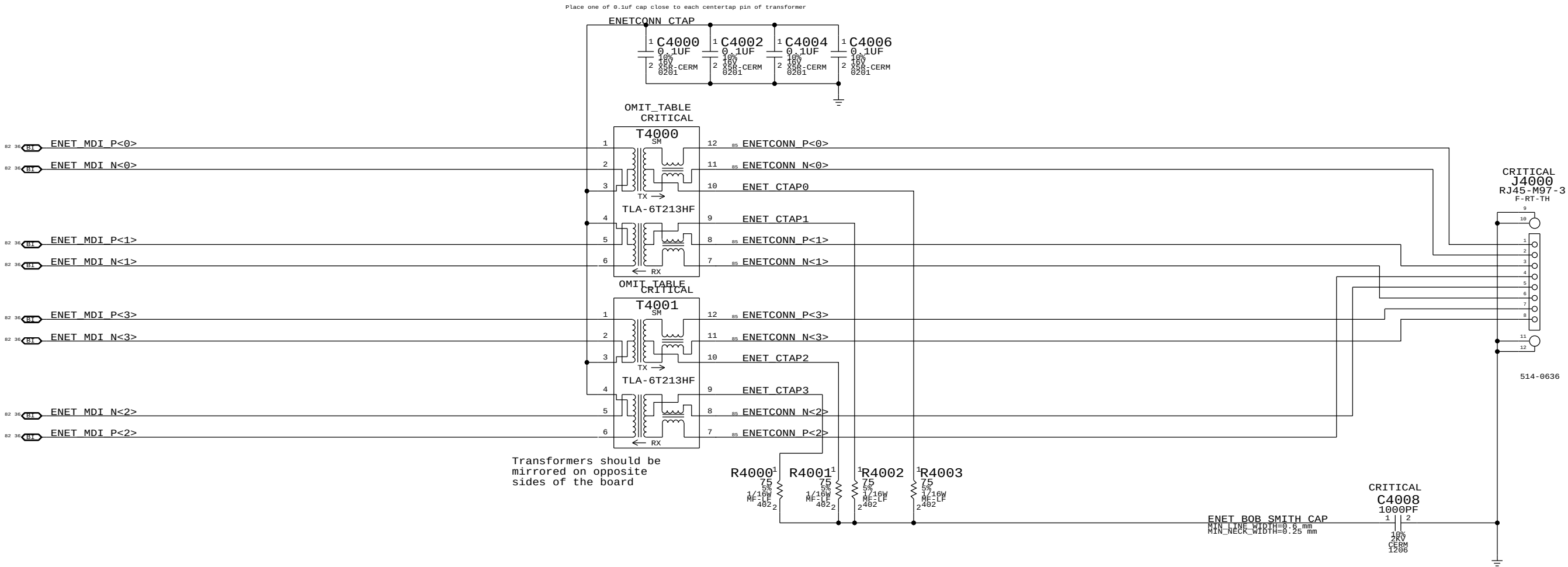


Page Notes

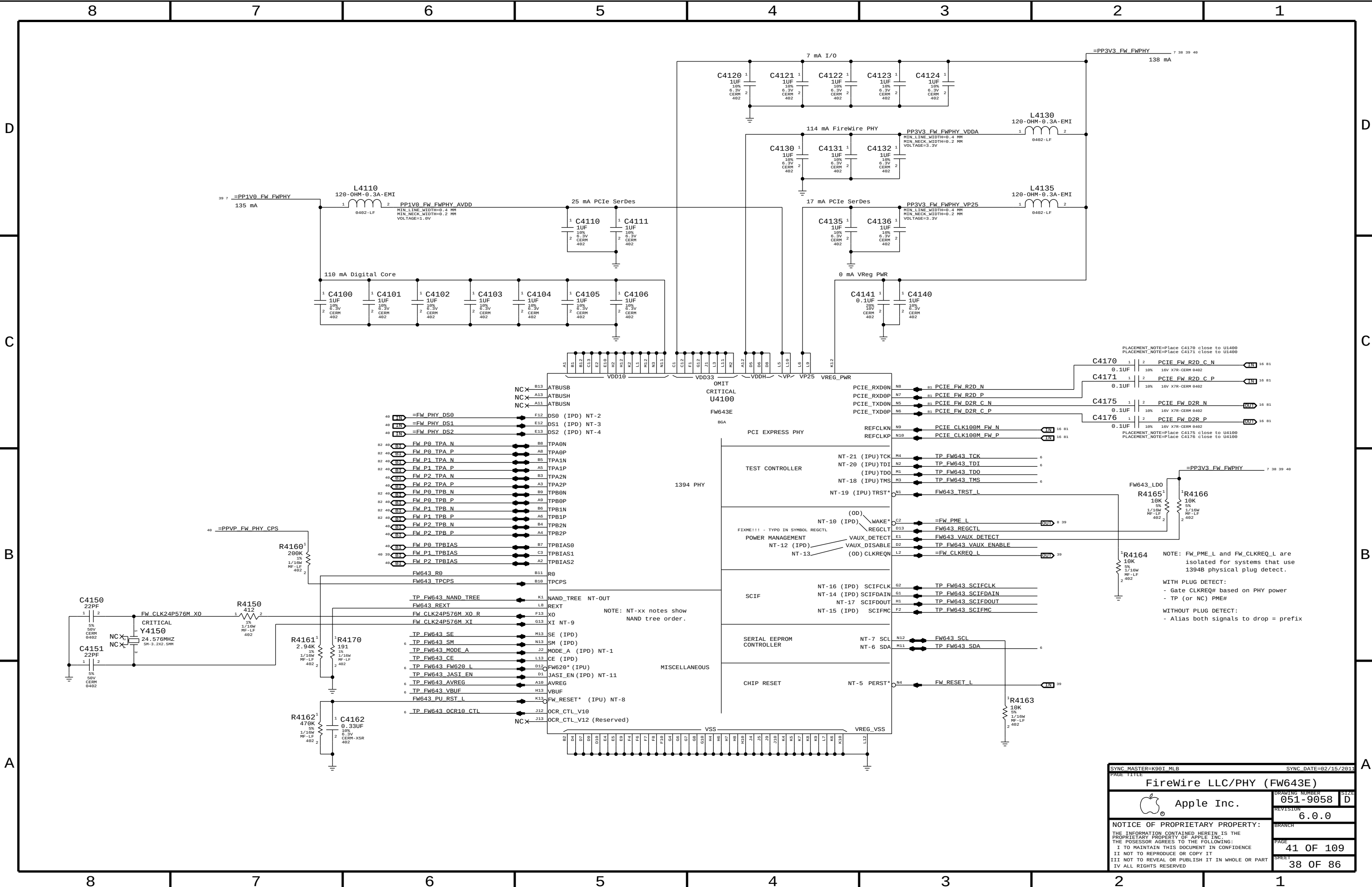
Power aliases required by this page:
(NONE)

Signal aliases required by this page:
(NONE)

BOM options provided by this page:
(NONE)



SYNC MASTER=K901 MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
Ethernet Connector			
 Apple Inc.	DRAWING NUMBER		SIZE
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		40 OF 109	
		SHEET	
		37 OF 86	



Page Notes

Power aliases required by this page:

- =PPBUS_S5_FWPWRSW (FW VP FET Input)
- =PPBUS_FW_FET (FW VP FET Output)
- =PP3V3_FW_P3V3FWFET (3.3V FET Input)
- =PP3V3_FW_FET (3.3V FET Output)
- =PP3V3_FW_FWPHY (PHY 3.3V Power)
- =PP3V3_S0_FWLATEVG
- =PP3V3_S0_FWPWRCTL
- =PP1V05_S0_FWPWRCTL (5KPD Bias Rail)
- =PP1V05_FW_P1V0FWFET (1.0V FET Input)
- =PP1V0_FW_FET_R (1.0V FET Output)
- =PP1V0_FW_FWPHY (PHY 1.0V)

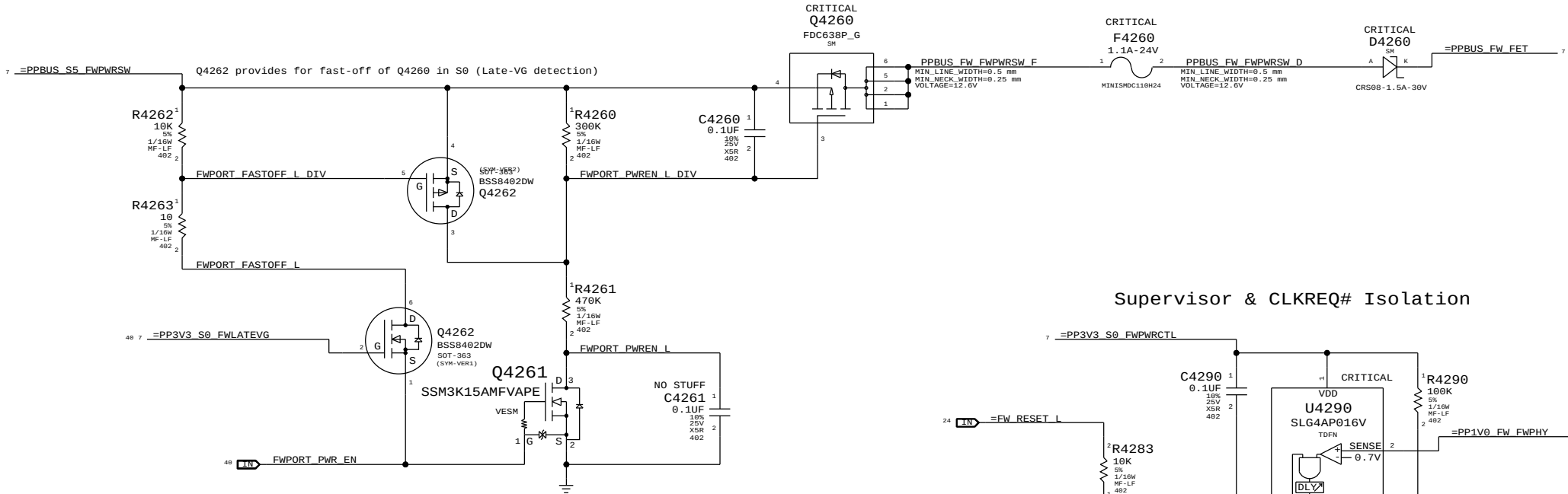
Signal aliases required by this page:

- =FW_CLKREQ_L
- =FW_PME_L

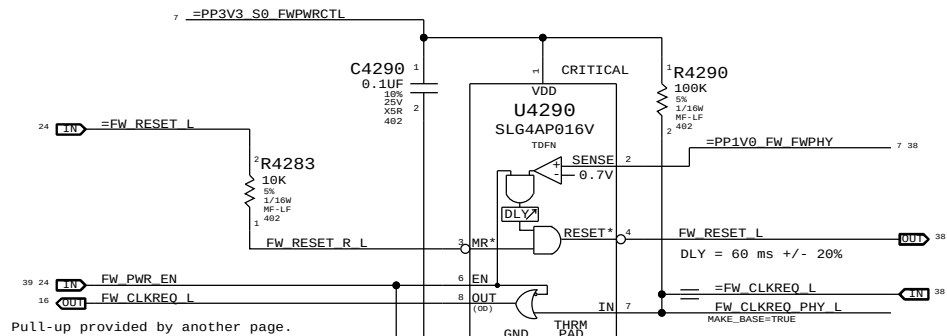
BOM options provided by this page:

(NONE)

FireWire Port Power Switch

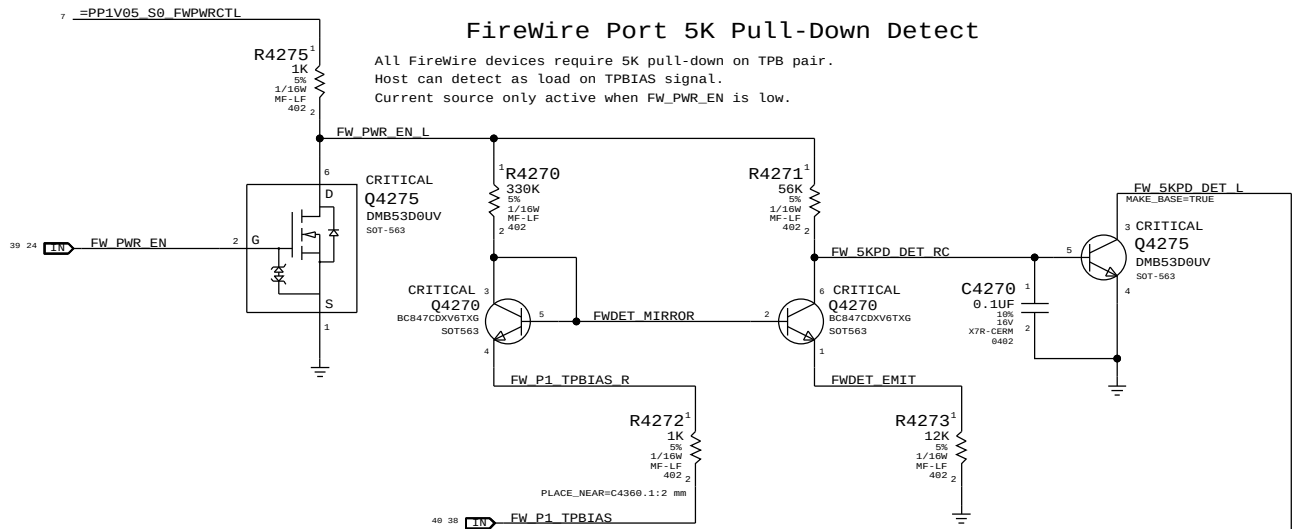


Supervisor & CLKREQ# Isolation



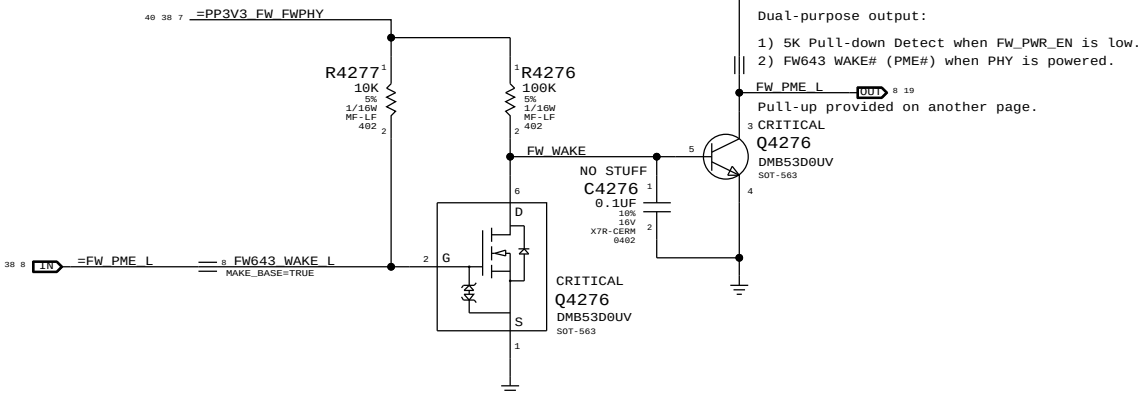
FireWire Port 5K Pull-Down Detect

All FireWire devices require 5K pull-down on TPB pair.
Host can detect as load on TPBIAS signal.
Current source only active when FW_PWR_EN is low.



FireWire PHY WAKE# Support

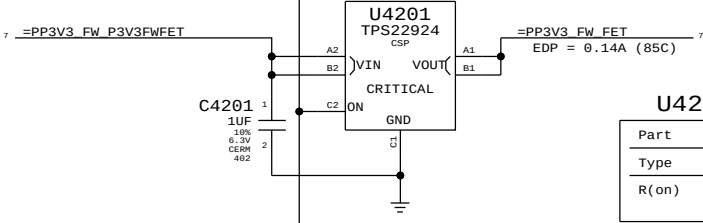
When PHY is powered, FW_5KPD_DET_ acts as legacy PME# signal.



Dual-purpose output:
1) 5K Pull-down Detect when FW_PWR_EN is low.
2) FW643 WAKE# (PME#) when PHY is powered.

Pull-up provided on another page.

3.3V FW Switch

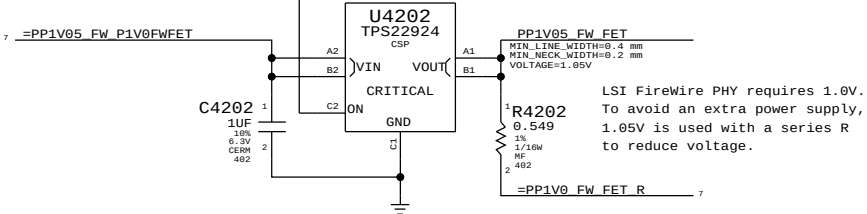


U4201 & U4202

Part	TPS22924C
Type	Load Switch
R(on)	18 mOhm Typ 50 mOhm Max

Max Output: 2A

1.0V FW Switch



LSI FireWire PHY requires 1.0V.
To avoid an extra power supply,
1.05V is used with a series R
to reduce voltage.

TEXT NOTE FOR 3.3V RAIL CURRENT CHANGED TO EDP NUMBER.

SYNC MASTER=K90T MLB		SYNC DATE=06/23/2011	
PAGE TITLE		FireWire Port & PHY Power	
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Page Notes

Power aliases required by this page:

- PPVP_FW_PORT1
- PPVP_FW_PHY_CPS_FET (From Port)
- PPVP_FW_PHY_CPS (To PHY)
- PP3V3_FW_FWPHY
- PP3V3_S0_FWLATEVG

Signal aliases required by this page:

- FW_PHY_DS0
- FW_PHY_DS1
- FW_PHY_DS2

NOTE: This page is expected to contain the necessary aliases to map the FireWire TPA/TPB pairs to their appropriate connectors and/or to properly terminate unused signals.

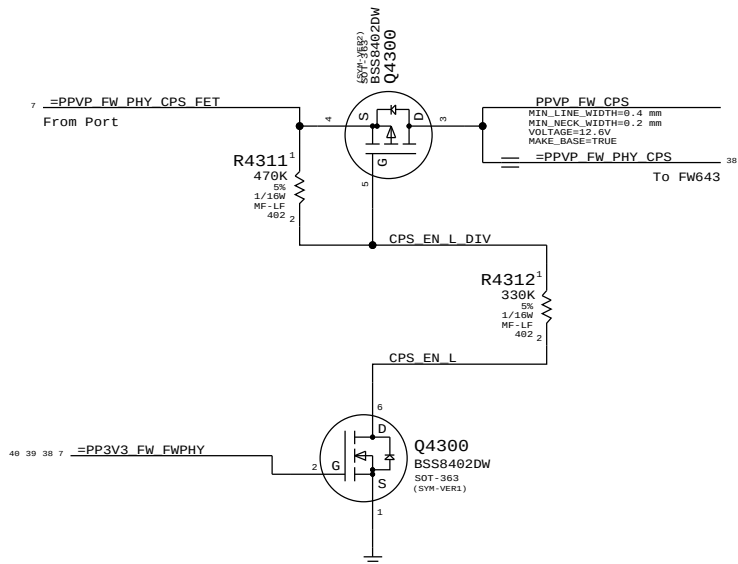
BOM options provided by this page:

(NONE)

1394b implementation based on Apple FireWire Design Guide (FWDG 0.6, 5/14/03)

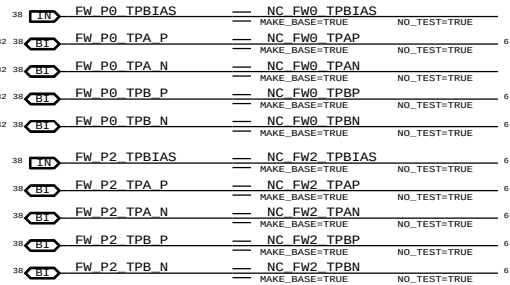
FW643 TPCPS Leakage Protection

FW643 has internal leakage path from TPCPS pin to VDD33.
FET blocks current to TPCPS until VDD33 is powered.



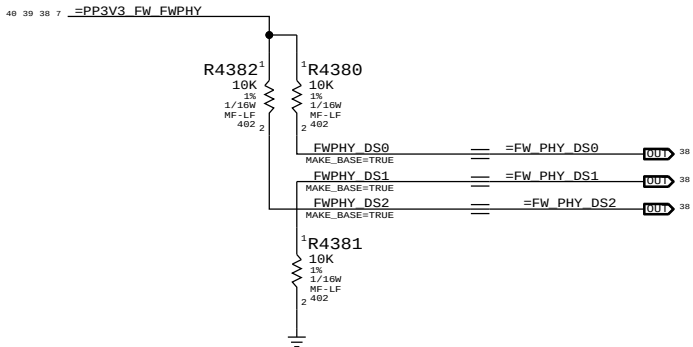
Unused FireWire Ports

Disabled per LSI instructions
(All unused port signals TP/NC)



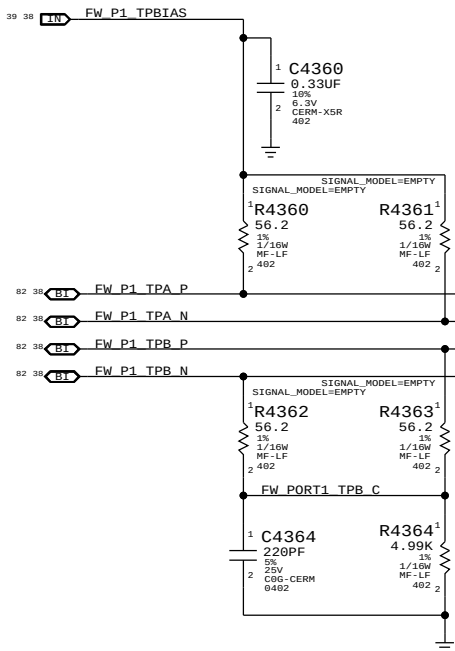
FireWire PHY Config Straps

Configures PHY for:
- Port "1" Bilingual (1394B)



Termination

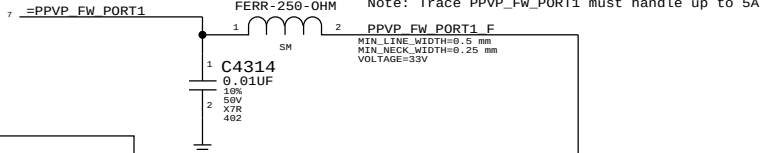
Place close to FireWire PHY



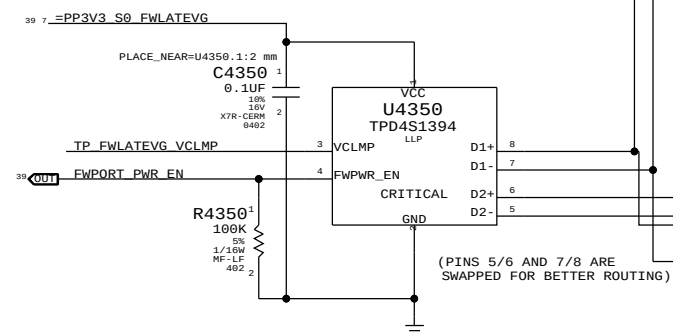
Cable Power

CRITICAL
L4310
FERR-250-OHM

Note: Trace PPVP_FW_PORT1 must handle up to 5A

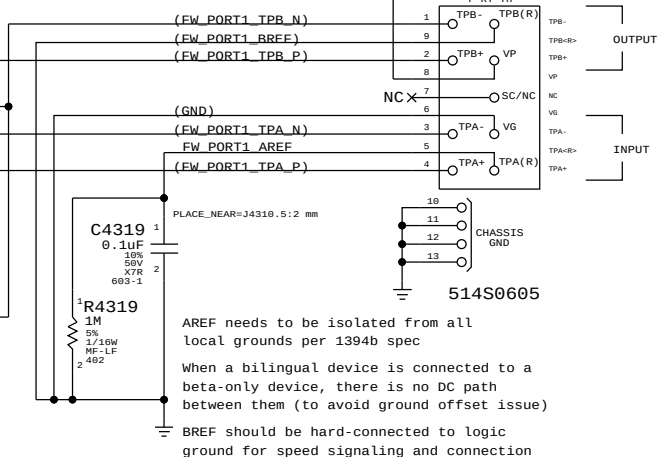


"Snapback" & "Late VG" Protection



PORT 1 BILINGUAL

CRITICAL
J4310
1394B-M97
F-RT-TH



AREF needs to be isolated from all local grounds per 1394b spec
When a bilingual device is connected to a beta-only device, there is no DC path between them (to avoid ground offset issue)
BREF should be hard-connected to logic ground for speed signaling and connection

CANNOT SYNC THIS PAGE FROM T27, TPA AND TPB FOR U4350 IS SWAPPED

FireWire Connector

Apple Inc.

DRAWING NUMBER
051-9058

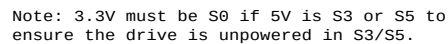
REVISION
6.0.0

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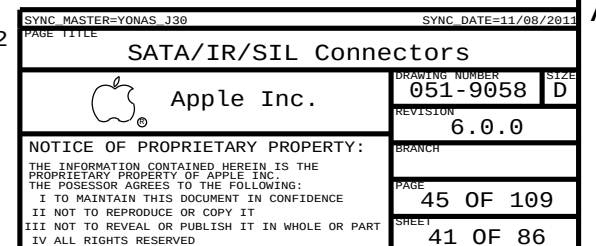
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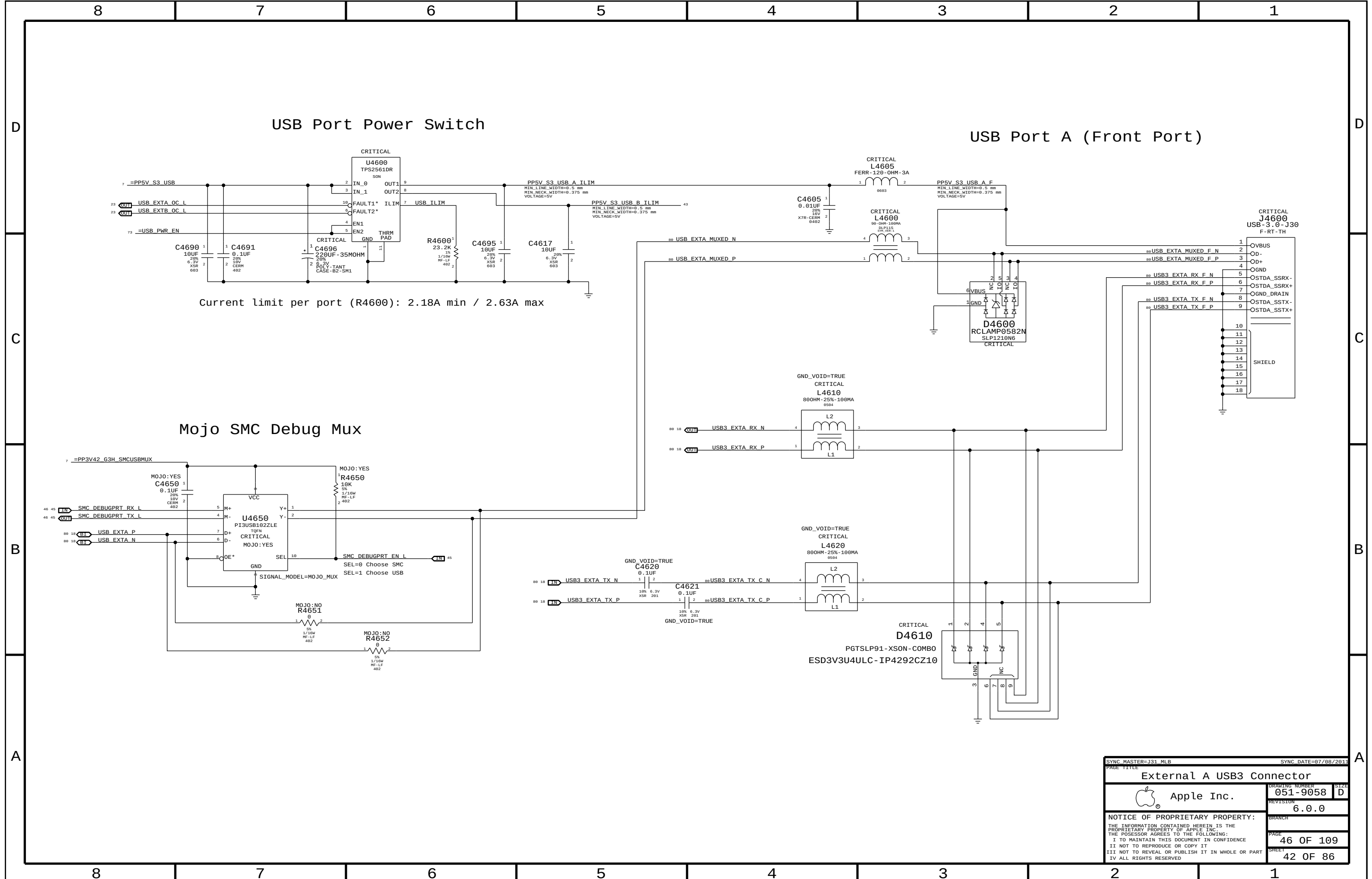
PAGE
43 OF 109

SHEET
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ADDR1	ADD0	Address (R/W)
L	L	0x96/0x97
L	H	0x98/0x99
H	L	0xB6/0xB7
H	H	0xB8/0xB9





Current limit per port (R4600): 2.18A min / 2.63A max

SYNC MASTER=J31 MLB		SYNC DATE=07/08/2011	
PAGE TITLE			
External A USB3 Connector			
 Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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		PAGE	46 OF 109
		SHEET	42 OF 86

PP5V_S3_USB_B_ILIM
MIN_LINE_WIDTH=0.5 mm
MIN_NECK_WIDTH=0.375 mm
VOLTAGE=5V

PP5V_S3_USB_B_F
MIN_LINE_WIDTH=0.5 mm
MIN_NECK_WIDTH=0.375 mm
VOLTAGE=5V

C4705
0.01uF
20%
10V
X7R-CERM
0402

CRITICAL L4705
90-OHM-100MA
0V P115
DNR-VER-1

USB_EXTB_MUX_N
USB_EXTB_MUX_P

USB3_EXTB_RX_N
USB3_EXTB_RX_P

USB3_EXTB_TX_N
USB3_EXTB_TX_P

USB3_EXTB_TX_C_N
USB3_EXTB_TX_C_P

GND_VOID=TRUE
CRITICAL L4710
800OHM-25%-100MA
0504

GND_VOID=TRUE
CRITICAL L4720
800OHM-25%-100MA
0504

GND_VOID=TRUE
C4720
0.1uF
10%
6.3V
X5R 201

GND_VOID=TRUE
C4721
0.1uF
10%
6.3V
X5R 201

CRITICAL D4700
RCLAMP0582N
SLP1210N6
CRITICAL

CRITICAL D4710
PGTSLP91-XSON-COMBO
ESD3V3U4ULC-IP4292CZ10

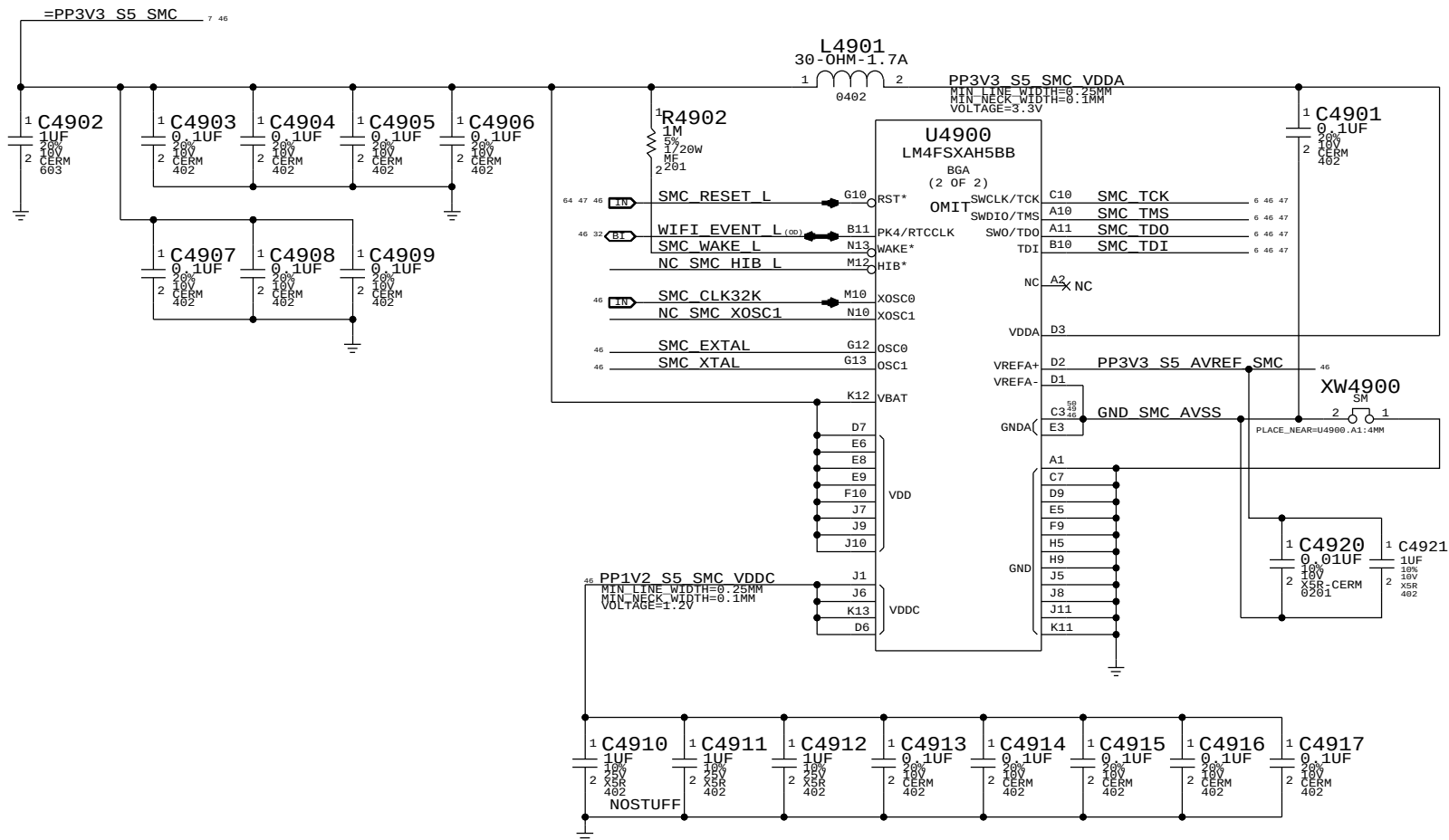
CRITICAL J4700
USB-3.0-J30
F-RT-TH

SHIELD

Pin	Signal
1	OVBUS
2	OD-
3	OD+
4	OGND
5	OSTDA_SSRX-
6	OSTDA_SSRX+
7	OGND_DRAIN
8	OSTDA_SSTX-
9	OSTDA_SSTX+
10	
11	
12	
13	
14	
15	
16	
17	
18	

SYNC MASTER=J31 MLB		SYNC DATE=07/08/2013	
PAGE TITLE			
External B USB3 Connector			
 Apple Inc.		DRAWING NUMBER	051-9058
		SIZE	D
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		BRANCH	
		PAGE	47 OF 109
		SHEET	43 OF 86

SYNC MASTER=K901 MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
Front Flex Support			
	Apple Inc.		DRAWING NUMBER 051-9058
			SIZE D
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NOTE:
Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



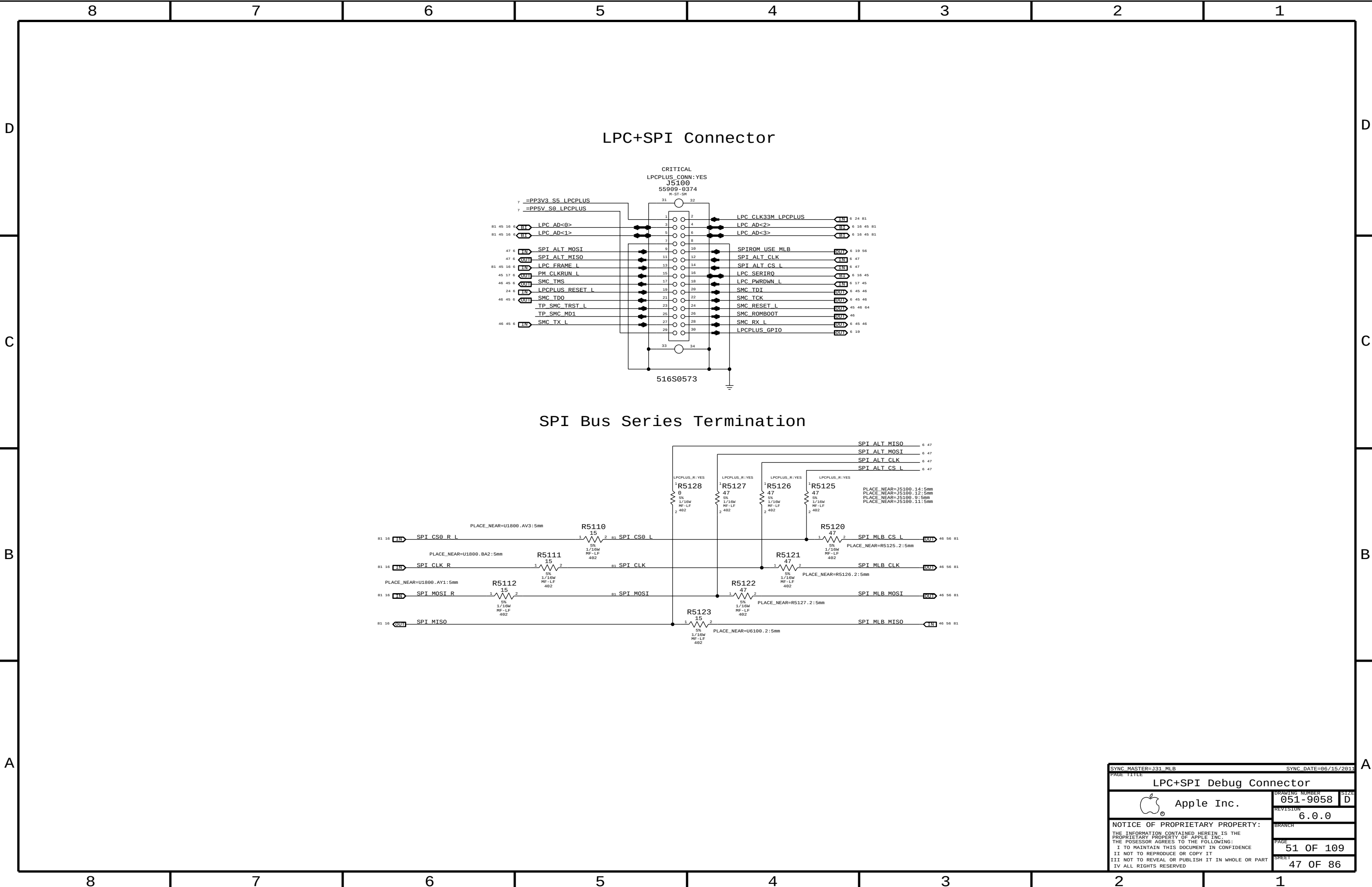
C

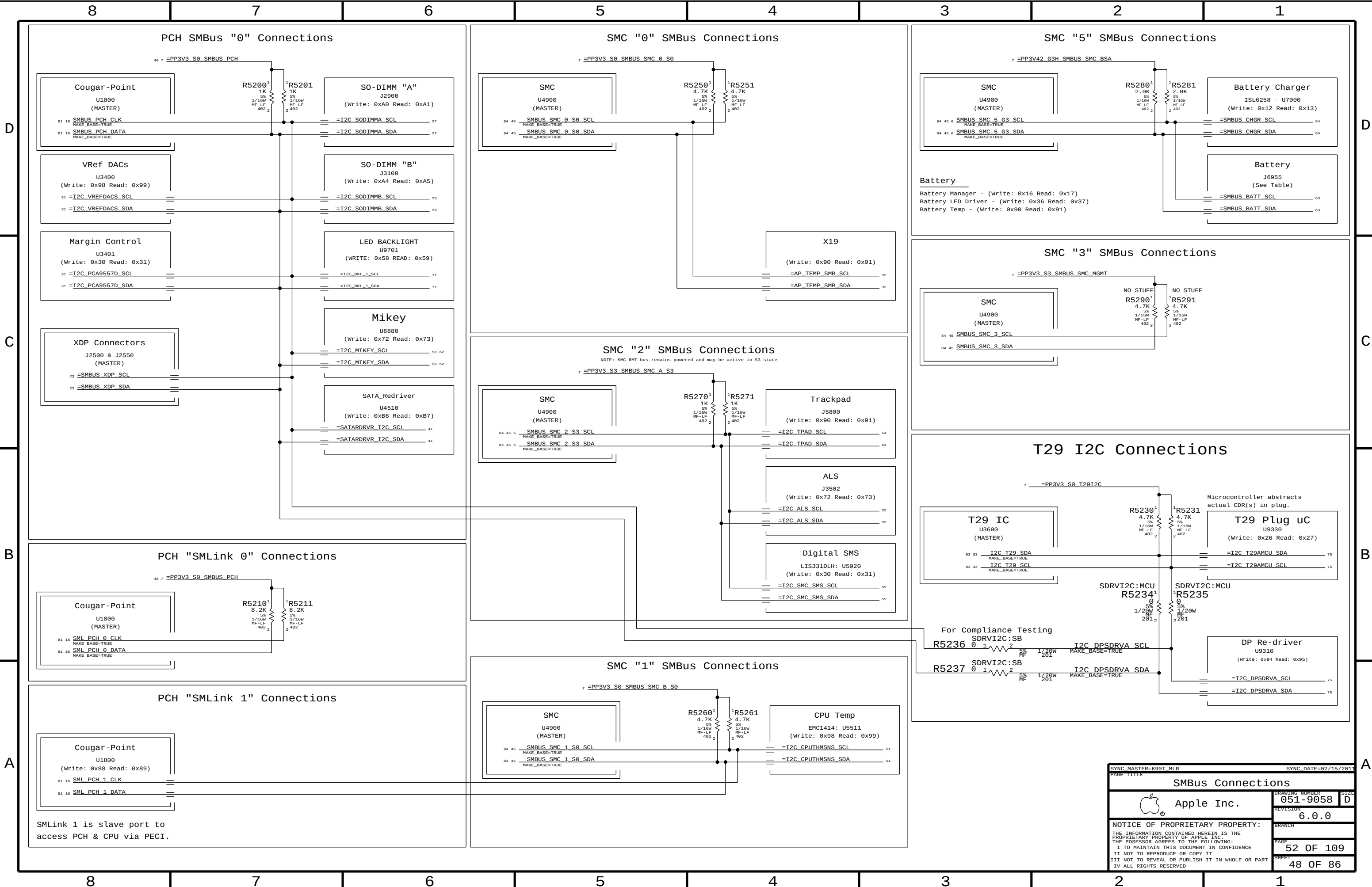


C

B

A

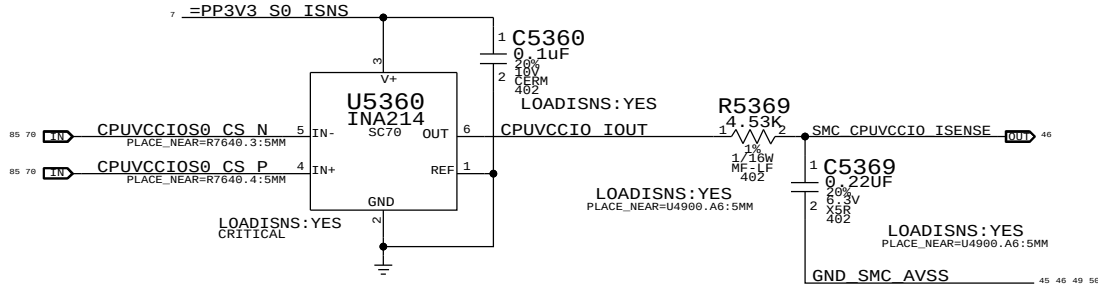




87654321

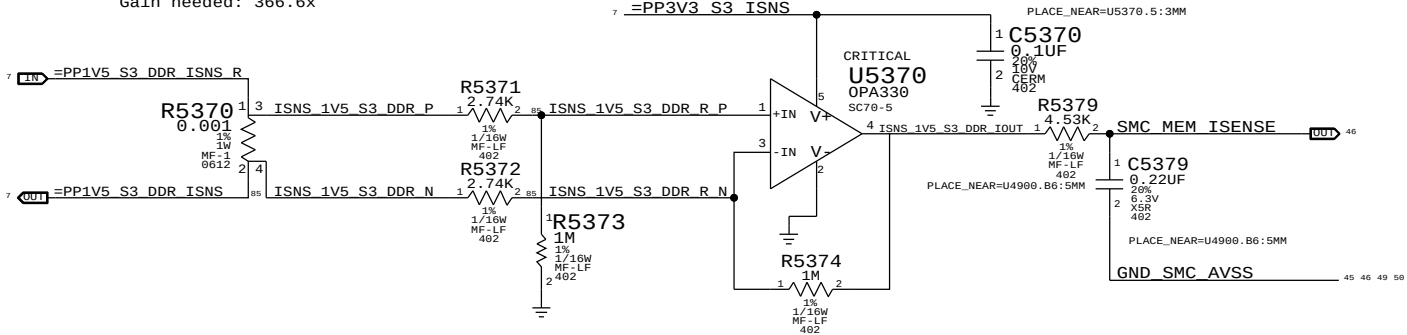
CPU VCCIO 1.05V Load Side Current Sense (IC1C)

Gain: 100x, EDP: 20.1 A
Rsense: 0.001 (R7640)
V across Rsense: 20.1 mV
Gain needed: 164.2x



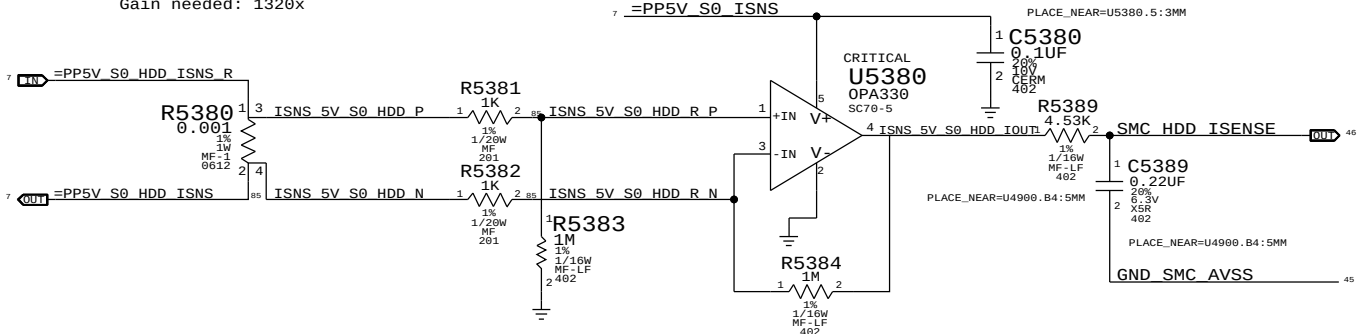
DDR 1.5V S3 (Memory) Current Sense (IM0C)

Gain: 364.9x, EDP: 9 A
Rsense: 0.001 (R5370)
V across Rsense: 9 mV
Gain needed: 366.6x

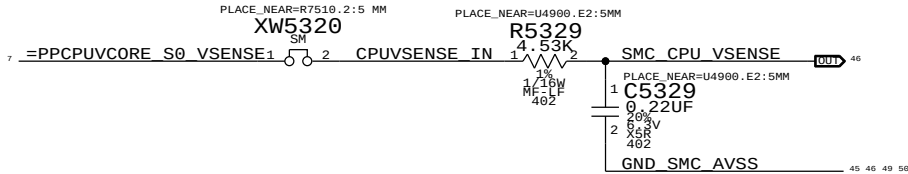


HDD Current Sense (IHDC)

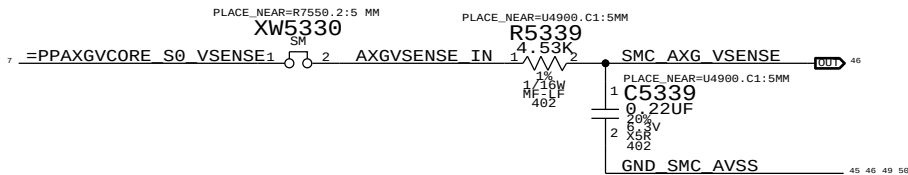
Gain: 1000x, EDP: 2.5 A (12.5 W)
Rsense: 0.001 (R5380)
V across Rsense: 2.5 mV
Gain needed: 1320x



CPU Core Voltage Sense (VC0C)

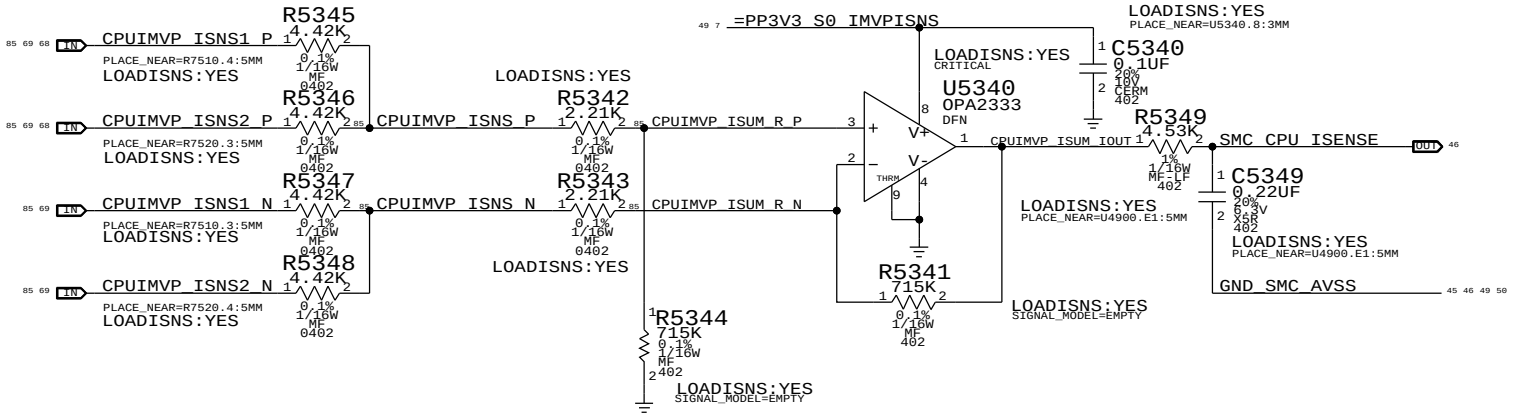


AXG Core Voltage Sense (VN0C)



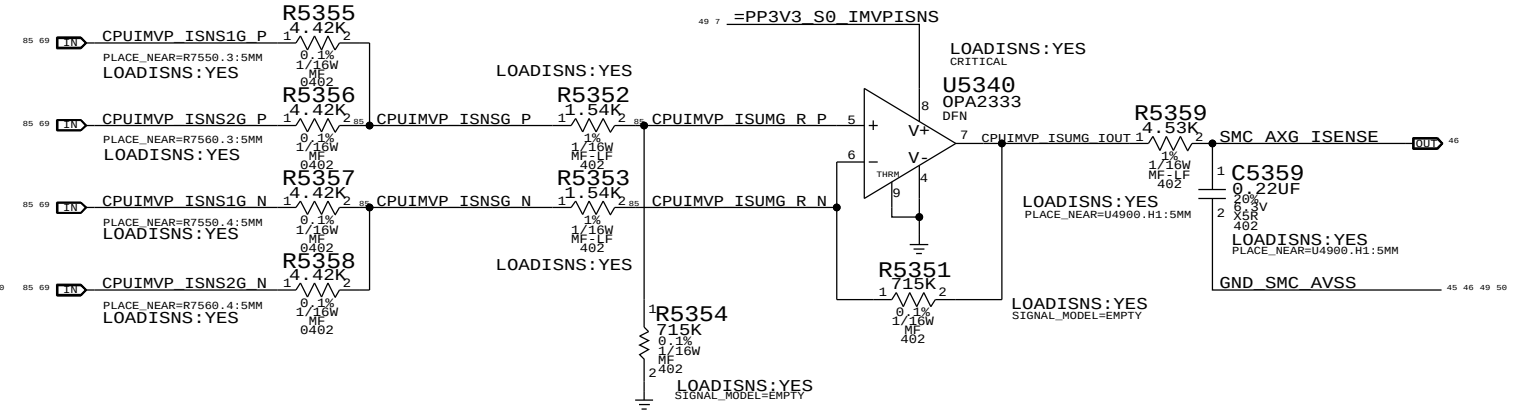
CPU Core Load Side Current Sense (IC0C)

Gain: 161.5x, EDP: 53 A
Rsense: 2x of 0.00075 (R7510, R7520), Rsum: 0.000375
V across Rsense: 19.8 mV
Gain needed: 166.1x



AXG Core Load Side Current Sense (IN0C)

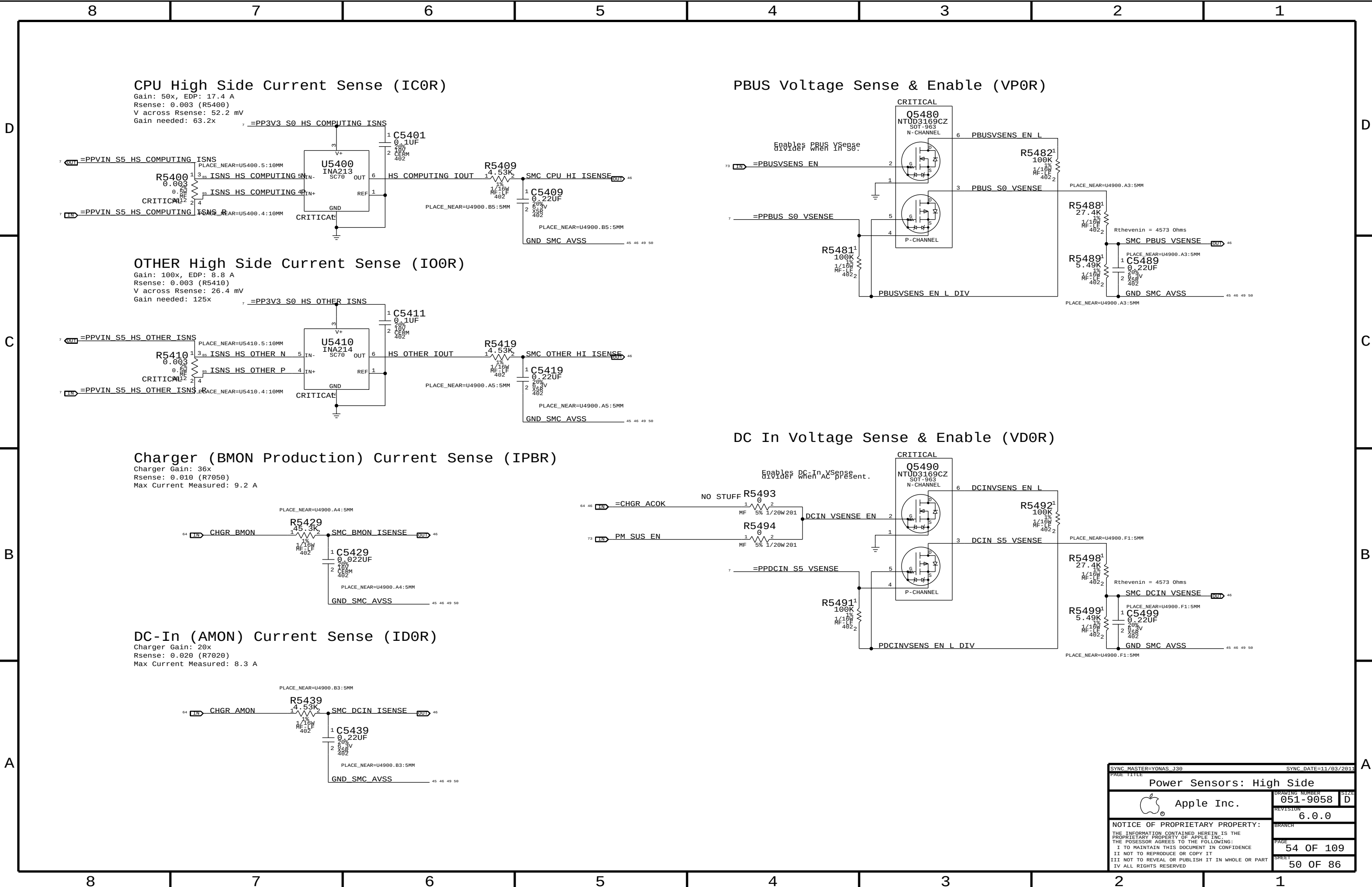
Gain: 190.6x, EDP: 46 A
Rsense: 2x of 0.00075 (R7550, R7560), Rsum: 0.000375
V across Rsense: 17.25 mV
Gain needed: 191.3x



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
116S0114	3	RES, MTL, FLIM, 100K, 1/16W, 0402, SMD, LF	C5349, C5359, C5369		LOADISNS:NO

SYNC MASTER=LINDA J30		SYNC DATE=09/28/2011	
PAGE TITLE			
Power Sensors: Load Side			
 Apple Inc.		DRAWING NUMBER	051-9058
		SIZE	D
		REVISION	6.0.0
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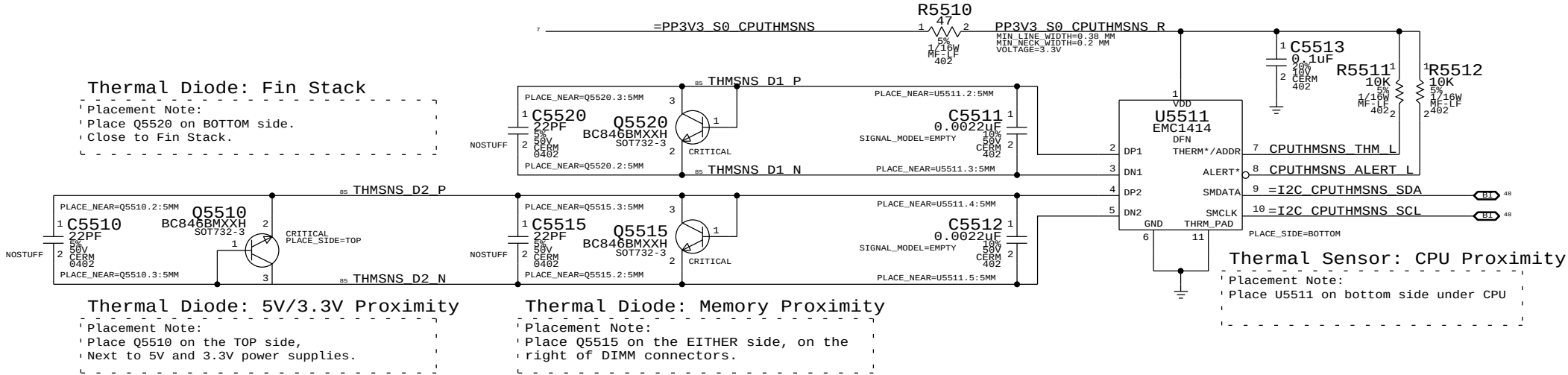
87654321



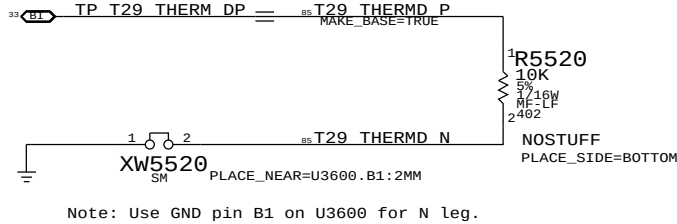
SYNC MASTER=YONAS J30		SYNC DATE=11/03/2011	
PAGE TITLE		Power Sensors: High Side	
		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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		PAGE	54 OF 109
		SHEET	50 OF 86

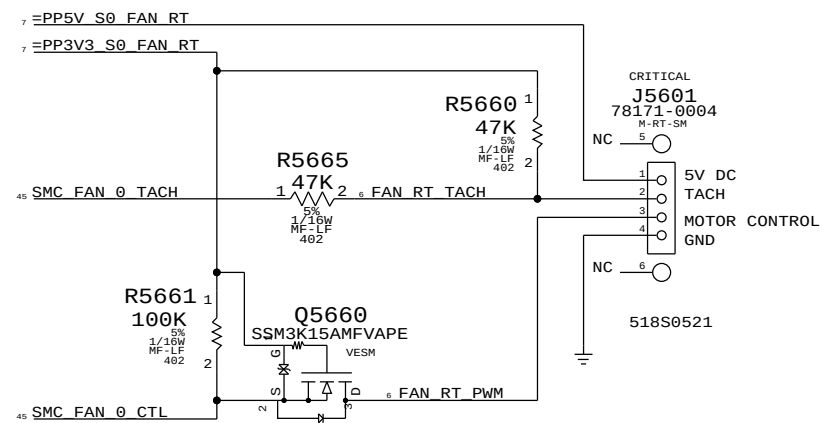
Thermal Sensor:
CPU Proximity, Fin Stack, Memory Proximity, 5V/3.3V Proximity

I2C Write: 0x98, I2C Read: 0x99



Thermal Sensor: T29 Die





SYNC MASTER-K90I MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
Fan			
	Apple Inc.	DRAWING NUMBER	051-9058
		SIZE	D
		REVISION	6.0.0
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		56 OF 109	
		SHEET	
		52 OF 86	

D

C

B

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D

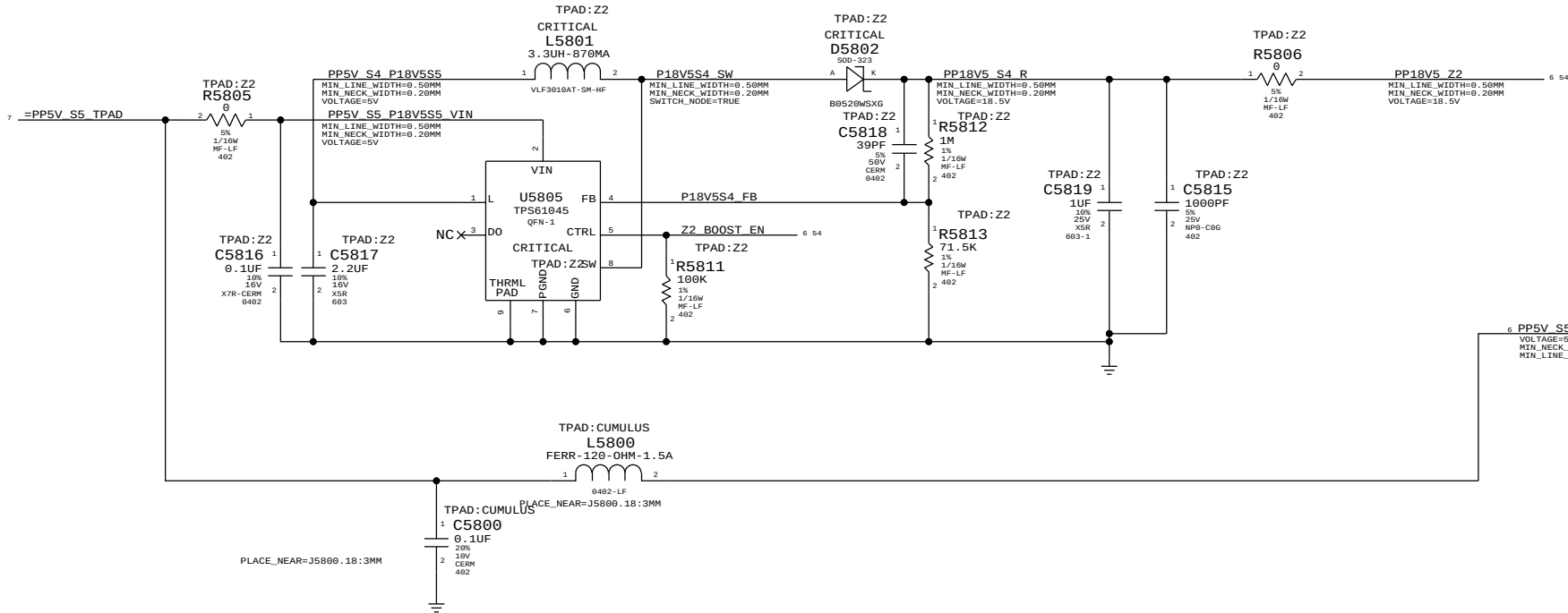
C

B

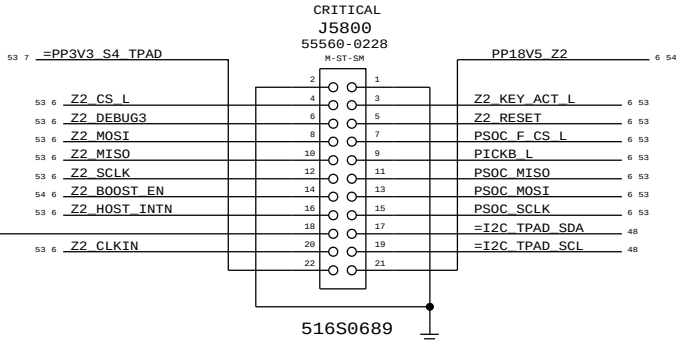
A

BOOSTER +18.5VDC FOR SENSORS

- BOOSTER DESIGN CONSIDERATION:
- POWER CONSUMPTION
 - DROOP LINE REGULATION
 - RIPPLE TO MEET ERS
 - 100-300 KHZ CLEAN SPECTRUM
 - STARTUP TIME LESS THAN 2MS
 - R5812, R5813, C5818 MODIFIED

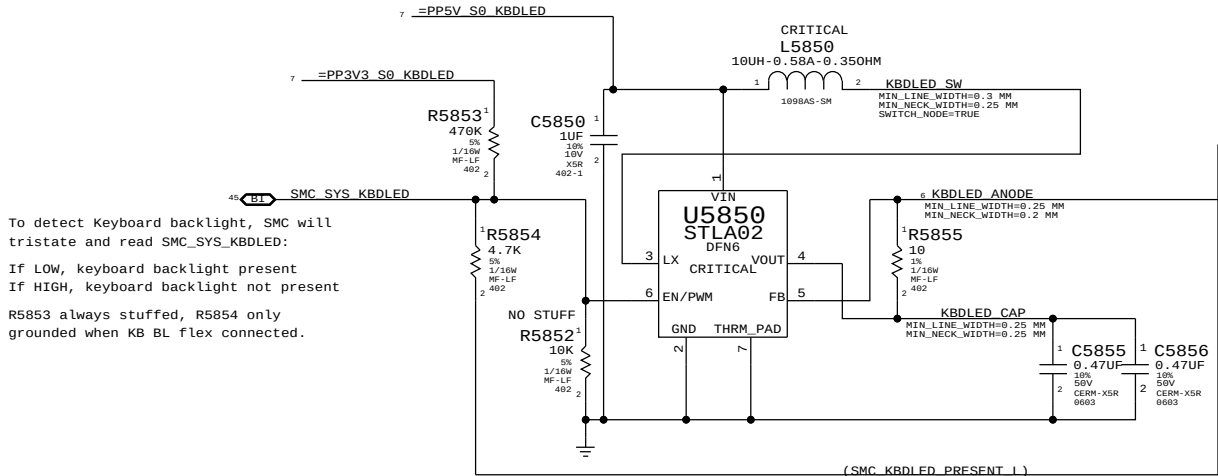


IPD Flex Connector



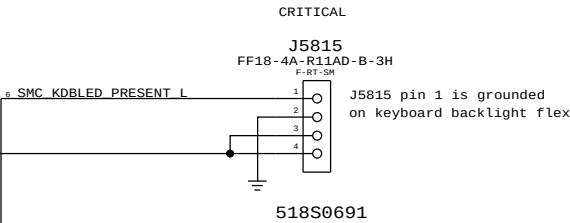
PIN 21 IS NC ON CUMULUS FLEX
PIN 18 IS NC ON Z2 FLEX

Keyboard Backlight Driver & Detection

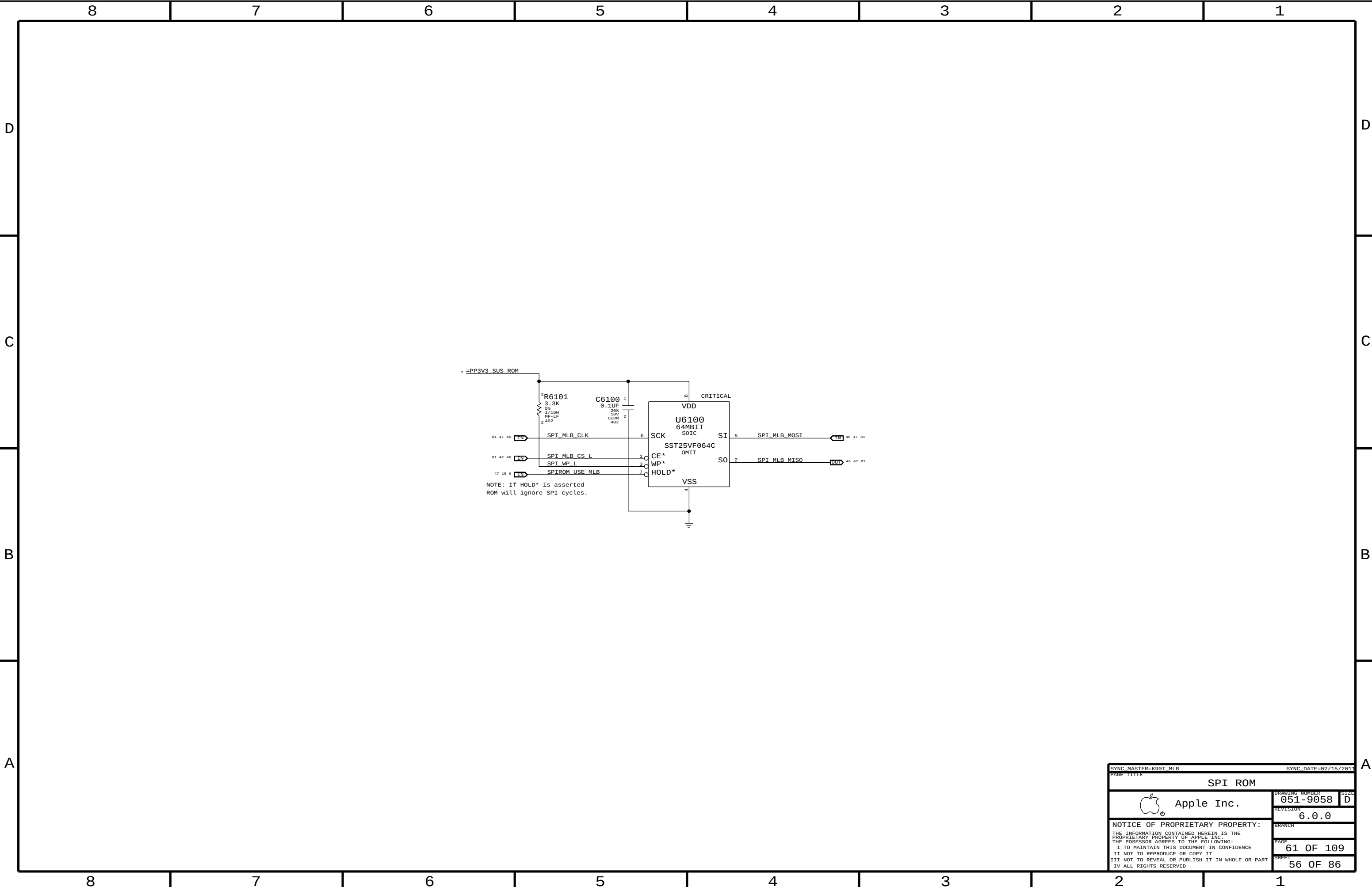


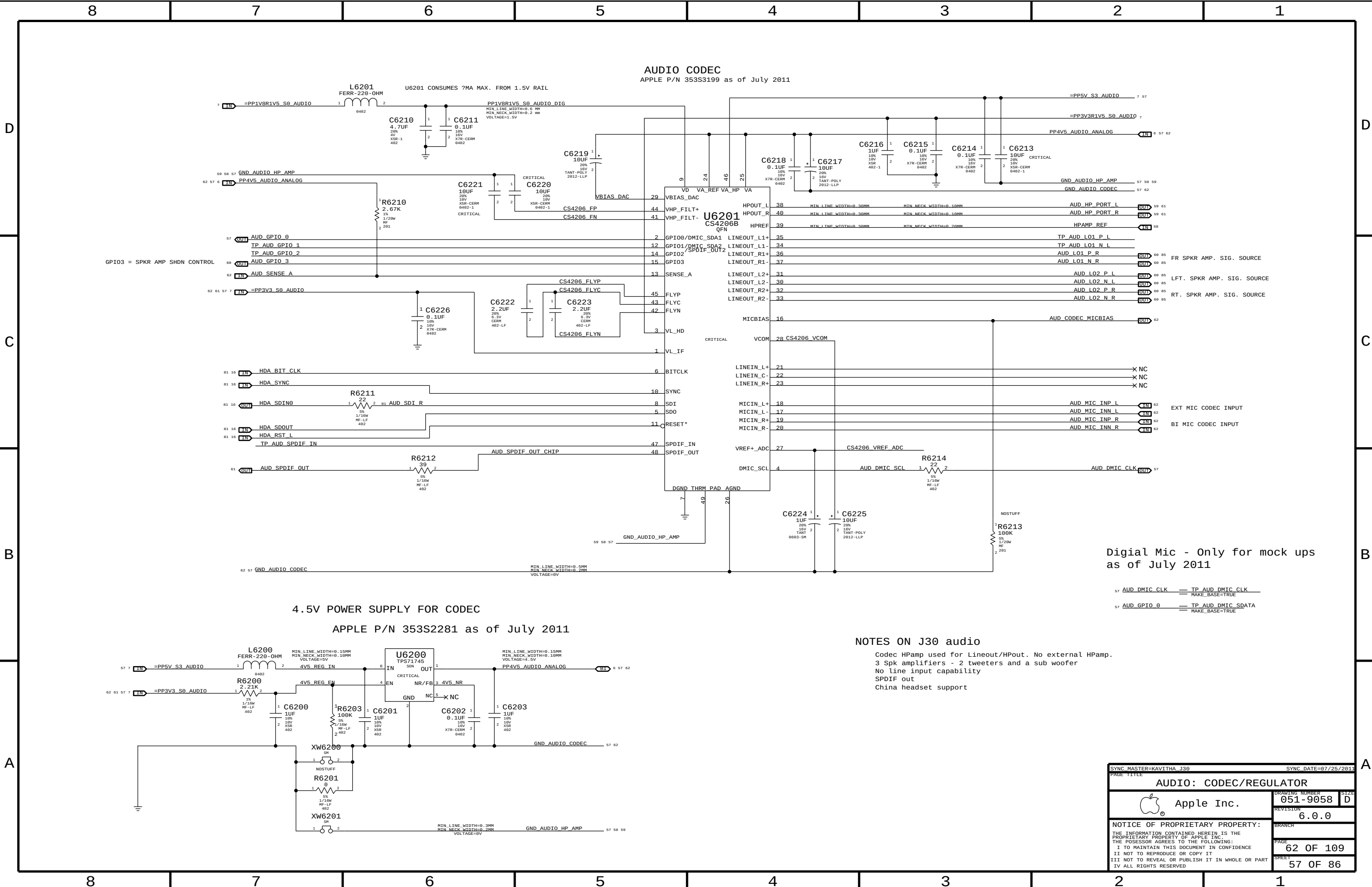
To detect Keyboard backlight, SMC will tristate and read SMC_SYS_KBDLED:
If LOW, keyboard backlight present
If HIGH, keyboard backlight not present
R5853 always stuffed, R5854 only grounded when KB BL flex connected.

Keyboard Backlight Connector



WELLSPRING 2		DRAWING NUMBER		SIZE
Apple Inc.		051-9058		D
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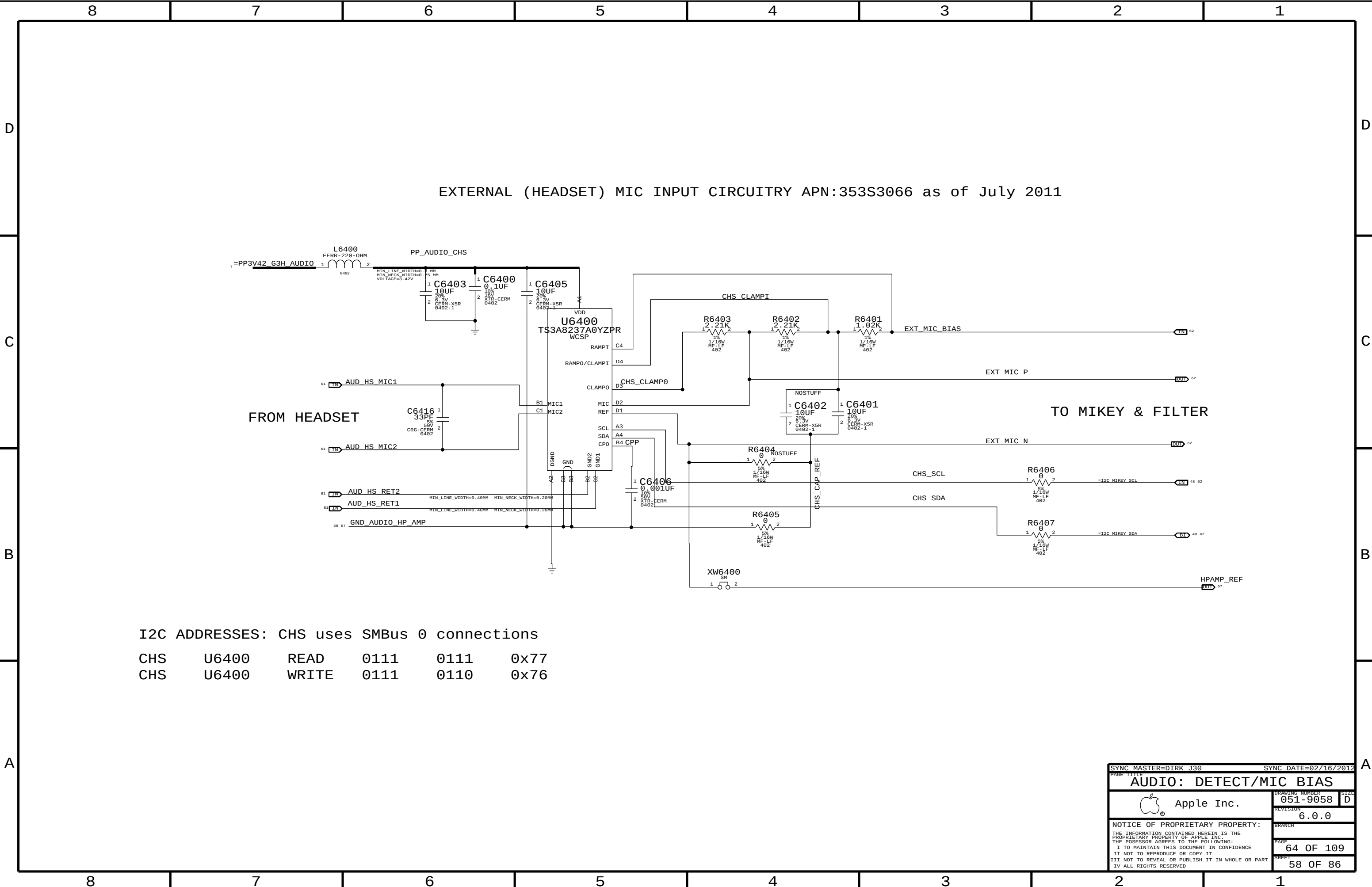
AUDIO CODEC
APPLE P/N 353S3199 as of July 2011

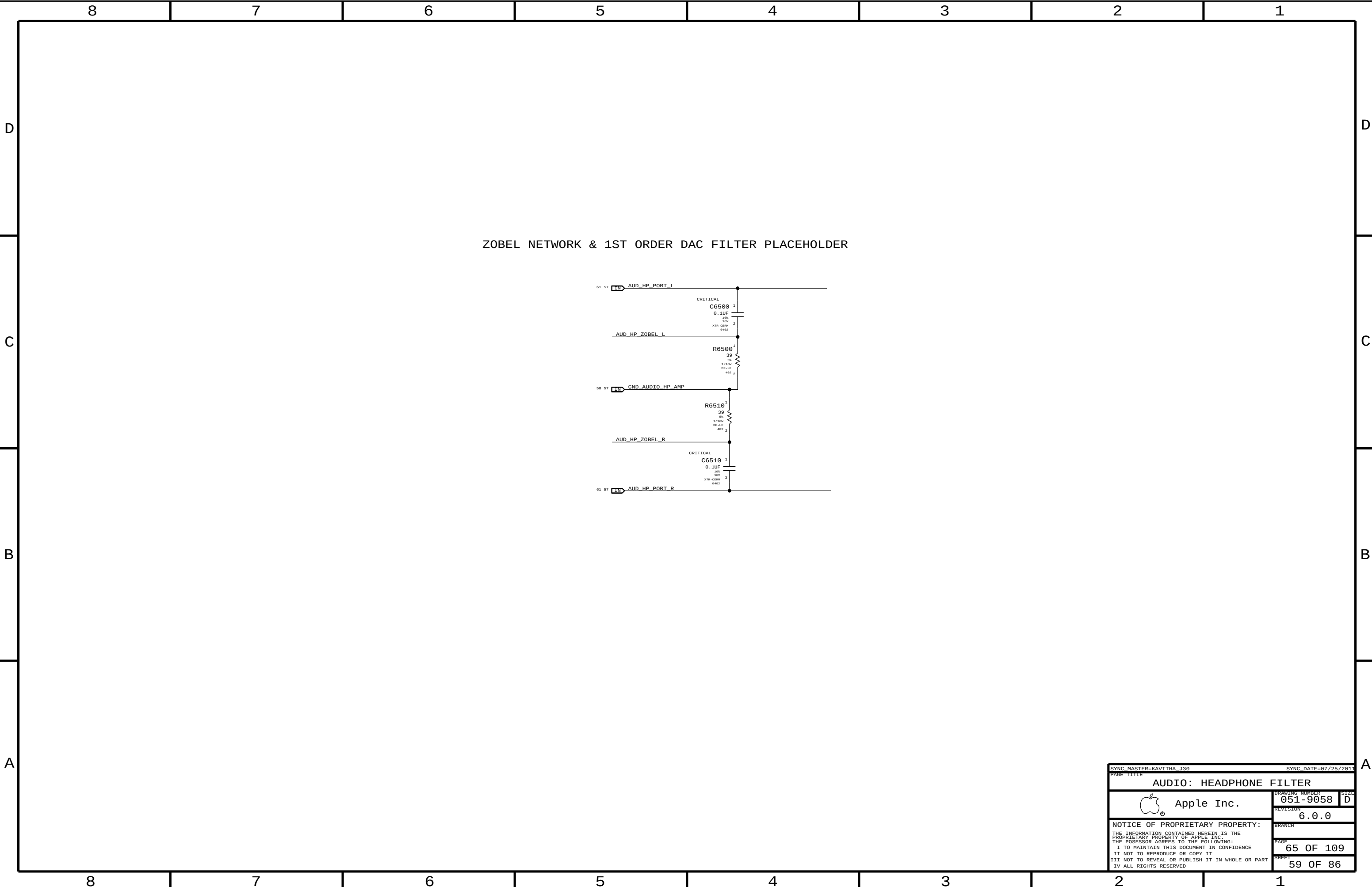
4.5V POWER SUPPLY FOR CODEC
APPLE P/N 353S2281 as of July 2011

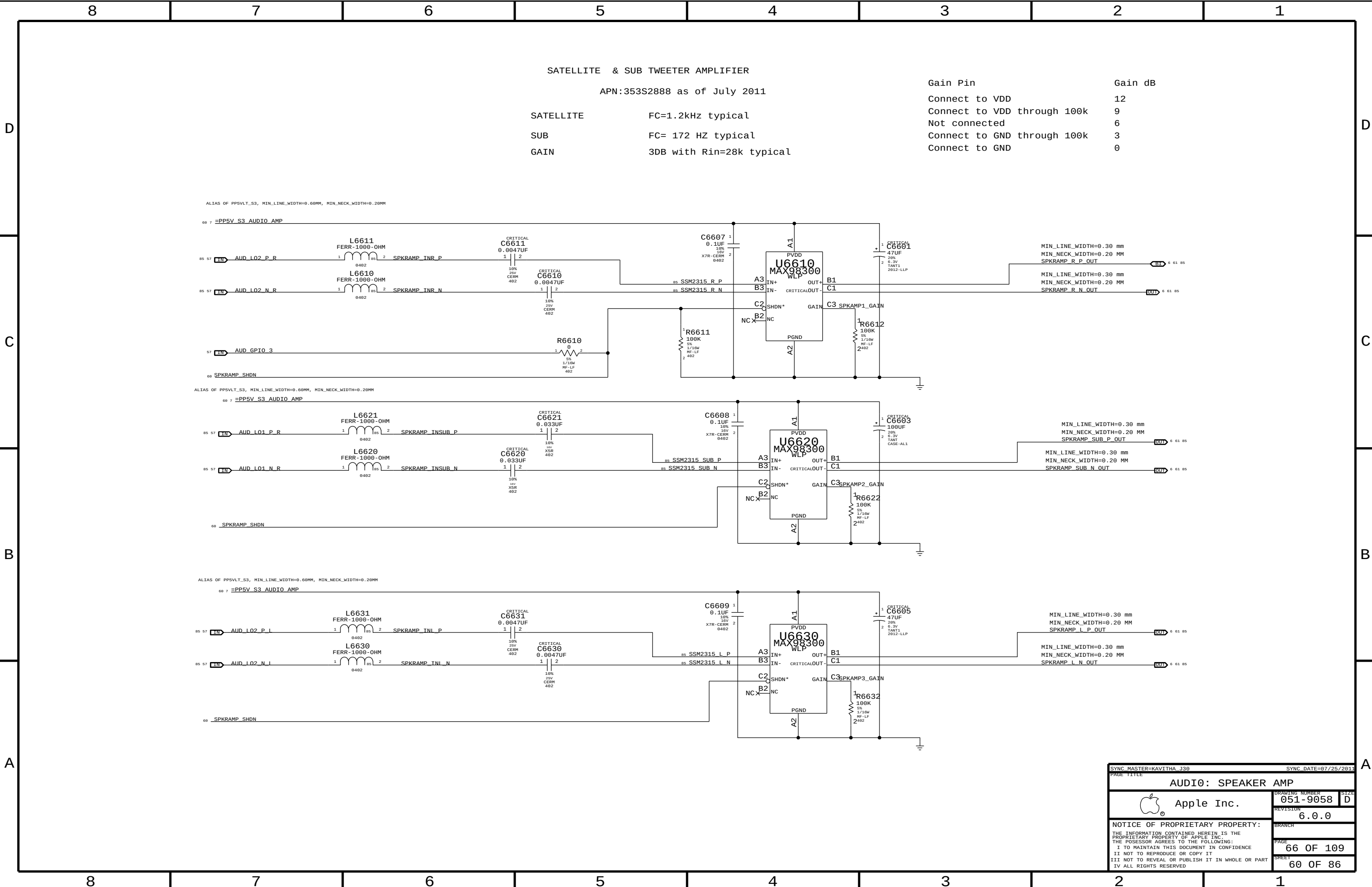
Digital Mic - Only for mock ups
as of July 2011

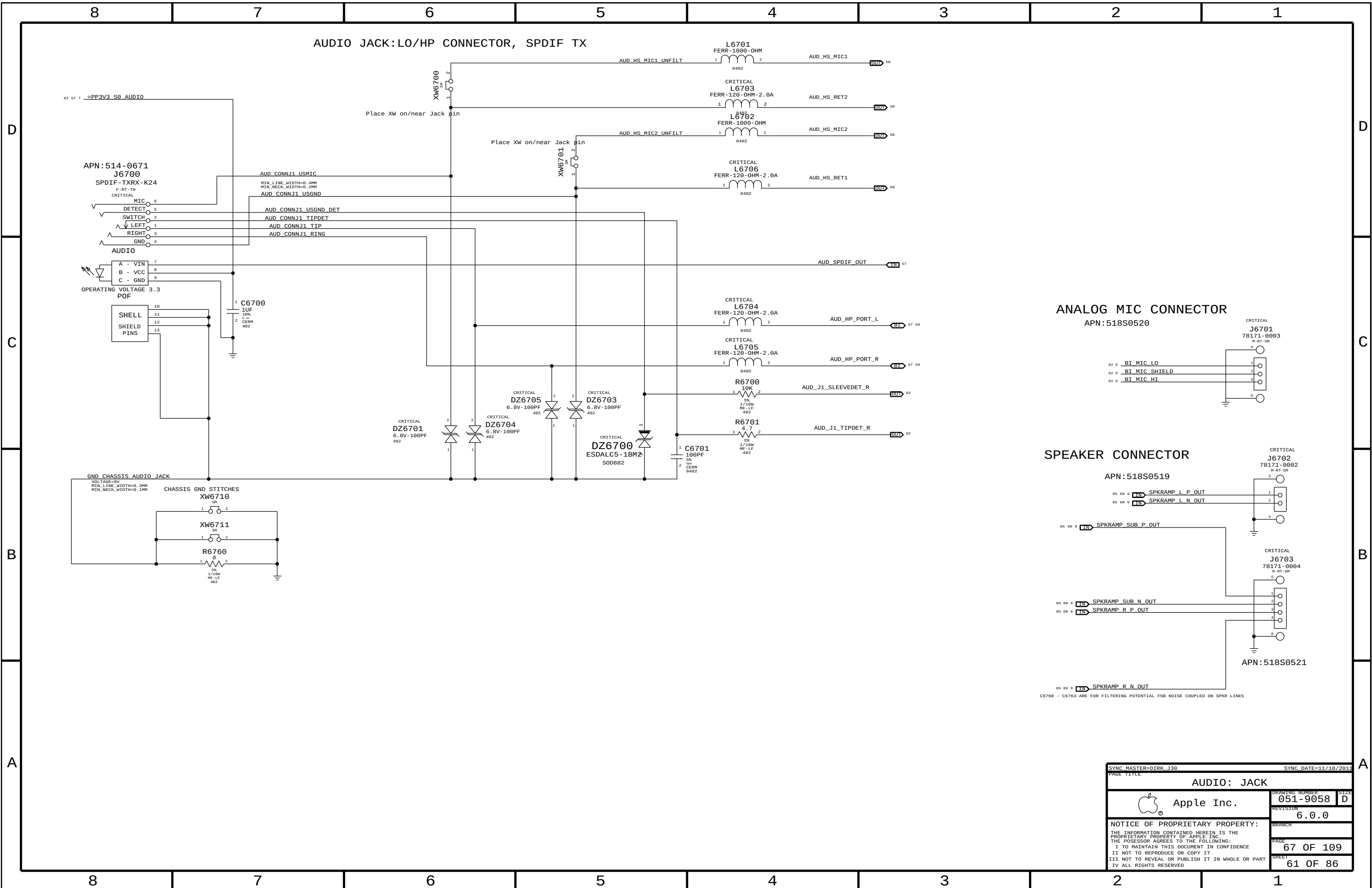
NOTES ON J30 audio
Codec HPamp used for Lineout/HPout. No external HPamp.
3 Spk amplifiers - 2 tweeters and a sub woofer
No line input capability
SPDIF out
China headset support

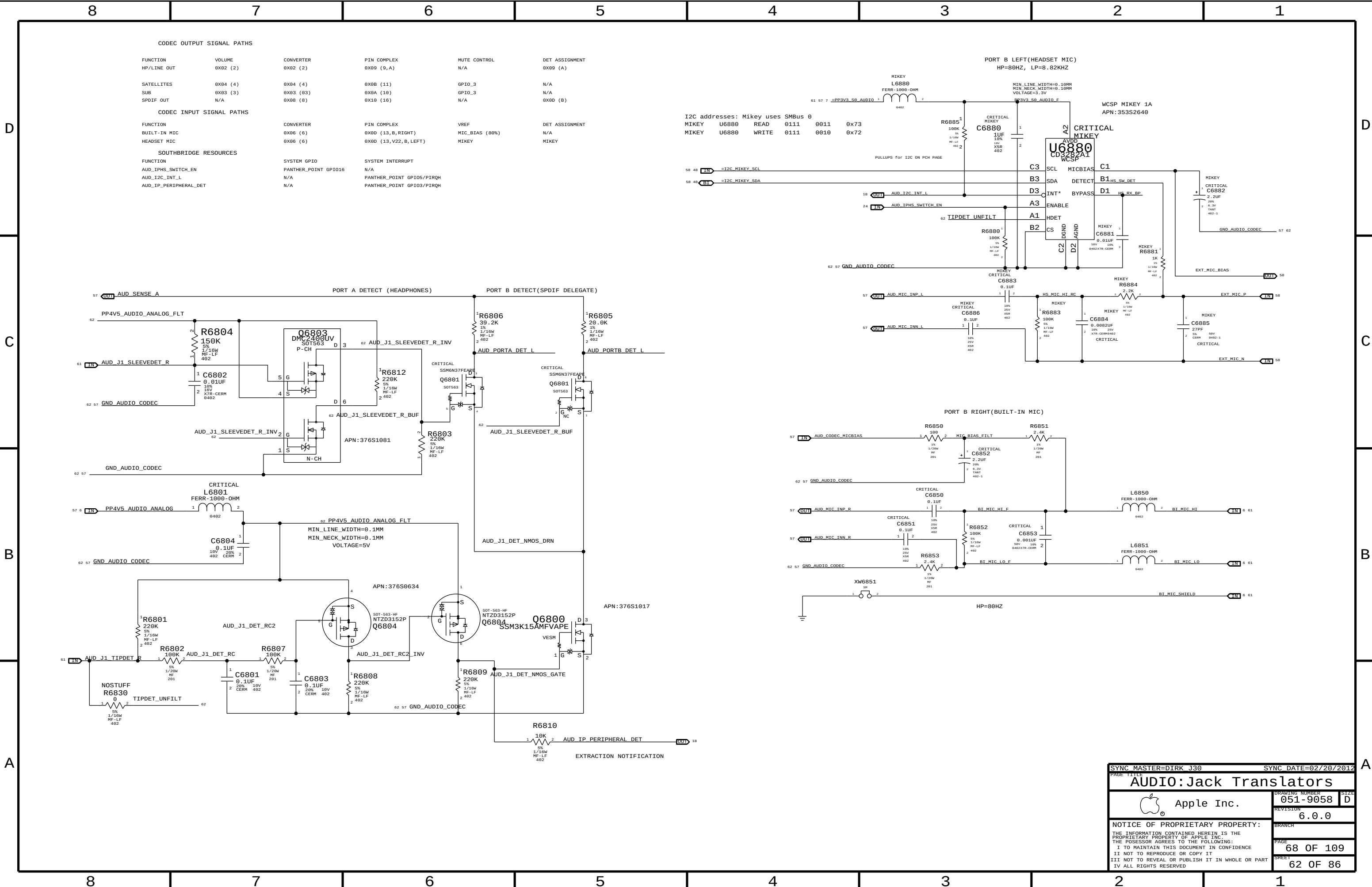
SYNC MASTER=KAVITHA J30		SYNC DATE=07/25/2011	
PAGE TITLE		AUDIO: CODEC/REGULATOR	
Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL	DET ASSIGNMENT
HP/LINE OUT	0x02 (2)	0x02 (2)	0x09 (9, A)	N/A	0x09 (A)
SATELLITES	0x04 (4)	0x04 (4)	0x0B (11)	GPIO_3	N/A
SUB	0x03 (3)	0x03 (03)	0x0A (10)	GPIO_3	N/A
SPDIF OUT	N/A	0x08 (8)	0x10 (16)	N/A	0x0D (B)

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF	DET ASSIGNMENT
BUILT-IN MIC	0x06 (6)	0x0D (13, B, RIGHT)	MIC_BIAS (80%)	N/A
HEADSET MIC	0x06 (6)	0x0D (13, V22, B, LEFT)	MIKEY	MIKEY

SOUTHBRIDGE RESOURCES

FUNCTION	SYSTEM GPIO	SYSTEM INTERRUPT
AUD_IPHS_SWITCH_EN	PANTHER_POINT GPIO16	N/A
AUD_I2C_INT_L	N/A	PANTHER_POINT GPIO5/PIRQH
AUD_IP_PERIPHERAL_DET	N/A	PANTHER_POINT GPIO3/PIRQH

I2C addresses: Mikey uses SMBus 0
MIKEY U6880 READ 0111 0011 0x73
MIKEY U6880 WRITE 0111 0010 0x72

PORT B LEFT(HEADSET MIC)
HP=80HZ, LP=B.82KHZ

WCSF MIKEY 1A
APN:353S2640

PORT B RIGHT(BUILT-IN MIC)
HP=80HZ

SYNC MASTER=DIRK J30

SYNC DATE=02/20/2012

AUDIO:Jack Translators

Apple Inc.

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DRAWING NUMBER

051-9058

REVISION

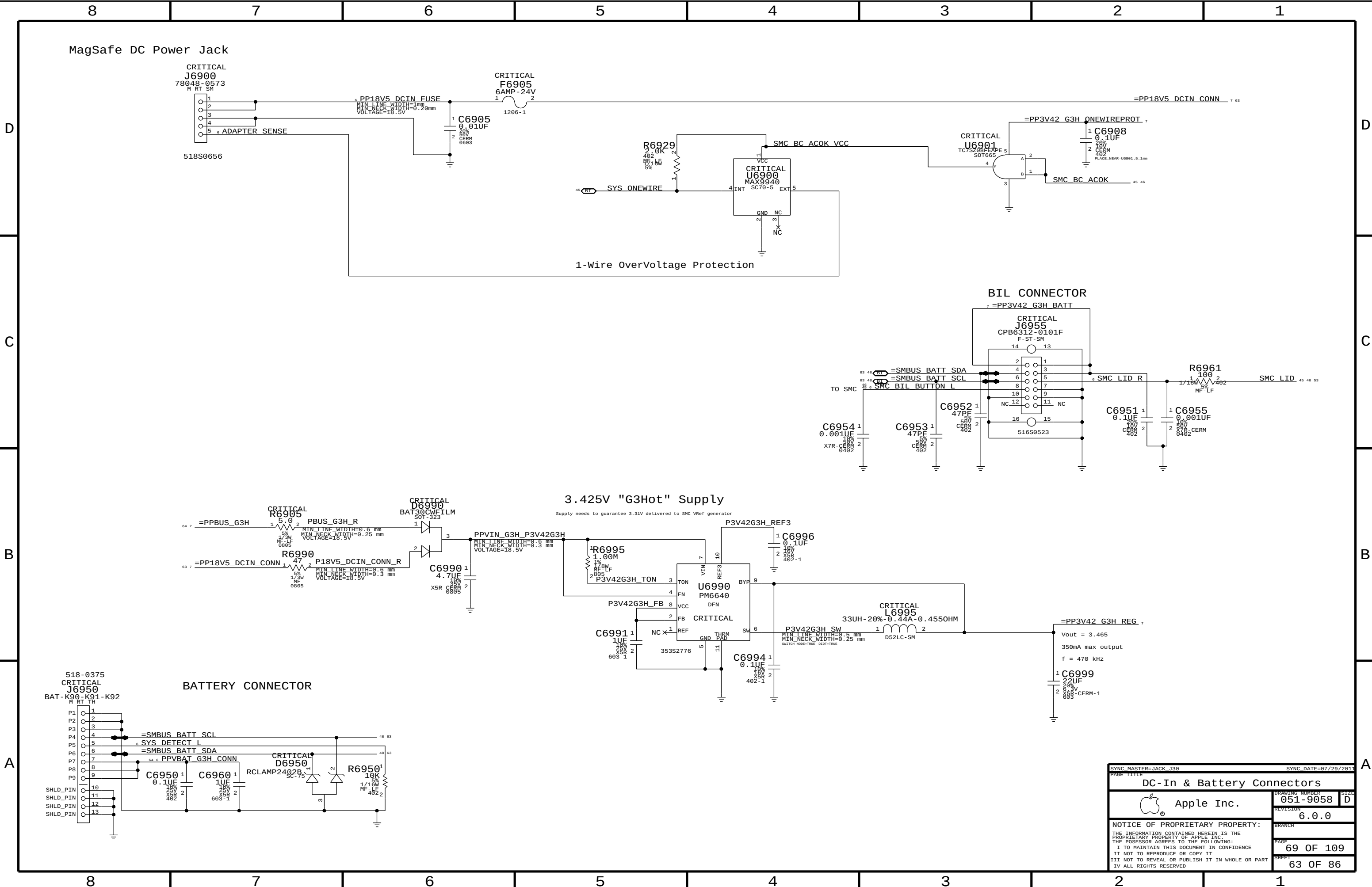
6.0.0

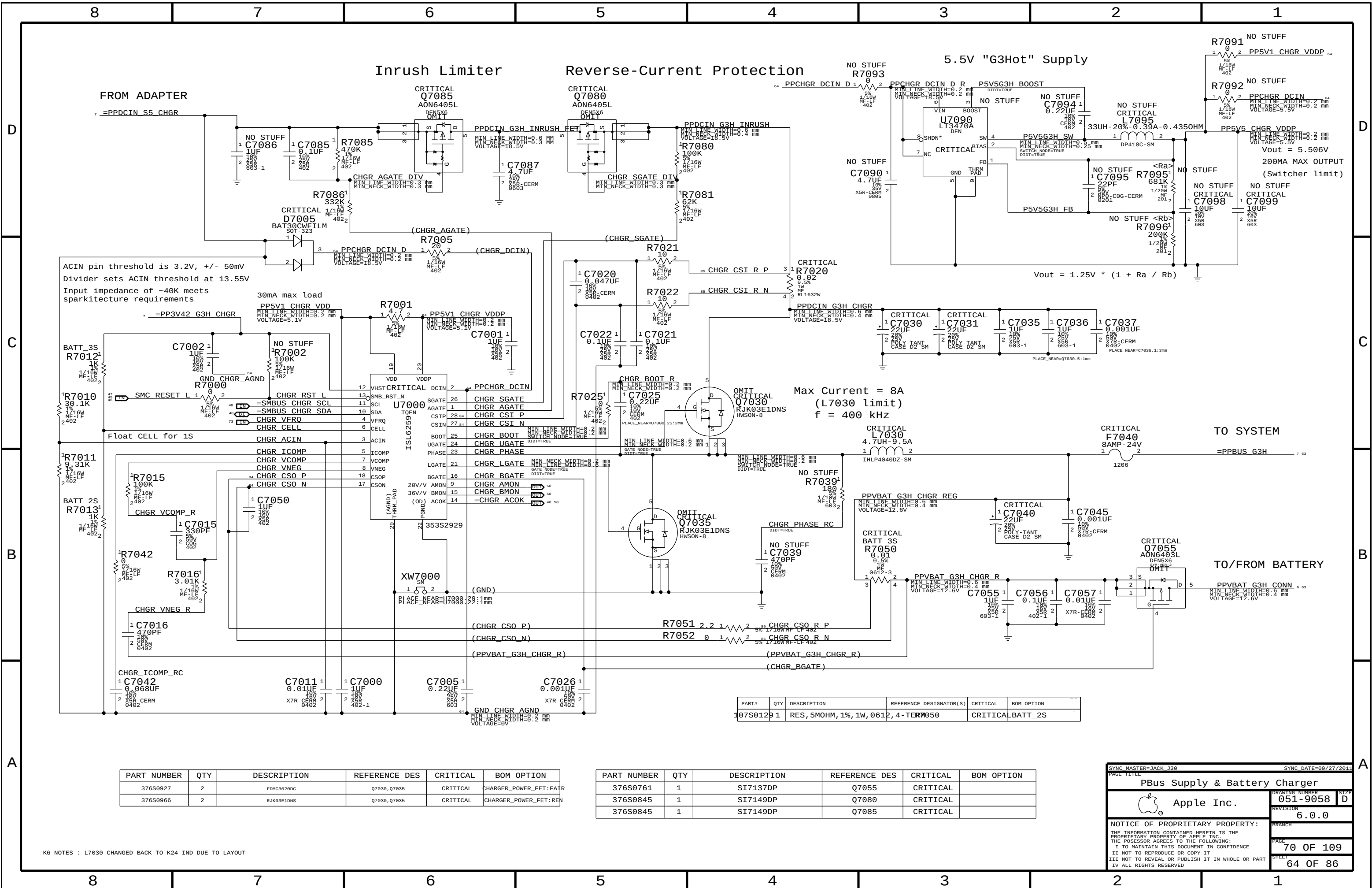
PAGE

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0927	2	F0MC3020DC	Q7030, Q7035	CRITICAL	CHARGER_POWER_FET:FAJR
376S0966	2	RJK03E1DNS	Q7030, Q7035	CRITICAL	CHARGER_POWER_FET:REN

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
376S0761	1	SI7137DP	Q7055	CRITICAL	
376S0845	1	SI7149DP	Q7080	CRITICAL	
376S0845	1	SI7149DP	Q7085	CRITICAL	

SYNC MASTER=JACK 330

SYNC DATE=09/27/2011

PAGE TITLE

PBus Supply & Battery Charger

Apple Inc.

Apple Inc.

DRAWING NUMBER

051-9058

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K6 NOTES : L7030 CHANGED BACK TO K24 IND DUE TO LAYOUT

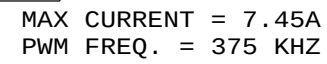
D

BBA

5V_S3/3.3V_S5 POWER SUPPLY

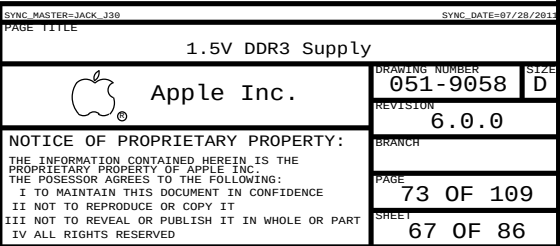
$V_{OUT} = (2 * R_A / R_B) + 2$ $V_{OUT} = (2 * R_C / R_D) + 2$

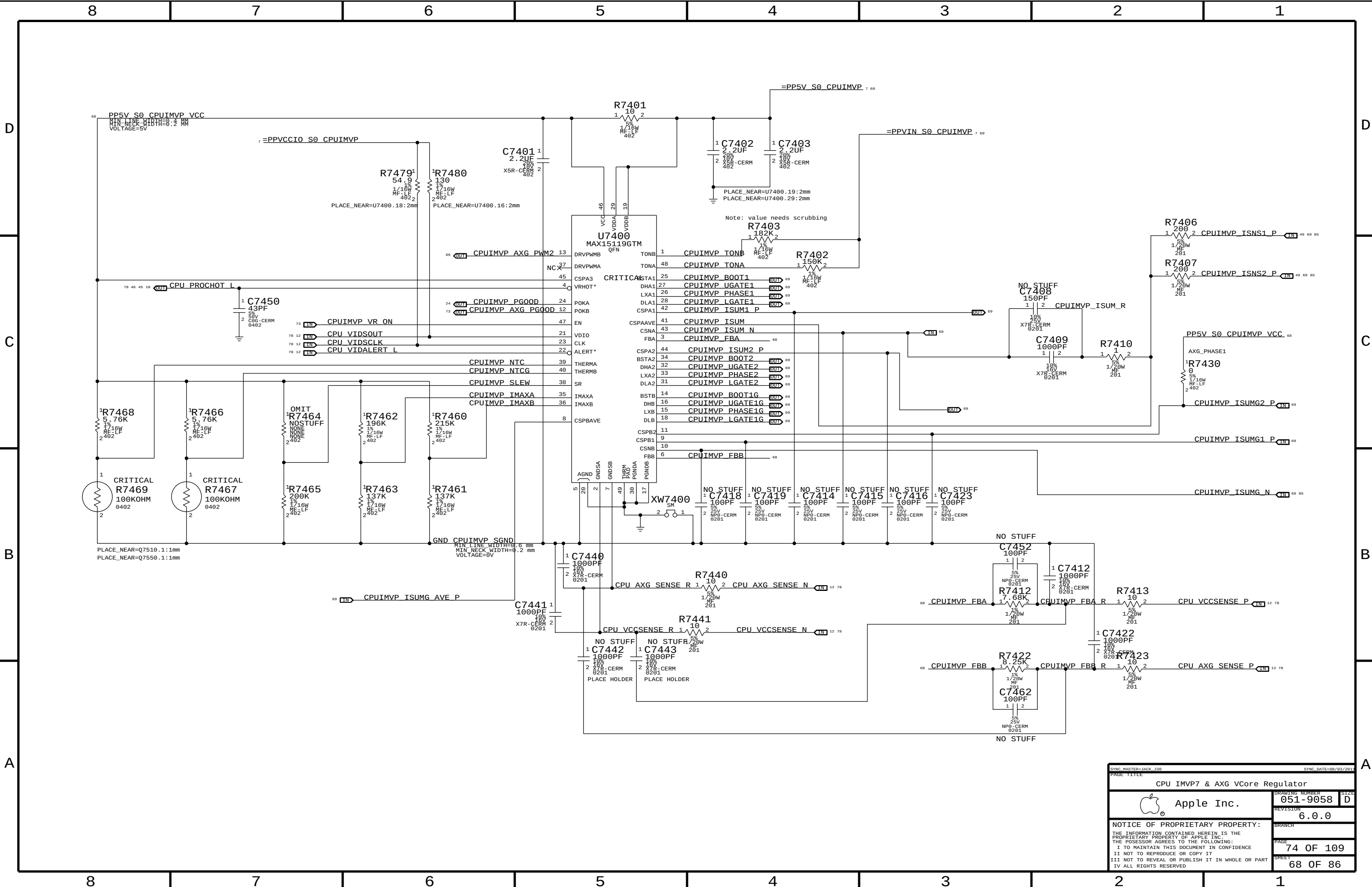
$$V_{OUT} = (2 * RC / RD) + 2$$

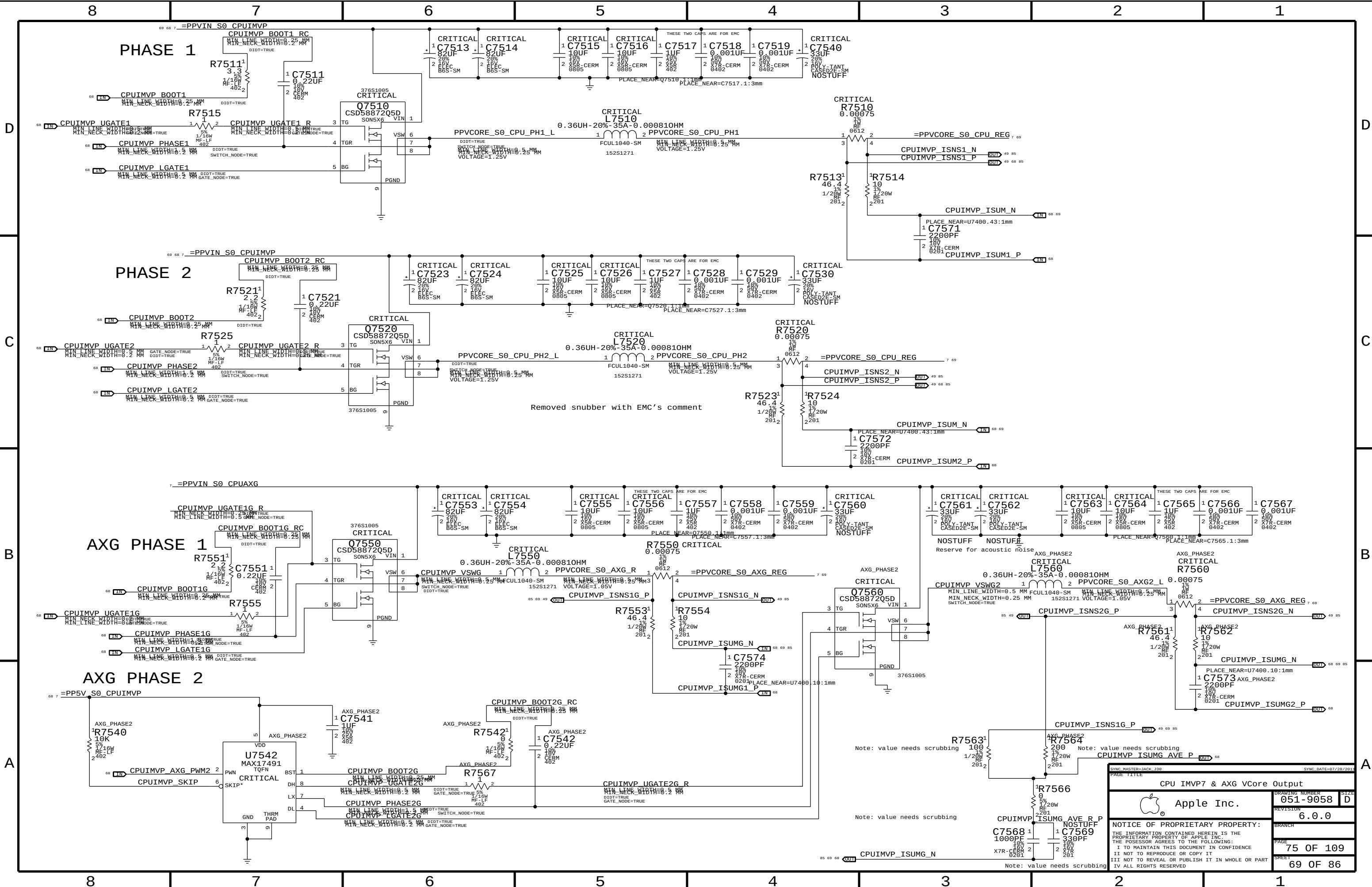


SYMC MASTER=JACK J30		SYMC DATE=08/22/2011	
PAGE TITLE			
5V/3.3V SUPPLY			
 Apple Inc.		DRAWING NUMBER 051-9058	
		SIZE D	
		REVISION 6.0.0	
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SEPERATED MASTER PG00D FOR BOTH 5V AND 3V3







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DRAWING NUMBER
051-9058

REVISION
6.0.0

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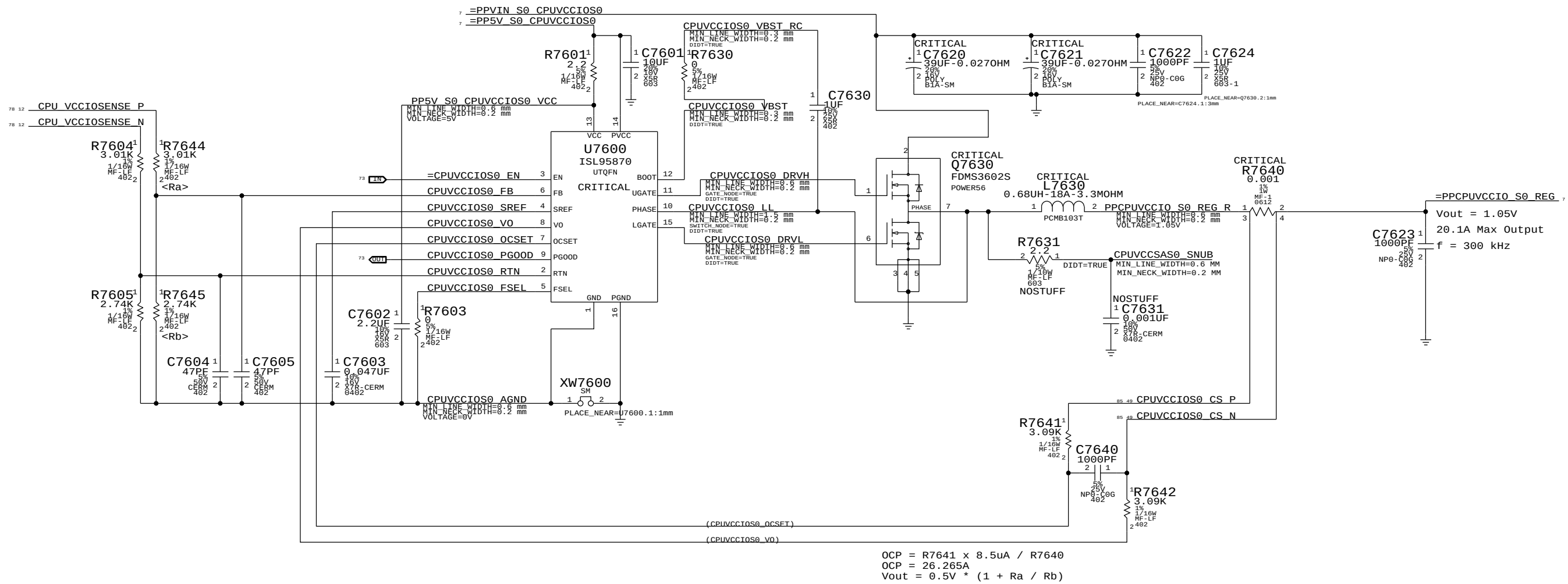
SYNCHMASTER-JACK_339

SYNCHDATE=07/28/2011

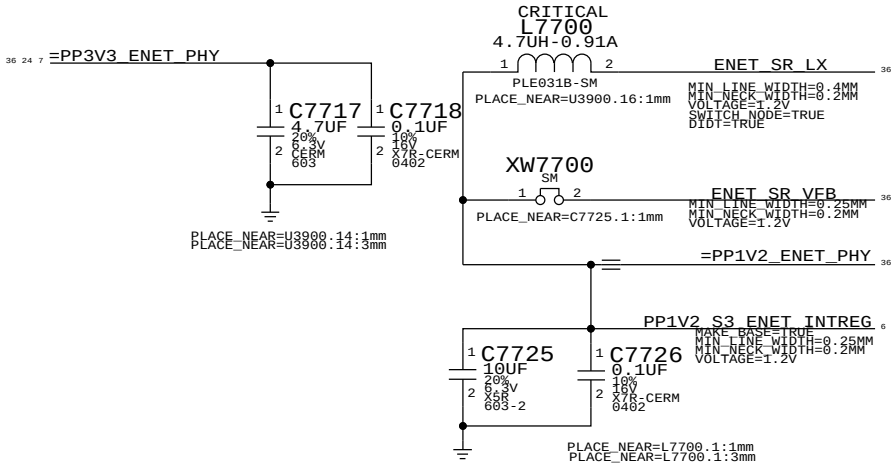
CPU IMVP7 & AXG VCore Output

CPU IMVP7 & AXG VCore Output

CPU VCCIO (1.05V S0) Regulator

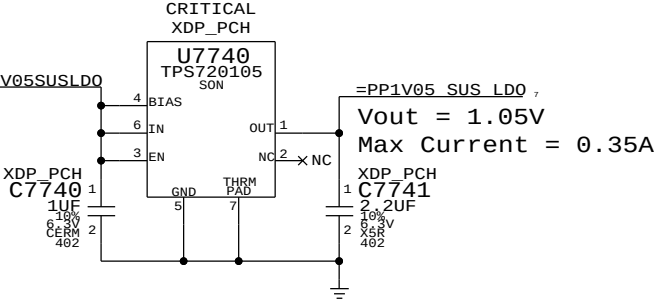


CAESAR IV 1.2V INT.VR CMPTS



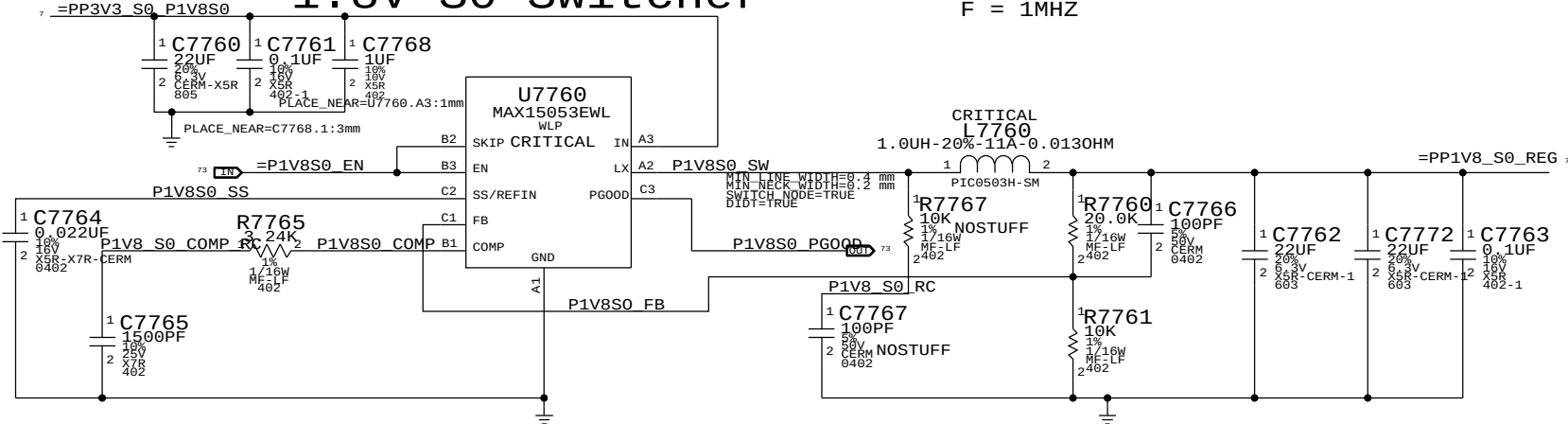
1.05V SUS LDO

Cougar Point requires JTAG pull-ups to be powered at 1.05V in SUS. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.



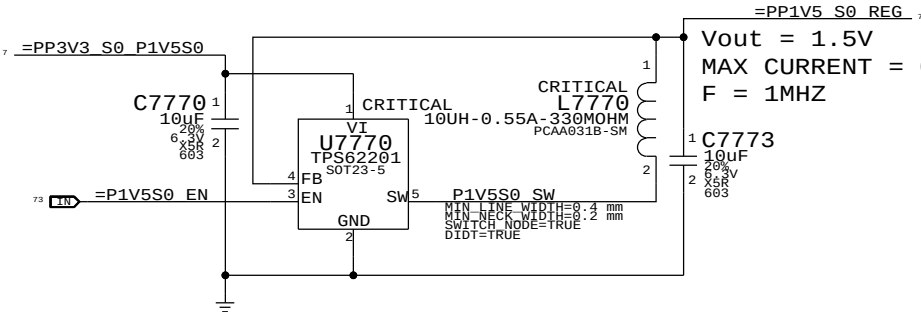
1.8V S0 Switcher

Vout = 1.8V
MAX CURRENT = 2A
F = 1MHZ

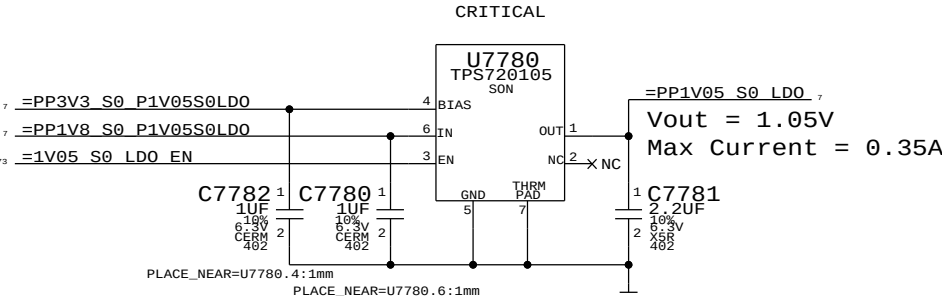


1.5V S0 Switcher

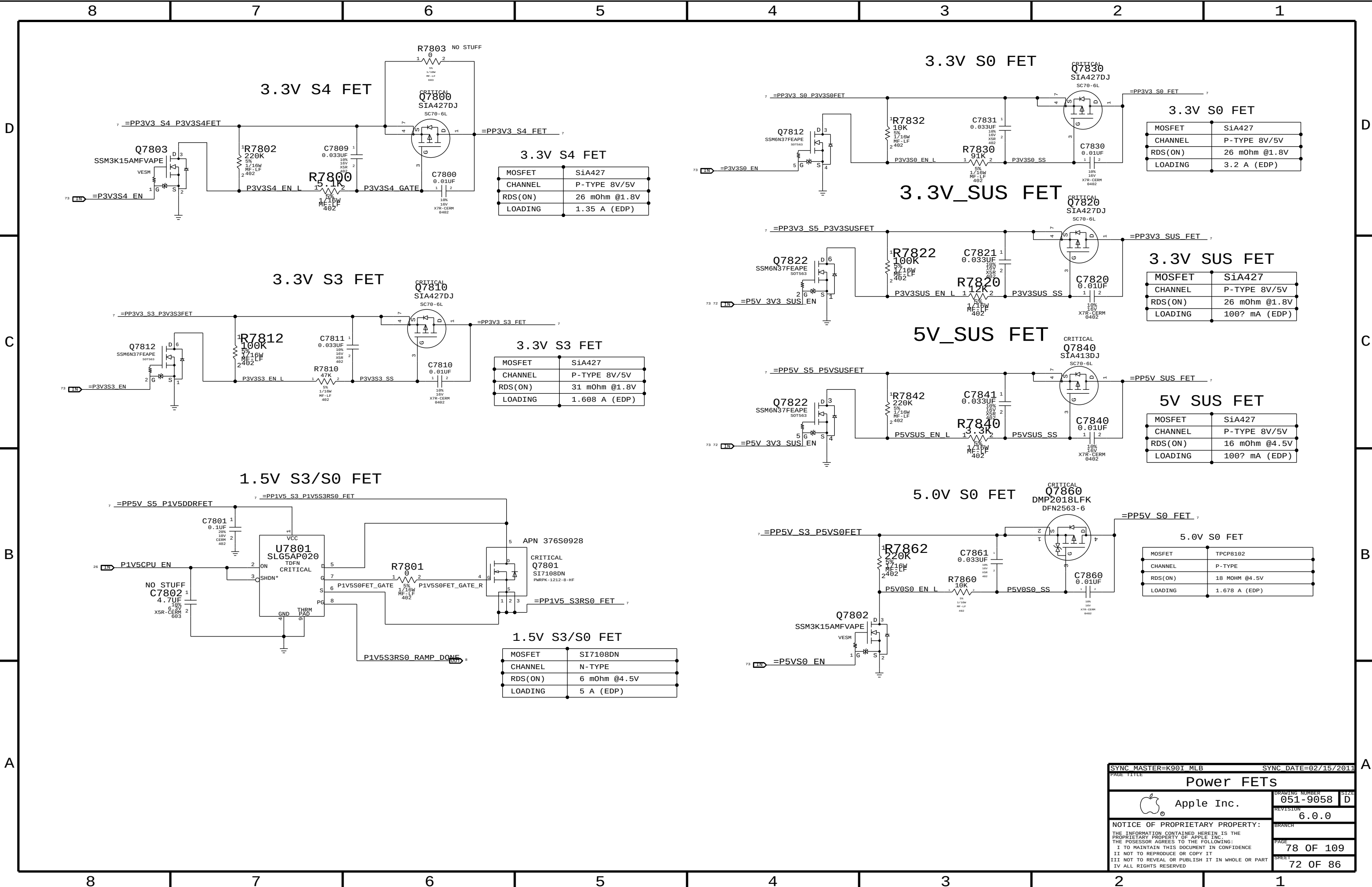
Vout = 1.5V
MAX CURRENT = 0.3A
F = 1MHZ



1.05V S0 LDO



SYNC MASTER=JACK J30		SYNC DATE=07/28/2011	
PAGE TITLE		Misc Power Supplies	
Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	1.35 A (EDP)

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	3.2 A (EDP)

MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	31 mOhm @1.8V
LOADING	1.608 A (EDP)

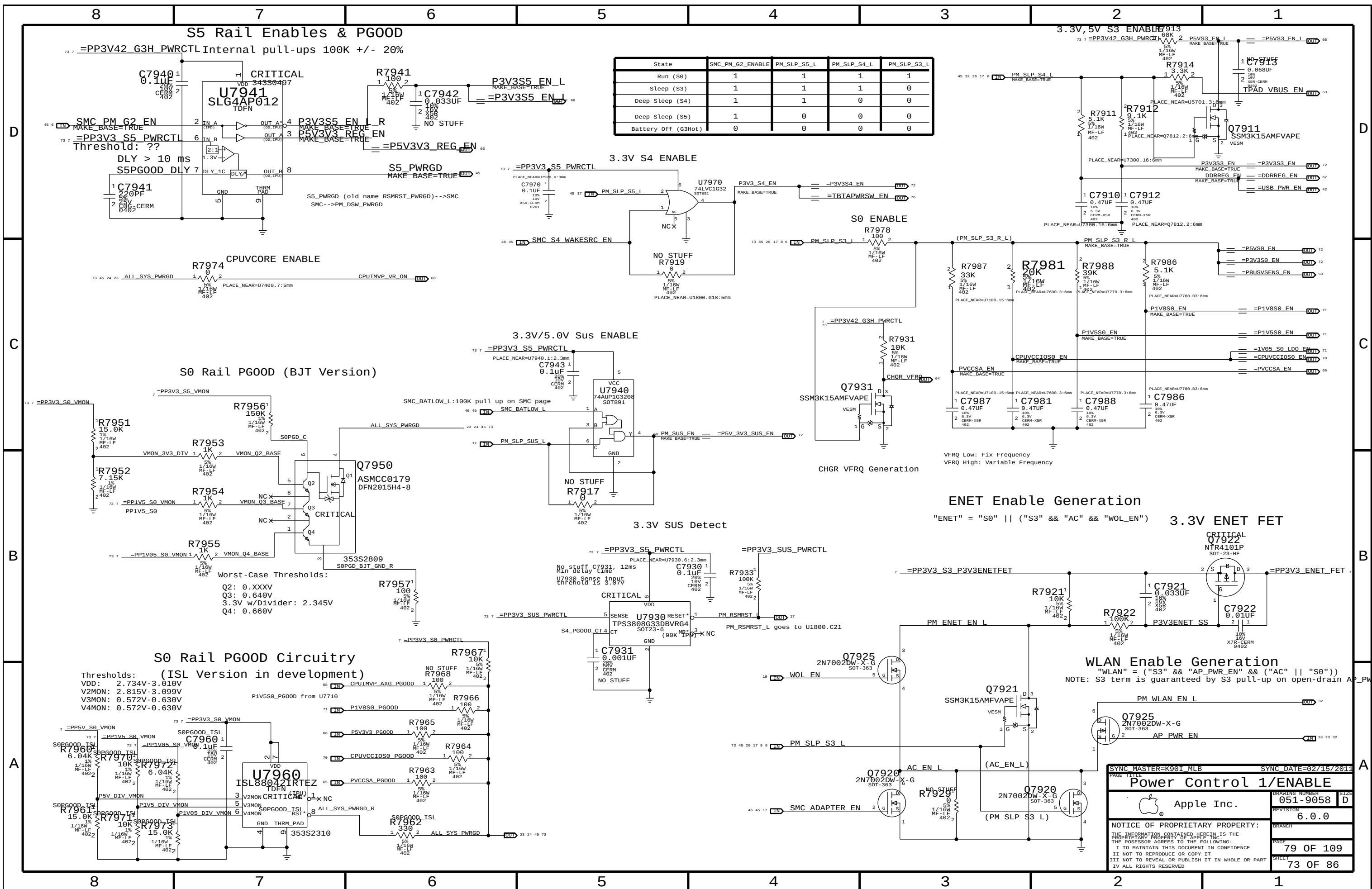
MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	26 mOhm @1.8V
LOADING	100? mA (EDP)

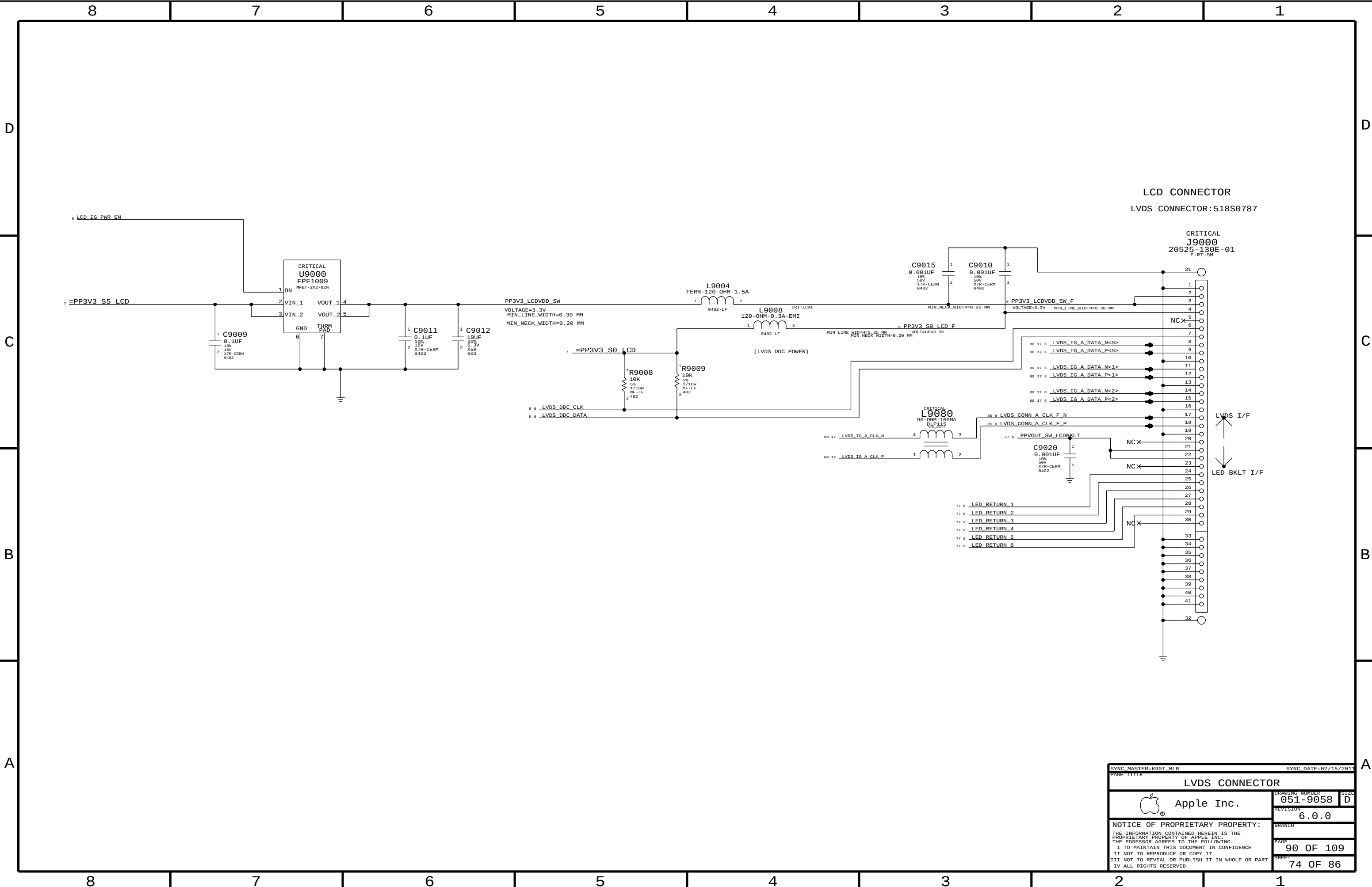
MOSFET	SiA427
CHANNEL	P-TYPE 8V/5V
RDS(ON)	16 mOhm @4.5V
LOADING	100? mA (EDP)

MOSFET	SI7108DN
CHANNEL	N-TYPE
RDS(ON)	6 mOhm @4.5V
LOADING	5 A (EDP)

MOSFET	TPCP8102
CHANNEL	P-TYPE
RDS(ON)	18 MOHM @4.5V
LOADING	1.678 A (EDP)

SYNC MASTER=K90I MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
Power FETs			
 Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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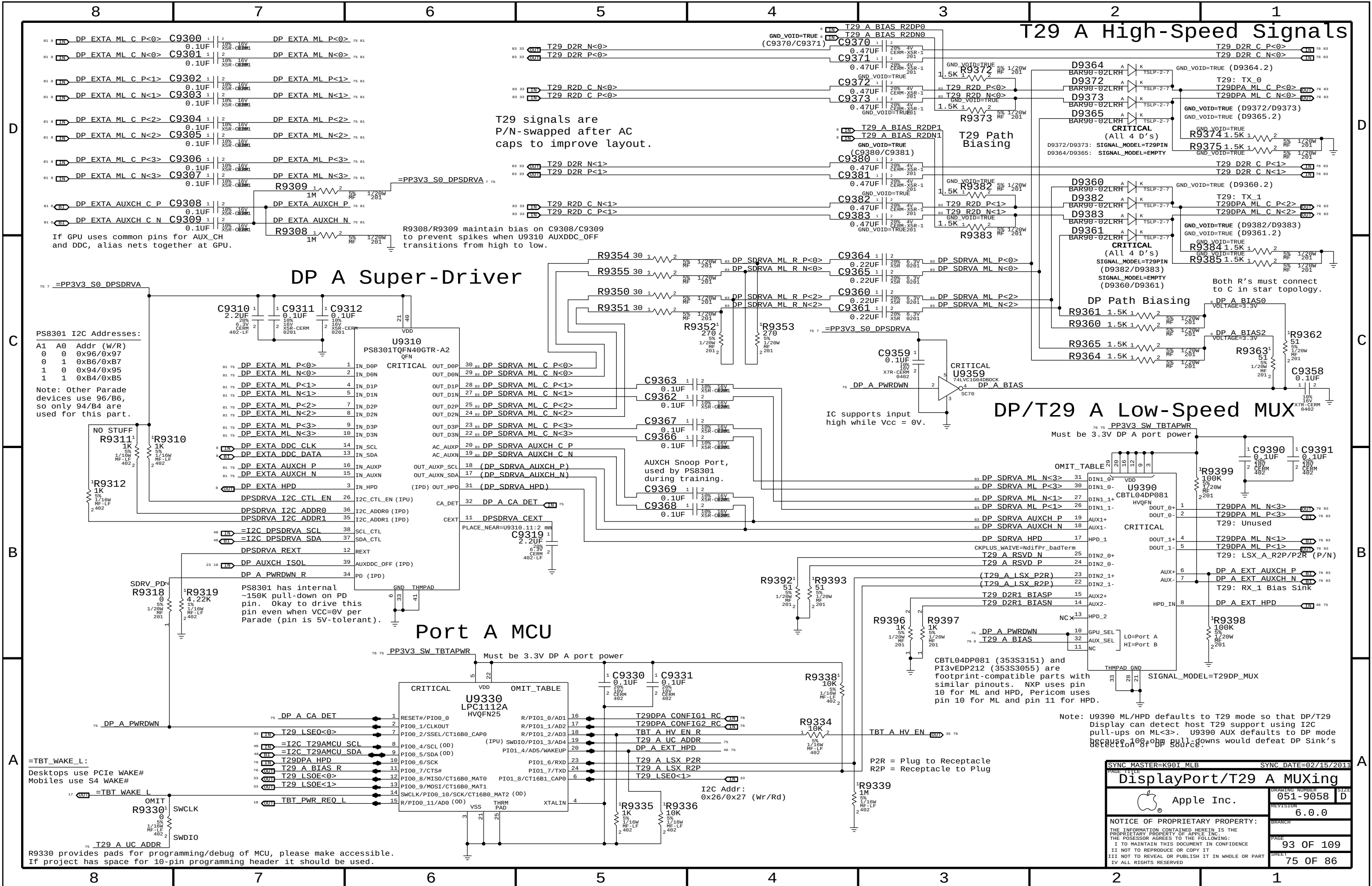


LCD CONNECTOR
LVDS CONNECTOR:518S0787

CRITICAL
J9000
20525-130E-01
F-RT-SM

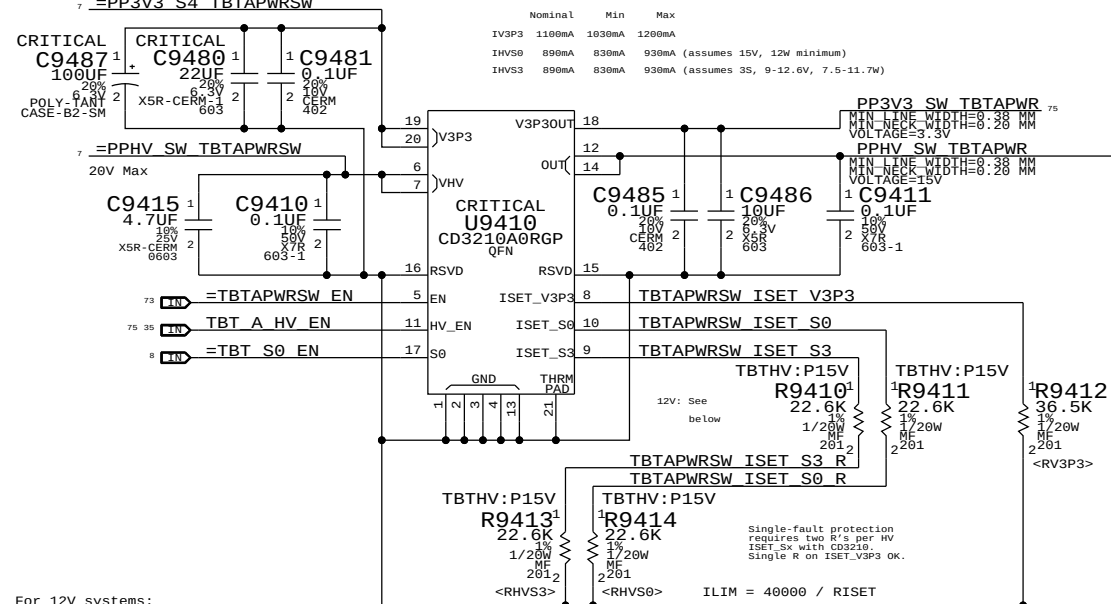
LVDS I/F
LED BKLT I/F

SYNC MASTER=K901 MLB		SYNC DATE=02/15/2011	
PAGE TITLE		LVDS CONNECTOR	
 Apple Inc.		DRAWING NUMBER	051-9058
		REVISION	6.0.0
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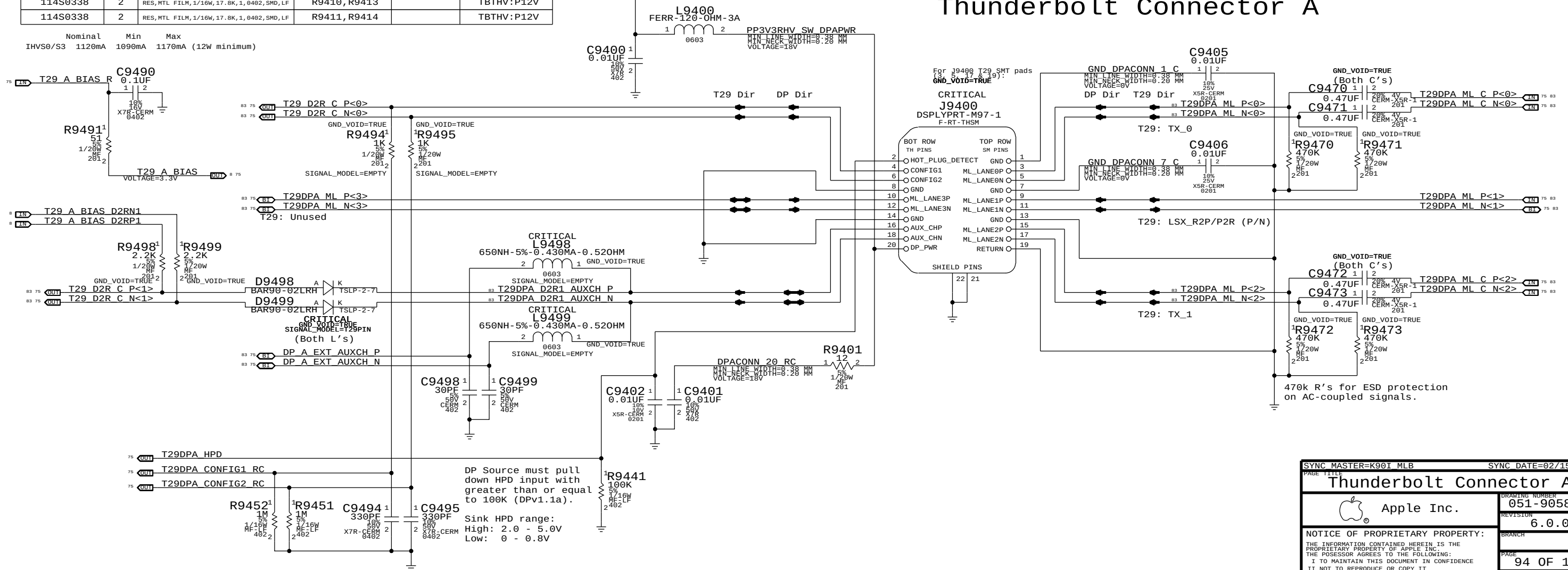


3.3V/HV Power MUX

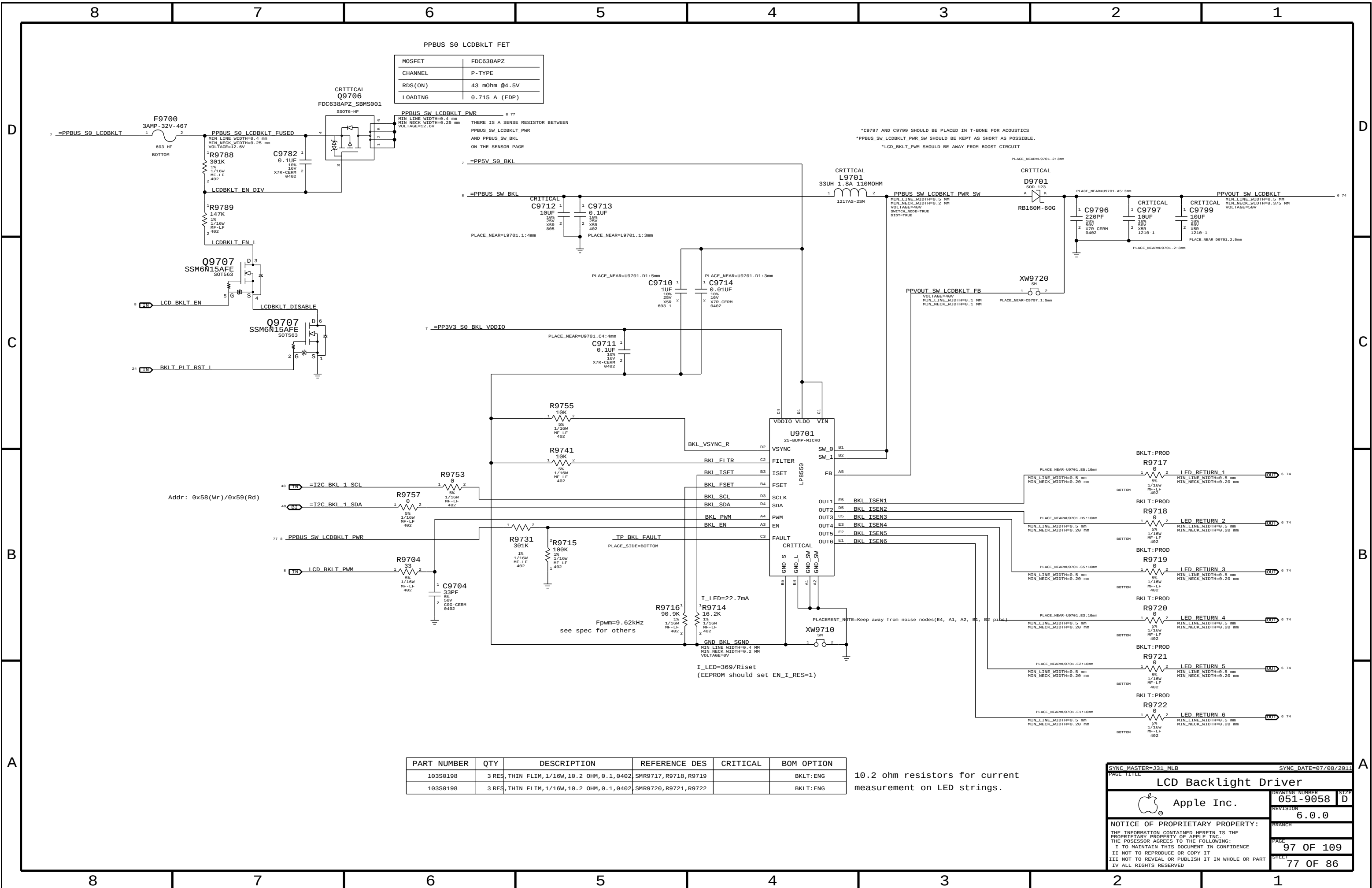
V3P3 must be S4 to support wake from Thunderbolt devices.



Thunderbolt Connector A



SYNC MASTER=K90I MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
Thunderbolt Connector A			
Apple Inc.		DRAWING NUMBER	051-9058
REVISION		6.0.0	
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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3 RES	THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402	SMR9717, R9718, R9719		BKLT:ENG
103S0198	3 RES	THIN FLIM, 1/16W, 10.2 OHM, 0.1, 0402	SMR9720, R9721, R9722		BKLT:ENG

10.2 ohm resistors for current measurement on LED strings.

SYNC MASTER=J31 MLB		SYNC DATE=07/08/2011	
PAGE TITLE			
LCD Backlight Driver			
 Apple Inc.		DRAWING NUMBER	051-9058
		SIZE	D
		REVISION	6.0.0
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Digital Video Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_850	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
LVDS_900	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_PCH	*	=3x_DIELECTRIC	?	DP_PCH	TOP, BOTTOM	=4x_DIELECTRIC	?
DP_PCH_TX	*	=3x_DIELECTRIC	?	DP_PCH_TX	TOP, BOTTOM	=4x_DIELECTRIC	?
LVDS_PCH_TX	*	=3x_DIELECTRIC	?	LVDS_PCH_TX	TOP, BOTTOM	=4x_DIELECTRIC	?

SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_90D	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_PCH_TX	*	=3x_DIELECTRIC	?
SATA_PCH_RX	*	=3x_DIELECTRIC	?
SATA_ICOMP	*	8 MIL	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA3_PCH_TX2TX	*	=4X_DIELECTRIC	?	SATA3_PCH_TX2TX	TOP, BOTTOM	=5X_DIELECTRIC	?
SATA3_PCH_TX2RX	*	=5X_DIELECTRIC	?	SATA3_PCH_TX2RX	TOP, BOTTOM	=6X_DIELECTRIC	?
SATA3_PCH_RX2RX	*	=4x_DIELECTRIC	?	SATA3_PCH_RX2RX	TOP, BOTTOM	=5x_DIELECTRIC	?
SATA3_PCH_RX2TX	*	=5X_DIELECTRIC	?	SATA3_PCH_RX2TX	TOP, BOTTOM	=6X_DIELECTRIC	?
SATA3_PCH_20THER	*	=4x_DIELECTRIC	?	SATA3_PCH_20THER	TOP, BOTTOM	=5x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA3_PCH_TX	*_PCH_TX	*	SATA3_PCH_TX2TX
SATA3_PCH_TX	*_PCH_RX	*	SATA3_PCH_TX2RX
SATA3_PCH_RX	*_PCH_RX	*	SATA3_PCH_RX2RX
SATA3_PCH_RX	*_PCH_TX	*	SATA3_PCH_RX2TX
SATA3_PCH_TX	*	*	SATA3_PCH_20THER
SATA3_PCH_RX	*	*	SATA3_PCH_20THER

SOURCE: 471984_Cheif_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP,BOTTOM	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.8

USB 3.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB_850	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_PCH_TX2TX	*	=4X_DIELECTRIC	?	USB3_PCH_TX2TX	TOP,BOTTOM	=5X_DIELECTRIC	?
USB3_PCH_TX2RX	*	=5X_DIELECTRIC	?	USB3_PCH_TX2RX	TOP,BOTTOM	=6X_DIELECTRIC	?
USB3_PCH_RX2RX	*	=4x_DIELECTRIC	?	USB3_PCH_RX2RX	TOP,BOTTOM	=5x_DIELECTRIC	?
USB3_PCH_RX2TX	*	=5x_DIELECTRIC	?	USB3_PCH_RX2TX	TOP,BOTTOM	=6x_DIELECTRIC	?
USB3_PCH_20THER	*	=4x_DIELECTRIC	?	USB3_PCH_20THER	TOP,BOTTOM	=5x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_PCH_TX	*_PCH_TX	*	USB3_PCH_TX2TX
USB3_PCH_TX	*_PCH_RX	*	USB3_PCH_TX2RX
USB3_PCH_RX	*_PCH_RX	*	USB3_PCH_RX2RX
USB3_PCH_RX	*_PCH_TX	*	USB3_PCH_RX2TX
USB3_PCH_TX	*	*	USB3_PCH_20THER
USB3_PCH_RX	*	*	USB3_PCH_20THER

SOURCE: 471984_Cheif_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	LVDS_I6_A_CLK	LVDS_960	LVDS_PCH_TX	LVDS_I6_A_CLK_P	17 74
	LVDS_I6_A_CLK	LVDS_960	LVDS_PCH_TX	LVDS_I6_A_CLK_N	17 74
	LVDS_I6_A_DATA	LVDS_960	LVDS_PCH_TX	LVDS_I6_A_DATA_P<2..0>	6 17 74
	LVDS_I6_A_DATA	LVDS_960	LVDS_PCH_TX	LVDS_I6_A_DATA_N<2..0>	6 17 74
		LVDS_960	LVDS_PCH_TX	LVDS_I6_A_DATA_P<3>	8 17
		LVDS_960	LVDS_PCH_TX	LVDS_I6_A_DATA_N<3>	8 17
		LVDS_960	LVDS_PCH_TX	LVDS_I6_B_DATA_P<3..0>	8 17
		LVDS_960	LVDS_PCH_TX	LVDS_I6_B_DATA_N<3..0>	8 17
		LVDS_960	LVDS_PCH_TX	LVDS_I6_B_CLK_P	8 17
		LVDS_960	LVDS_PCH_TX	LVDS_I6_B_CLK_N	8 17
	SATA_HDD_R2D	SATA_960	SATA3_PCH_TX	SATA_HDD_R2D_C_P	16 41
	SATA_HDD_R2D	SATA_960	SATA3_PCH_TX	SATA_HDD_R2D_C_N	16 41
	SATA_HDD_R2D_CONN	SATA_960	SATA3_PCH_TX	SATA_HDD_R2D_P	6 41
	SATA_HDD_R2D	SATA_960	SATA3_PCH_RX	SATA_HDD_D2R_N	16 41
	SATA_HDD_D2R	SATA_960	SATA3_PCH_RX	SATA_HDD_D2R_N	16 41
	SATA_HDD_D2R_CONN	SATA_960	SATA3_PCH_RX	SATA_HDD_D2R_C_P	6 41
	SATA_HDD_D2R_CONN	SATA_960	SATA3_PCH_RX	SATA_HDD_D2R_C_N	6 41
	SATA_ODD_R2D	SATA_960	SATA_PCH_TX	SATA_ODD_R2D_C_P	16 41
	SATA_ODD_R2D	SATA_960	SATA_PCH_TX	SATA_ODD_R2D_C_N	16 41
	SATA_ODD_R2D	SATA_960	SATA_PCH_TX	SATA_ODD_R2D_P	6 41
	SATA_ODD_D2R	SATA_960	SATA_PCH_RX	SATA_ODD_D2R_P	16 41
	SATA_ODD_D2R	SATA_960	SATA_PCH_RX	SATA_ODD_D2R_N	16 41
	SATA_HDD_R2D_CONN	SATA_960	SATA3_PCH_TX	SATA_HDD_R2D_RC_P	41
	SATA_HDD_R2D_CONN	SATA_960	SATA3_PCH_TX	SATA_HDD_R2D_RC_N	41
	SATA_HDD_D2R_CONN	SATA_960	SATA3_PCH_RX	SATA_HDD_D2R_RC_P	41
	SATA_HDD_D2R_CONN	SATA_960	SATA3_PCH_RX	SATA_HDD_D2R_RC_N	41
	PCH_SATA_ICOMP		SATA_ICOMP	PCH_SATAICOMP	16
	USB_HUB1_UP	USB_850	USB	USB_HUB_UP_P	18 25
		USB_850	USB	USB_HUB_UP_N	18 25
	USB_EXT_A	USB_850	USB	USB_EXT_A_P	18 42
	USB_EXT_A	USB_850	USB	USB_EXT_A_N	18 42
	USB_EXT_B	USB_850	USB	USB_EXTB_MUX_P	25 43
		USB_850	USB	USB_EXTB_MUX_N	25 43
		USB_850	USB	USB_EXTB_MUXED_F_P	42
		USB_850	USB	USB_EXTB_MUXED_F_N	42
		USB_850	USB	USB_EXTB_F_P	43
		USB_850	USB	USB_EXTB_F_N	43
		USB_850	USB	USB_EXTB_MIXED_P	42
		USB_850	USB	USB_EXTB_MIXED_N	42
		USB_850	USB	USB_EXTD_XHCI_P	8 18
		USB_850	USB	USB_EXTD_XHCI_N	8 18
		USB_850	USB	USB_EXTB_EHCI_P	18 25
		USB_850	USB	USB_EXTB_EHCI_N	18 25
		USB_850	USB	USB_EXTB_XHCI_P	18 25
		USB_850	USB	USB_EXTB_XHCI_N	18 25
	USB3_EXT_RX	USB_850	USB3_PCH_RX	USB3_EXTB_RX_P	18 42
		USB_850	USB3_PCH_RX	USB3_EXTB_RX_N	18 42
	USB3_EXT_TX	USB_850	USB3_PCH_TX	USB3_EXTB_TX_P	18 42
		USB_850	USB3_PCH_TX	USB3_EXTB_TX_N	18 42
	USB3_EXT_RX	USB_850	USB3_PCH_RX	USB3_EXTB_RX_P	18 43
		USB_850	USB3_PCH_RX	USB3_EXTB_RX_N	18 43
	USB3_EXT_TX	USB_850	USB3_PCH_TX	USB3_EXTB_TX_P	18 43
		USB_850	USB3_PCH_TX	USB3_EXTB_TX_N	18 43
		USB_850	USB3_PCH_RX	USB3_EXTB_RX_F_P	42
		USB_850	USB3_PCH_RX	USB3_EXTB_RX_F_N	42
		USB_850	USB3_PCH_TX	USB3_EXTB_TX_F_P	42
		USB_850	USB3_PCH_TX	USB3_EXTB_TX_F_N	42
		USB_850	USB3_PCH_RX	USB3_EXTB_RX_C_P	43
		USB_850	USB3_PCH_TX	USB3_EXTB_TX_C_N	42
		USB_850	USB3_PCH_TX	USB3_EXTB_TX_C_P	43
		USB_850	USB3_PCH_TX	USB3_EXTB_TX_C_N	43
	USB_EXT_A	USB_850	USB	USB_SMC_P	8 45
		USB_850	USB	USB_SMC_N	8 45
	USB_EXT_C	USB_850	USB	USB_EXTC_P	8 18
		USB_850	USB	USB_EXTC_N	8 18
	USB_CAMERA	USB_850	USB	USB_CAMERA_P	18 32
		USB_850	USB	USB_CAMERA_N	18 32
	USB_CAMERA	USB_850	USB	USB_CAMERA_CONN_P	6 32
		USB_850	USB	USB_CAMERA_CONN_N	6 32
	USB_BT	USB_850	USB	USB_BT_P	8 32
	USB_BT	USB_850	USB	USB_BT_N	8 32
	USB_BT	USB_850	USB	USB_BT_CONN_P	6 32
	USB_BT	USB_850	USB	USB_BT_CONN_N	6 32
	USB_TPAD	USB_850	USB	USB_TPAD_P	8 53
		USB_850	USB	USB_TPAD_N	8 53
	USB_IR	USB_850	USB	USB_IR_P	8 44
		USB_850	USB	USB_IR_N	8 44
	PCH_USB_RBIAS	PCH_USB_RBIAS		PCH_USB_RBIAS	18
	PCH_D1EECLK_UNUSED	CLK_PCTE_960	CLK_PCTE	PCIE_CLK106M_PCH_P	16
	PCH_D1EECLK_UNUSED	CLK_PCTE_960	CLK_PCTE	PCIE_CLK106M_PCH_N	16
	PCH_D1EECLK_UNUSED	CLK_PCTE_960	CLK_PCTE	PCH_CLK96M_DOT_P	16
	PCH_D1EECLK_UNUSED	CLK_PCTE_960	CLK_PCTE	PCH_CLK96M_DOT_N	16
	PCH_D1EECLK_UNUSED	CLK_PCTE_960	CLK_PCTE	PCH_CLK106M_SATA_P	16
	PCH_D1EECLK_UNUSED	CLK_PCTE_960	CLK_PCTE	PCH_CLK106M_SATA_N	16
	CPU_50S	CPU_50S	CLK_PCTE	PCH_CLK143M_REFCLK	16
	LPC_CLK33M	CPU_50S	CLK_PCTE	PCH_CLK33M_PCTIN	16 24

SYNC MASTER-K90I MLB		SYNC DATE=02/15/2011	
PAGE TITLE			
PCH Constraints 1			
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DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

T29 I2C Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
T29_I2C_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
T29_I2C	*	=2x_DIELECTRIC	?

T29 SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
T29_SPI_55S	*	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
T29_SPI	*	=2x_DIELECTRIC	?

T29/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
T29DP_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
T29DP_100D	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
T29DP	*	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
T29DP	TOP,BOTTOM	=7x_DIELECTRIC	?

SOURCE: Bill Cornelius’s T29 Routing Notes

T29 IC Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
DP_T29SNK0_MI	DP_85D	DP_PCH_TX	DP T29SNK0 ML C P<3..0>
DP_T29SNK0_MI	DP_85D	DP_PCH_TX	DP T29SNK0 ML C N<3..0>
DP_T29SNK0_MI	DP_85D	DP_PCH_TX	DP T29SNK0 ML P<3..0>
DP_T29SNK0_MI	DP_85D	DP_PCH_TX	DP T29SNK0 ML N<3..0>
DP_T29SNK0_AUXCH	DP_85D	DP_PCH	DP T29SNK0 AUXCH C P
DP_T29SNK0_AUXCH	DP_85D	DP_PCH	DP T29SNK0 AUXCH C N
DP_T29SNK0_AUXCH	DP_85D	DP_PCH	DP T29SNK0 AUXCH P
DP_T29SNK0_AUXCH	DP_85D	DP_PCH	DP T29SNK0 AUXCH N
DP_T29SNK1_MI	DP_85D	DP_PCH_TX	DP T29SNK1 ML C P<3..0>
DP_T29SNK1_MI	DP_85D	DP_PCH_TX	DP T29SNK1 ML C N<3..0>
DP_T29SNK1_MI	DP_85D	DP_PCH_TX	DP T29SNK1 ML P<3..0>
DP_T29SNK1_MI	DP_85D	DP_PCH_TX	DP T29SNK1 ML N<3..0>
DP_T29SNK1_AUXCH	DP_85D	DP_PCH	DP T29SNK1 AUXCH C P
DP_T29SNK1_AUXCH	DP_85D	DP_PCH	DP T29SNK1 AUXCH C N
DP_T29SNK1_AUXCH	DP_85D	DP_PCH	DP T29SNK1 AUXCH P
DP_T29SNK1_AUXCH	DP_85D	DP_PCH	DP T29SNK1 AUXCH N
DP_T29SRC	DP_85D	DISPLAYPORT	DP T29SRC ML C P<3..0>
DP_T29SRC	DP_85D	DISPLAYPORT	DP T29SRC ML C N<3..0>
DP_T29SRC	DP_85D	DISPLAYPORT	DP T29SRC AUXCH C P
DP_T29SRC	DP_85D	DISPLAYPORT	DP T29SRC AUXCH C N
T29_I2C_55S	T29_I2C_55S	T29_I2C	I2C T29_SCL
T29_I2C_55S	T29_I2C_55S	T29_I2C	I2C T29_SDA
T29_SPI_CLK	T29_SPI_55S	T29_SPI	T29_SPI_CLK
T29_SPI_M0SI	T29_SPI_55S	T29_SPI	T29_SPI_M0SI
T29_SPI_MISO	T29_SPI_55S	T29_SPI	T29_SPI_MISO
T29_SPI_CS_L	T29_SPI_55S	T29_SPI	T29_SPI_CS_L
T29DP_80D	T29DP_80D	T29DP	T29 R2D C P<3..0>
T29DP_80D	T29DP_80D	T29DP	T29 R2D C N<3..0>
T29DP_100D	T29DP_100D	T29DP	T29 D2R P<3..0>
T29DP_100D	T29DP_100D	T29DP	T29 D2R N<3..0>

Only used on hosts supporting T29 video-in

T29/DP Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
T29_R2D0	T29DP_80D	T29DP	T29 R2D P<0>
T29_R2D0	T29DP_80D	T29DP	T29 R2D N<0>
T29_R2D1	T29DP_80D	T29DP	T29 R2D P<1>
T29_R2D1	T29DP_80D	T29DP	T29 R2D N<1>
T29_R2D1	T29DP_80D	T29DP	T29 R2D C F P<1..0>
T29_R2D1	T29DP_80D	T29DP	T29 R2D C F N<1..0>
T29_D2R0	T29DP_100D	T29DP	T29 D2R C P<0>
T29_D2R0	T29DP_100D	T29DP	T29 D2R C N<0>
T29_D2R1	T29DP_100D	T29DP	T29 D2R C P<1>
T29_D2R1	T29DP_100D	T29DP	T29 D2R C N<1>
T29_D2R1	T29DP_100D	T29DP	T29DPA D2R1_AUXCH_P
T29_D2R1	T29DP_100D	T29DP	T29DPA D2R1_AUXCH_N
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_ML_C_P<3..0>
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_ML_C_N<3..0>
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_ML_R_P<3..0>
T29DP_80D	T29DP_80D	T29DP	DP_SDRVA_ML_R_N<3..0>
DP_SDRVA_ML_EVEN	T29DP_80D	T29DP	DP_SDRVA_ML_P<2..0:2>
DP_SDRVA_ML_EVEN	T29DP_80D	T29DP	DP_SDRVA_ML_N<2..0:2>
DP_SDRVA_ML_ODD	T29DP_80D	T29DP	DP_SDRVA_ML_P<3..1:2>
DP_SDRVA_ML_ODD	T29DP_80D	T29DP	DP_SDRVA_ML_N<3..1:2>
DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP_SDRVA_AUXCH_P
DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP_SDRVA_AUXCH_N
DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP_SDRVA_AUXCH_C_P
DP_SDRVA_AUXCH	T29DP_80D	T29DP	DP_SDRVA_AUXCH_C_N
T29DP_80D	T29DP_80D	T29DP	T29DPA_ML_P<3..0>
T29DP_80D	T29DP_80D	T29DP	T29DPA_ML_N<3..0>
T29DP_80D	T29DP_80D	T29DP	T29DPA_ML_C_P<3..0>
T29DP_80D	T29DP_80D	T29DP	T29DPA_ML_C_N<3..0>
T29DP_80D	T29DP_80D	T29DP	DP_A_EXT_AUXCH_P
T29DP_80D	T29DP_80D	T29DP	DP_A_EXT_AUXCH_N
T29_R2D2	T29DP_80D	T29DP	T29 R2D P<2>
T29_R2D2	T29DP_80D	T29DP	T29 R2D N<2>
T29_R2D3	T29DP_80D	T29DP	T29 R2D P<3>
T29_R2D3	T29DP_80D	T29DP	T29 R2D N<3>
T29_R2D3	T29DP_80D	T29DP	T29 R2D C F P<3..2>
T29_R2D3	T29DP_80D	T29DP	T29 R2D C F N<3..2>
T29_D2R2	T29DP_100D	T29DP	T29 D2R C P<2>
T29_D2R2	T29DP_100D	T29DP	T29 D2R C N<2>
T29_D2R3	T29DP_100D	T29DP	T29 D2R C P<3>
T29_D2R3	T29DP_100D	T29DP	T29 D2R C N<3>
T29_D2R3	T29DP_100D	T29DP	T29DPB_D2R3_AUXCH_P
T29_D2R3	T29DP_100D	T29DP	T29DPB_D2R3_AUXCH_N
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_ML_C_P<3..0>
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_ML_C_N<3..0>
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_ML_R_P<3..0>
T29DP_80D	T29DP_80D	T29DP	DP_SDRVB_ML_R_N<3..0>
DP_SDRVB_ML_EVEN	T29DP_80D	T29DP	DP_SDRVB_ML_P<2..0:2>
DP_SDRVB_ML_EVEN	T29DP_80D	T29DP	DP_SDRVB_ML_N<2..0:2>
DP_SDRVB_ML_ODD	T29DP_80D	T29DP	DP_SDRVB_ML_P<3..1:2>
DP_SDRVB_ML_ODD	T29DP_80D	T29DP	DP_SDRVB_ML_N<3..1:2>
DP_SDRVB_AUXCH	T29DP_80D	T29DP	DP_SDRVB_AUXCH_P
DP_SDRVB_AUXCH	T29DP_80D	T29DP	DP_SDRVB_AUXCH_N
DP_SDRVB_AUXCH	T29DP_80D	T29DP	DP_SDRVB_AUXCH_C_P
DP_SDRVB_AUXCH	T29DP_80D	T29DP	DP_SDRVB_AUXCH_C_N
T29DP_80D	T29DP_80D	T29DP	T29DPB_ML_P<3..0>
T29DP_80D	T29DP_80D	T29DP	T29DPB_ML_N<3..0>
T29DP_80D	T29DP_80D	T29DP	T29DPB_ML_C_P<3..0>
T29DP_80D	T29DP_80D	T29DP	T29DPB_ML_C_N<3..0>
T29DP_80D	T29DP_80D	T29DP	DP_B_EXT_AUXCH_P
T29DP_80D	T29DP_80D	T29DP	DP_B_EXT_AUXCH_N

Only used on dual-port hosts.

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T29 Constraints

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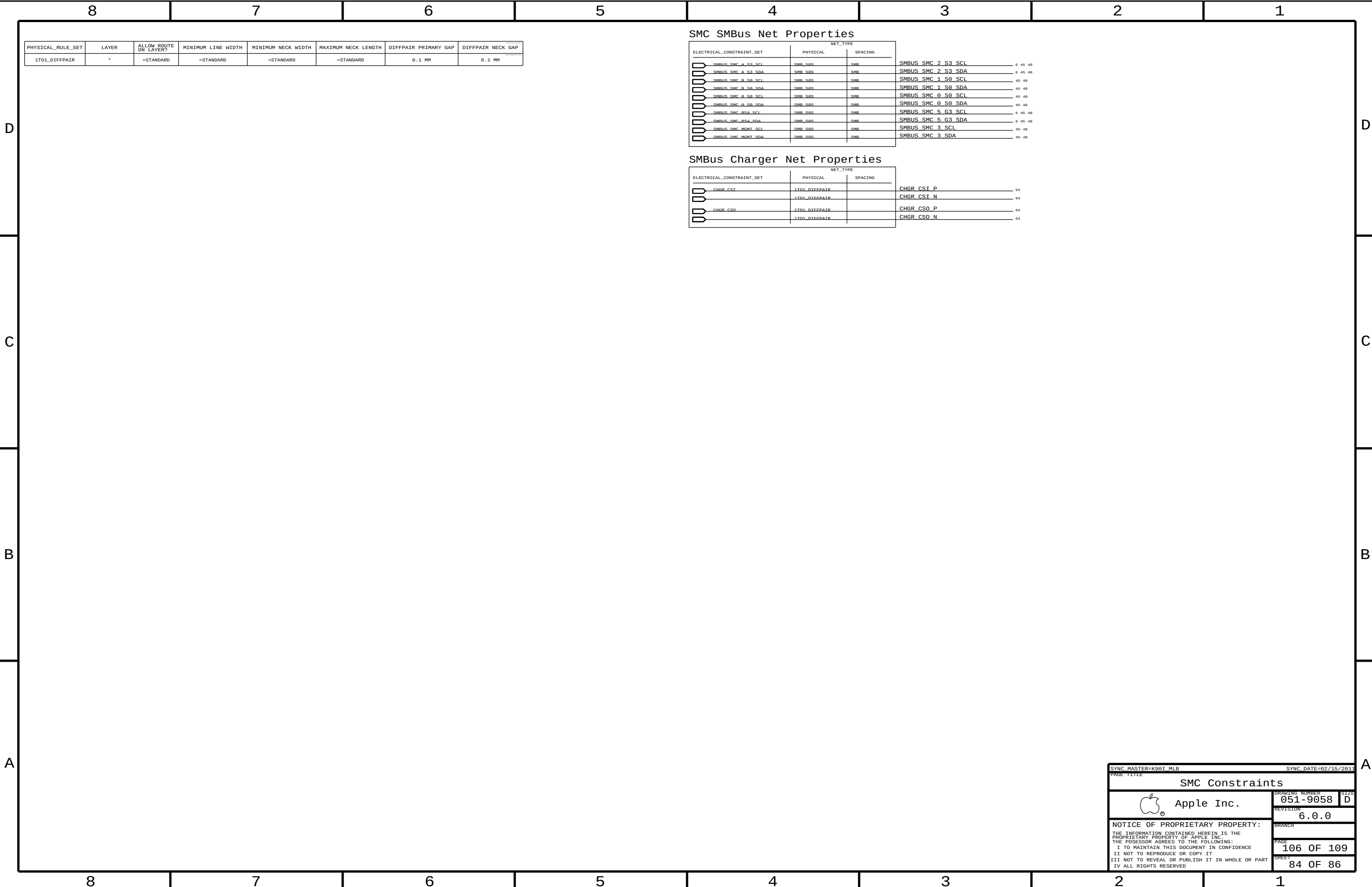
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PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_I101_555	"	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
THERM_I101_555	"	=1:1_DIFFPAIR	=55_OHM_SE	=55_OHM_SE	=55_OHM_SE	=1:1_DIFFPAIR	=1:1_DIFFPAIR
DIFFPAIR	"	=1:1_DIFFPAIR			=1:1_DIFFPAIR	=1:1_DIFFPAIR	=1:1_DIFFPAIR

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2:1_SPACING	?
THERM	*	=2:1_SPACING	?
AUDIO	*	=2:1_SPACING	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
ENETCONN	*	25 MILS	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	GND	*	GND_P2MH
MEM_CMD	GND	*	GND_P2MH
MEM_CTRL	GND	*	GND_P2MH
MEM_DATA	GND	*	GND_P2MH
MEM_DQS	GND	*	GND_P2MH

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	GND	*	GND_P2MM
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
ENET_MDI	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	CLK_PCTIE	*	GND_P2961
GND	PCIE*	*	GND_P2961
GND	SATA*	*	GND_P2961
GND	USB*	*	GND_P2961
SB_POWER	CLK_PCTIE	*	PwR_P2961
SB_POWER	SATA*	*	PwR_P2961
SB_POWER	SATA*	*	PwR_P2961

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	LVDS*	*	GND_F2MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_4QS OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_37S OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	*	OVERRIDE	OVERRIDE	0.076 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.1 MM OVERRIDE	500 MIL OVERRIDE	OVERRIDE	OVERRIDE
CPU_27P4S OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE
CLK_PCIE_90D OVERRIDE	TOP	OVERRIDE	OVERRIDE	0.09 MM OVERRIDE	400 MIL OVERRIDE	OVERRIDE	OVERRIDE

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEP_72D	BOTTOM			0.127 MM	6.35 MM		
MEP_85D	TOP			0.1 MM	6.35 MM		

J30 Specific Net Properties

[illegible]

J30 Specific Net Properties

[illegible]

Memory Constraint Relaxations

Allow 0.127 mm necks for >0.127 mm lines for ARD fanout.

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Project Specific Constraints			
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K90i Board-Specific Spacing & Physical Constraints

BOARD LAYERS					BOARD AREAS			BOARD UNITS (MIL OR MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM					NO_TYPE, BGA			MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	*	Y	=50_OHR_SE	=50_OHR_SE	10 MM	0 MM	0 MM
STANDARD	*	Y	=DEFAULT	=DEFAULT	10 MM	=DEFAULT	=DEFAULT

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHR_SE	TOP, BOTTOM	Y	0.110 MM	0.090 MM			
50_OHR_SE	*	Y	0.080 MM	0.080 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHR_SE	TOP, BOTTOM	Y	0.165 MM	0.165 MM			
40_OHR_SE	ISL10	N	0.126 MM	0.126 MM	=STANDARD	=STANDARD	=STANDARD
40_OHR_SE	ISL3, ISL4, ISL9	Y	0.126 MM	0.126 MM	=STANDARD	=STANDARD	=STANDARD
40_OHR_SE	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
37_OHR_SE	TOP, BOTTOM	Y	0.190 MM	0.1 MM			
37_OHR_SE	ISL10	N	0.145 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD
37_OHR_SE	ISL3, ISL4, ISL9	Y	0.145 MM	0.1 MM	=STANDARD	=STANDARD	=STANDARD
37_OHR_SE	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHR_SE	TOP, BOTTOM	Y	0.310 MM	0.2 MM			
27P4_OHR_SE	*	Y	0.235 MM	0.2 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHR_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM			
55_OHR_SE	*	Y	0.070 MM	0.070 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
72_OHR_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
72_OHR_DIFF	ISL3, ISL4, ISL9	Y	0.140 MM	0.140 MM		0.190 MM	0.190 MM
72_OHR_DIFF	ISL10	N	0.140MM	0.140 MM		0.190 MM	0.190 MM
72_OHR_DIFF	TOP, BOTTOM	Y	0.175 MM	0.175 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
85_OHR_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
85_OHR_DIFF	ISL3, ISL4	Y	0.101 MM	0.1 MM		0.170 MM	0.170 MM
85_OHR_DIFF	ISL9, ISL10	Y	0.101 MM	0.1 MM		0.170 MM	0.170 MM
85_OHR_DIFF	TOP, BOTTOM	Y	0.125 MM	0.1 MM		0.190 MM	0.190 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHR_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
90_OHR_DIFF	ISL3, ISL4	Y	0.091 MM	0.091 MM		0.180 MM	0.180 MM
90_OHR_DIFF	ISL9, ISL10	Y	0.091 MM	0.091 MM		0.180 MM	0.180 MM
90_OHR_DIFF	TOP, BOTTOM	Y	0.111 MM	0.111 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_OHR_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
100_OHR_DIFF	ISL3, ISL4	Y	0.076 MM	0.076 MM		0.250 MM	0.250 MM
100_OHR_DIFF	ISL9, ISL10	Y	0.076 MM	0.076 MM		0.250 MM	0.250 MM
100_OHR_DIFF	TOP, BOTTOM	Y	0.085 MM	0.085 MM		0.200 MM	0.200 MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
110_OHR_DIFF	*	N	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
110_OHR_DIFF	ISL3, ISL4	Y	0.068 MM	0.068 MM		0.250 MM	0.250 MM
110_OHR_DIFF	ISL9, ISL10	Y	0.068 MM	0.068 MM		0.250 MM	0.250 MM
110_OHR_DIFF	TOP, BOTTOM	Y	0.081 MM	0.081 MM		0.250 MM	0.250 MM

NOTE: These are Intel recommended impedances for PEG, unused on K901.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHR_SE	TOP, BOTTOM	Y	0.165 MM	0.165 MM			
40_OHR_SE	*	Y	0.090 MM	0.090 MM	=STANDARD	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P1MM	*	=DEFAULT	?
BGA_P2MM	*	=DEFAULT	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:5:1_SPACING	*	0.15 MM	7
2:1:1_SPACING	*	0.2 MM	7
2.5:1_SPACING	*	0.25 MM	7
3:1:1_SPACING	*	0.3 MM	7
4:1:1_SPACING	*	0.4 MM	7

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P1M1 Minimum spacing rule set for BGA package
NEH_CLK	*	BGA	BGA_P2M1 Minimum spacing rule set for BGA package
CLK_PCIE	*	BGA	BGA_P2M1 Minimum spacing rule set for BGA package
CLK_SLOW	*	BGA	BGA_P2M1 Minimum spacing rule set for BGA package

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
2X_DIELECTRIC	"	0.140 MM	?
3X_DIELECTRIC	"	0.210 MM	?
4X_DIELECTRIC	"	0.280 MM	?
5X_DIELECTRIC	"	0.350 MM	?
6X_DIELECTRIC	"	0.420 MM	?
7X_DIELECTRIC	"	0.490 MM	?

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
1:1_DIFFPAIR	*	Y	=STANDARD	=STANDARD	=STANDARD	0.1 MM	0.1 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
BS_DIFF_BGA	*	=S_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF	=BS_OHM_DIFF
BS_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
BS_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 85_DIFF_BGA is 85-ohms differential impedance on outer layers and 80-ohms on inner layers.

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_DIFF_BGA	*	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF	=90_OHM_DIFF
90_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
90_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 90_DIFF_BGA is 90-ohms differential impedance on outer layers and 85-ohms on inner layers.

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
100_DIFF_BGA	*	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF	=100_OHM_DIFF
100_DIFF_BGA	ISL3, ISL4	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM
100_DIFF_BGA	ISL9, ISL10	Y	0.075 MM	0.075 MM		0.125 MM	0.125 MM

NOTE: 100_DIFF_BGA is 100-ohms differential impedance on outer layers and 95-ohms on inner layers.

NOTE: 110_DIFF is 110-ohms differential impedance on outer layers and 105-ohms on inner layers

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